

WP2 Deliverable 2.2

Preliminary Sa4CPS concept description

Sa4CPS

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Status: Proposal

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TABLE OF CONTENTS

Abstract.....	5
List of Abbreviations	6
1 Introduction.....	7
2 An overview of the Sa4CPS concept scope.....	8
2.1 <i>Situational awareness related challenges</i>	<i>8</i>
2.2 <i>Use cases set the scope of the Sa4CPS concept development</i>	<i>9</i>
3 Analysis of commonalities in requirements towards the Sa4CPS concept	11
3.1 <i>Data Spaces and Communication</i>	<i>12</i>
3.2 <i>Identity and Access Management (IAM)</i>	<i>15</i>
3.3 <i>Authorization.....</i>	<i>18</i>
3.4 <i>Perception</i>	<i>21</i>
3.5 <i>Comprehension.....</i>	<i>23</i>
3.6 <i>Projection.....</i>	<i>25</i>
3.7 <i>Monitoring.....</i>	<i>27</i>
3.8 <i>Zero Trust System</i>	<i>28</i>
3.9 <i>Discussion of Requirements.....</i>	<i>31</i>
4 Contribution(s) of use cases into the Sa4CPS concept	31
4.1 <i>Autonomous lightweight logistics (UC1)</i>	<i>31</i>
4.2 <i>People and machine position awareness in production or logistics area (UC2)</i>	<i>33</i>
4.3 <i>Application of advanced maps in public safety (UC3)</i>	<i>36</i>
4.4 <i>Critical energy distribution (UC4).....</i>	<i>38</i>
4.5 <i>Security and location awareness of persons (UC5).....</i>	<i>39</i>
4.6 <i>Multirouting secure awareness system for outdoors (UC7)</i>	<i>40</i>
4.6.1 <i>Natlink & Tracker: Enabling Technologies and Cross-Use Case Contributions</i>	<i>41</i>
4.6.2 <i>Polar data synchronization device for multiple wearables data collection</i>	<i>45</i>
4.7 <i>Drone operation control system (UC8)</i>	<i>48</i>
4.8 <i>Management of CPS assets (UC9).....</i>	<i>50</i>
5 Contribution(s) of each technical work package into the Sa4CPS concept	51
5.1 <i>Contributions related to perception and comprehension to the Sa4CPS concept (WP3).....</i>	<i>51</i>
5.1.1 <i>Perception and comprehension</i>	<i>51</i>
5.1.2 <i>Potential contributions from physical environment sensing (task 3.1)</i>	<i>51</i>
5.1.3 <i>Potential contributions from Interaction based sensing of information context (task 3.2)</i>	<i>51</i>
5.1.4 <i>Potential contributions from information and sensor fusion (task 3.3)</i>	<i>52</i>
5.1.5 <i>Potential contributions from modelling and tools for comprehension (task 3.4)</i>	<i>53</i>
5.2 <i>Contributions related to status, projections and actions to the Sa4CPS concept (WP4)</i>	<i>54</i>
5.2.1 <i>Status, projections and actions.....</i>	<i>54</i>
5.2.2 <i>Potential contributions from digital twins and dashboards (task 4.1)</i>	<i>55</i>
5.2.3 <i>Potential contributions from positioning and tracking of objects (task 4.2)</i>	<i>56</i>
5.2.4 <i>Potential contributions from autonomous operations (task 4.3)</i>	<i>56</i>
5.2.5 <i>Potential contributions from detection and responses (task 4.4).....</i>	<i>56</i>
5.2.6 <i>Potential contributions from real-time deep 3D maps and tools (task 4.5)</i>	<i>56</i>
5.3 <i>Contributions related to trust, security and privacy to the Sa4CPS concept (WP5).....</i>	<i>57</i>
5.3.1 <i>Trust, security and privacy.....</i>	<i>58</i>
5.3.2 <i>Potential contributions from identification, verification and access control (task 5.1)</i>	<i>60</i>
5.3.3 <i>Potential contributions from secure information spaces, sharing and overlayed routing (task 5.2)</i>	<i>61</i>
5.3.4 <i>Potential contributions from zero trust services and applications with DLT infrastructures (task 5.3)</i>	<i>61</i>
5.3.5 <i>Potential contributions from sensing of security situation (task 5.4).....</i>	<i>62</i>

6	Enhanced project story	63
7	Links of contributions into the enhanced story	65
	7.1 <i>Potential links of use case contributions into the project story</i>	65
	7.2 <i>Potential links of WP3-5 contributions into the project story</i>	73
8	Preliminary Sa4CPS concept framework description	81
9	Discussion and concluding remarks	85
	References	86

TABLE OF FIGURES

Figure 1. A view to the situational awareness - SA is the perception of environmental elements and events with respect to time or space, the comprehension of their meaning, and projection of their future status [1]. ...8	8
Figure 2. Sa4CPS concept with the application areas, use cases and their potential contribution areas.10	10
Figure 3. Sa4CPS concept, the work plan and technical enablers.11	11
Figure 4. A view of the Requ tool, with added requirement grouping columns.11	11
Figure 5 UC1 Concept.....32	32
Figure 6. High-level overview of UC2.33	33
Figure 7. Deployment architecture of UC235	35
Figure 8. The parts of the sensor nodes.35	35
Figure 9. An example how NSC3 will operate under Zero trust communication with other devices (camera in this example) and how this data will be utilized when planning further actions, if facing unwanted anomalies/objects.36	36
Figure 10. <i>UC4 concept</i>39	39
Figure 11. UC5 targeted ecosystem.....39	39
Figure 12. UC5 approach mapped in the research frame: perception, comprehension, projection & decision40	40
Figure 13. Natlink view to the UC7 concept.41	41
Figure 14. Setup for Raspberry PI Zero W / Raspberry PI 5 + 4G Module47	47
Figure 15. Structure of the experimental prototype.48	48
Figure 16. A view to the UC8 concept.49	49
Figure 17. A view to the UC9 concept.50	50
Figure 18. Scope and Integration of Status, Projections, and Actions in WP4 for Sa4CPS.55	55
Figure 19. A view to the WP5 content.58	58
Figure 20. Summarized description of the enhanced project story.63	63
Figure 21. NSC3 with ValoR AI in project context.68	68
Figure 22. A preliminary view to the Sa4CPS concept framework – main functional blocks.81	81

Abstract

This deliverable describes the preliminary secure situational awareness concept (Sa4CPS concept). First the commonalities of the previously identified requirements towards the concept were analysed, and a set of functional blocks (FBs) have been identified. These FBs, called as data spaces and communication, identity and access management, authorization, perception, comprehension, projection, monitoring and zero trust system, are then clarified relying on the referred requirements. The FBs have been used to structure the preliminary Sa4CPS concept in this report, so that the contributions of UCs and technological tasks towards the concept is a bit more clear. However, it is important to point out that these FBs are still quite abstract in nature, and plan is to continue and open their internal functional structure, relationships and interfaces to practical objects in which the real secure situational awareness capabilities are needed.

The content, focus and plans of the UCs have been described in this report so that their potential contribution areas into the FBs of the Sa4CPS concept are clarified. It is clear now that each UC has a specific role in the development of Sa4CPS concept, its FBs and links to the technological tasks of the project WPs3-5. However, there is still need to continue this work to further clarify the content of the identified contribution areas of the UCs into the concept. The technological tasks of WPs3-5 have been quite much still in the background, and therefore their relationships to the FBs are clarified in this report only in a preliminary way. It is expected that this deliverable provides a solid starting point and working framework for the R&D works in these technological tasks. In addition, the links to the project UCs and the Sa4CPS concept is now a bit clearer.

The project story has been structured and enhanced, and the links of the UCs and WP3-5 contributions towards solving the visions and challenges indicated by the story are now a bit more obvious. This is envisioned to help all of the project partners to adjust their focus and plans so that the resulting actions are beneficial to their business cases but also to the results deployment in the port environment.

As the main result, the preliminary Sa4CPS concept is described here so that the targets are now clearer and the technological WPs3-5 and the UCs of the project can now proceed in the technological R&D towards specification and design of their solutions. The plan is to let them now proceed deeper into the technological development, and then in the next iteration of the project come back and make a new version of the Sa4CPS concept description.

List of Abbreviations

Abbreviation	Explanation
FB	Functional block
DLT	Distributed ledger technology
Req	requirement
SA	situational awareness
Sa4CPS	Secure situational awareness for critical cyber-physical systems.

1 Introduction

The purpose of this deliverable is to represent the first preliminary description of the secure situational awareness concept (Sa4CPS concept) framework architecture. The preceding phase of the project has analysed the requirements identified by the project use cases (UC). These requirements have been taken as input for this work and further analysed in order to identify required functional blocks (FBs). In parallel, the plans of each UC have proceeded so that the functionalities that the UC could contribute towards the concept have been studied. Accordingly, the planning of WP tasks has proceeded towards identifying the related requirements and functionalities that the WP tasks solutions should support. In addition, the links of each contribution towards the project storyline around port environment have been refreshed. As the result, the 1st preliminary Sa4CPS concept description has been provided, and initially described in this deliverable.

A short description of the situational awareness related terminology is provided in the following. The term *situational awareness* considers multiple functional aspects such as perception, comprehension, projection, decision making and execution of actions. In addition, the situational awareness can be discussed also from the points of view of the entities, e.g. people, devices and service systems. For example, as a system of devices and information that aims to bring to a person's cognitive situational awareness those physiological variables that concern him or her and the environment, that have an impact on the health, well-being and safety of an individual or a group of people, physical objects and businesses, or digital twins. A digital twin of the virtual/real-world spaces of a company/organization continuously monitors the available resources for observation, understanding, projection, and decision-making in an intelligent way. In addition, it can be discussed from a business point of view, considering issues like productivity, profitability, and improving operational decision-making capacity in matters related to logistics, mobility, energy production and use, and security. Thus, the targeted secure situational awareness system includes various mechanical objects, sensors, measuring devices, intelligent algorithms and applications that are responsible for control and producing situational information on operations at the interfaces of humans and stakeholders to support decision-making. In the situational picture, decision-making is influenced by both the technical safety aspects of the systems and the individual's self-regulation mechanisms, which determine the understanding of the current state of health and well-being, long-term historical development and the future scenario according to which goals are set for the activities to achieve the desired results. Thus, the situational picture affects both concrete actions and mental models, where values, ideals, desires, needs, goals, behaviour, situational awareness and decision-making form a single entity that requires the processing of self-regulation mechanisms and the guidance of actions with the methods described by the organizations implementing this project.

The rest of this deliverable are organized as follows. An overview of the Sa4CPS concept scope is described in the section 2. The commonalities of the requirements towards the functionalities of the Sa4CPS concept are discussed in section 3. After that, the possible contribution areas of project UCs towards the concept are discussed in section 4. Respectively, the possible contributions of project tasks towards the concept are discussed in section 5. The project story has been refreshed in section 6, and the links of contributions into it are updated in section 6. After that, the preliminary Sa4CPS concept framework has been described. Finally, discussion about the next steps and concluding remarks are discussed in sections 9 and 10 respectively.

2 An overview of the Sa4CPS concept scope

2.1 Situational awareness related challenges

When looking at the critical areas such as ports, solutions related to several different sectors are needed for smart and secure situation aware operation - for example, logistics, security, mobility and energy. The focused use cases from these sectors share problems and technological needs related to *people, physical objects, and companies/digital twins*. An individual person needs to get information from the current state of the environment and from all the accessible data spaces of the currently active personal role for perception of elements in current situation. This information is needed for comprehension of the current situation, to reason projection of the future status and finally to make smart decisions and actions accordingly, Figure 1. A specific physical object may have local sensors, cameras, lidars etc. to sense what is happening nearby (local situational awareness). The physical object may also use ad hoc radio capabilities to get information from other local physical objects nearby. In addition, it is maybe possible to enrich the local situational awareness using all the accessible information sources from networks. All this information can then be used to make comprehension of the current situation, make projections of the future status and finally to make smart actions automatically. Respectively, the digital twins of the virtual/real world spaces of a company/organization continuously monitor all kinds of accessible resources available to make perception, comprehension, projection, and decision making for actions in a smart way. [1-5]

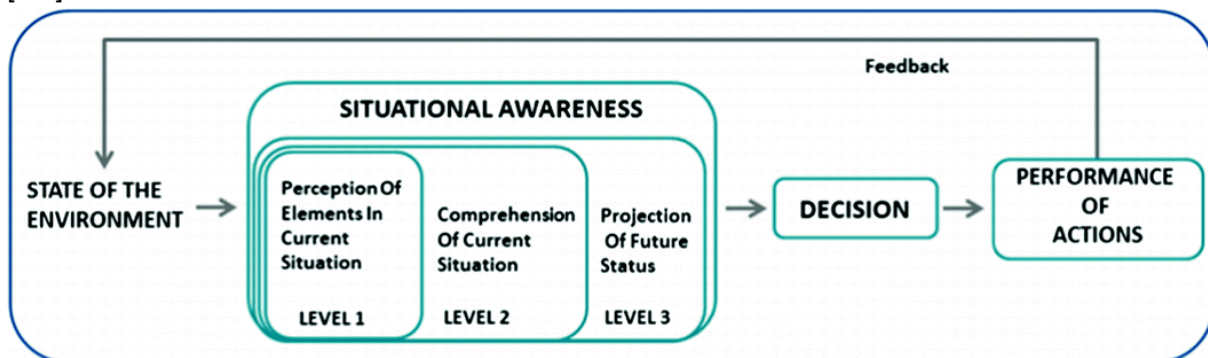


Figure 1. A view to the situational awareness - SA is the perception of environmental elements and events with respect to time or space, the comprehension of their meaning, and projection of their future status [1].

Today, these processes are usually executed using local processing and in tightly coupled company specific silos without sharing the information over stakeholder borders. For example, when goods proceed in a logistics chain from one stakeholder to another, crossing the border between data spaces of stakeholders is needed to track the product in the supply chain and finally to the retailer/end user. Another example can be found in the mobility environment: There are mobile physical assets from multiple stakeholders and disciplines and people acting in an ad hoc way. Each of these assets have an impact to the safety of the other entities and people, and therefore situational awareness crossing the stakeholder borders is inherently needed. A third example can be found in the energy sector: The digital twins of energy distribution systems need information from several sources hosted by different stakeholders to know the situation properly, e.g., frequency of the physical electricity grid, current demand response balance, future production and consumption estimations, flexibility capabilities of energy consumers, producers, and their distributed energy resources. Based on this situation awareness information, the system may be able to decide what actions should be done in the day-ahead markets, and what needs to be done in the intra-day real-time situations with the energy distribution network and with related energy flexibility markets.

The fourth example deals with public safety security systems, which have traditionally been quite separated from other industrial systems because of confidentiality reasons. However, today the critical industrial systems e.g. in logistics, mobility and energy have elements that trigger essential challenges compromising public safety. The situation awareness crossing the sectorial borders could establish a better view what is happening in the situation, and what should be done by the public authorities.

Another shared problem and technological needs arise from the security risks. The result from the realization of such risks may be e.g., accident, physical or virtual attacks, blocked use of critical resources, or misuse of privacy sensitive data, which can lead to loss of human lives, lack of critical human needed resources, bankruptcy of companies or even societal instability. The traditional perimeter-based network security model, applied today basically in all the focused use cases, has still serious risks to the assets of the enterprise system, because an attacker may have found a way inside the enterprise system. The trusted Internet connections and perimeter firewalls provide strong Internet gateways, which can block attackers from the Internet. However, they are less useful for detecting and blocking attacks coming from inside the network and they cannot protect subjects outside of the enterprise perimeter such as remote workers, cloud-based services, edge devices etc. Remotely working employees increase the risks for security threats. Phishing of credentials (such as username and password) is more likely to happen, and the likelihood for malicious user to be able to access the resources in the enterprise system increases. The Zero Trust security model has been developed to contribute towards solving these problems. It focuses on resource protection so that trust is never granted implicitly but must be continually evaluated [2]. It is an end-to-end approach to enterprise resource and data security that encompasses identity (person and non-person entities), credentials, access management, operations, endpoints, hosting environments, and the interconnecting infrastructure. The initial aim is on restricting resources only to those who have a need to access and grant only the minimum privileges (e.g., read, write, delete) needed to perform the required mission. When speaking about critical scenarios related to logistics, mobility, energy, and public safety, it is obvious that the traditional perimeter-based network security model is not enough, but the Zero Trust type of security model needs to be applied instead.

The focused critical use cases share these problems, and therefore the major objective of the project is to research and develop the secure situational awareness (SA4CPS) concept together. This has been planned to be done so that each use case has identified their contribution to the concept, which will be evaluated as the part of the referred use case experimentation. The critical port area of Oulu needs the results from these use cases to enable secure situation awareness for operation, and it can act as a potential pilot environment for conceptual validation.

2.2 Use cases set the scope of the Sa4CPS concept development

The Sa4CPS concept with the application areas, use cases and their potential contribution areas are depicted in Figure 2.. The project has eight use cases, that have both their own surrounding business ecosystem and a possible application in the port environment. The surrounding business ecosystems are represented as clouds in the figure. Each use case will focus on certain aspects of the Sa4CPS concept, represented as rectangular boxes. The Sa4CPS concept is represented in the centre of the figure, consisting all these aspects, or *technical enablers*, and the combined result will be Sa4CPS concept framework, the elements of which can be applied e.g. in the focused port environment.

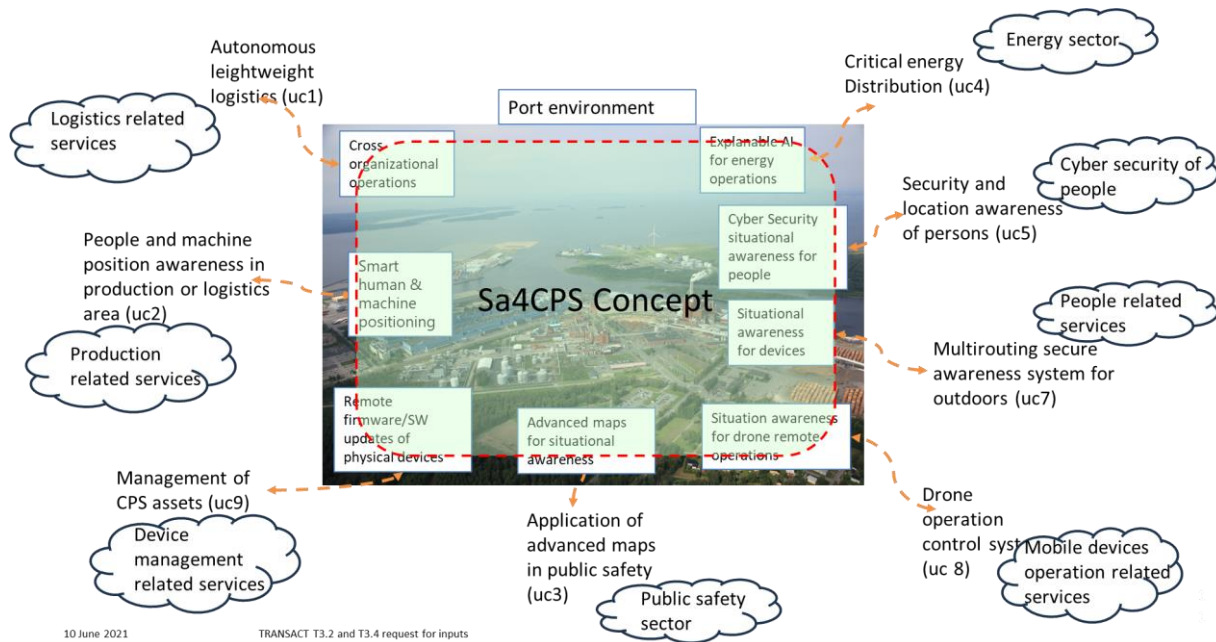


Figure 2. Sa4CPS concept with the application areas, use cases and their potential contribution areas.

The Sa4CPS concept, the work plan and technical enablers are depicted in Figure 3. The secure situational awareness functionalities are divided to work packages according to the functionalities that are required in the Sa4CPS system as follows:

- WP3: perception and comprehension
- WP4: status, projection and actions
- WP5: trust, security and privacy

Each work package area is envisioned to create a set of technical enablers, that are contributed by the project UC(s). Together these technical enablers establish the evaluated core of the secure situational awareness concept (Sa4CPS concept).

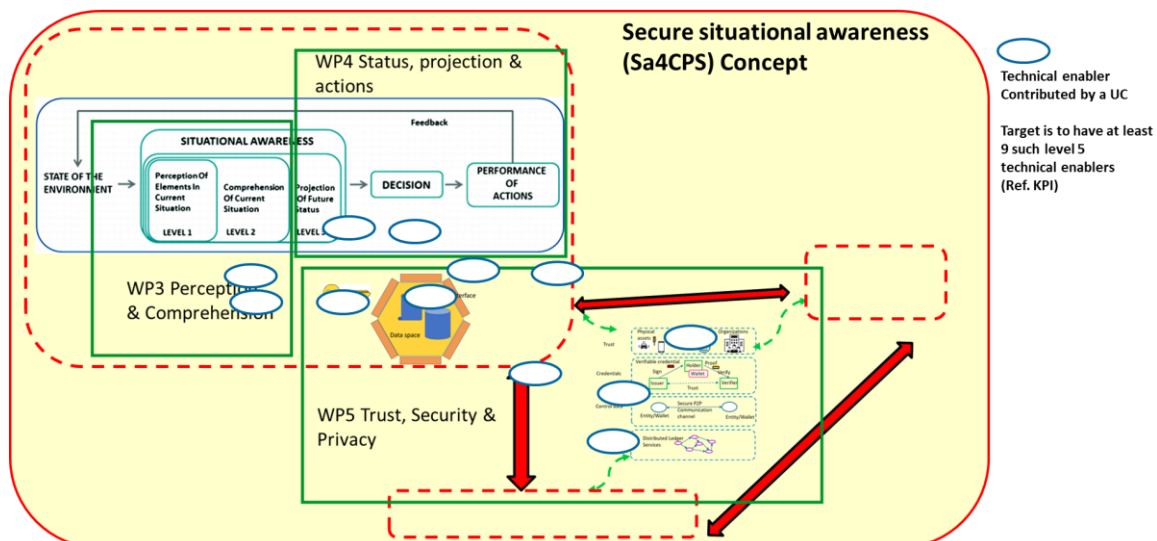


Figure 3. Sa4CPS concept, the work plan and technical enablers.

3 Analysis of commonalities in requirements towards the Sa4CPS concept

This section analyses the commonalities of the requirements towards the functionalities of the Sa4CPS concept. The input material for this work have been the requirements documented in the excel based Sa4CPS requ tool, and its version 1.0¹, and deliverable D2.1².

Based on requirements that were assigned by the use cases towards the Sa4CPS concept (specifically focused in the WP2 of the project), some common functionalities seem to arise, and they could be considered as basis for SA4CPS concept architecture functional blocks (FBs). The way these common FBs were identified was by multiple passes of examining the requirements. In each pass, the requirements grouping was refined, and the requirements assigned according to the new grouping. Figure 4 shows an example of requirements being grouped using the Requ tool.

A B C			R S T U V W X Y							
UPDATE ☐ Delete unreferenced ☒ Sort automatically										
WP 2: Sa4CPS concept architecture										
Req Id	Name	Description	DATA&COMM	IAM	AUTH	PERCEPT	COMPR	PROJECT	MONITOR	ZEROTR
REQ_STORY.1	Road and Weather Conditions Information	The system must provide information about road and weather conditions, such as slippery road surfaces, on the way to the port area and within the port area.	x			x	x			
REQ_STORY.10	Area Monitoring System	The area monitoring system must track the movements of vehicles, work machines, and people in the area and intervene if necessary.	x		x	x	x			
REQ_STORY.11	Hazard Alert and Communication	Zero trust authentication and authorization of video and audio material	x				x	x		
REQ_UC3.3	Unloading Time Notification	Use zero trust principles to verify authenticity of recorded material and authorize its usage	x		x					x
REQ_STORY.2	License Plate Recognition	The application must offer information about the next possible unloading time to the driver.	x			x	x	x		
REQ_STORY.3	Cabin Occupancy Check	The system must recognize the vehicle's license plate at the port gate upon arrival.	x			x	x			
REQ_STORY.4	Vehicle Condition Scanning	Cameras must check that there are no other persons in the truck cabin at the port gate.	x			x	x			
REQ_STORY.5	Driver Arrival Notification	Cameras and sensors must scan the vehicle at the port gate to detect any deviations in load securing, load condition, or vehicle condition before arrival.	x			x	x			
REQ_STORY.6	Drone Assistance for Route Instructions	The identification system must send information about the driver's arrival at the port area to the operator, who has prepared for the arrival based on the advance notice and ensure that the autonomous work machine is waiting in the warehouse.	x		x			x		
REQ_STORY.7	Stop Sign Projection	The system must enable the driver to order a drone for assistance, which projects route instructions and warnings onto the road.	x			x				
REQ_STORY.8	Correct Route Notification	The drone or fixed projector must project a stop sign onto the ground to make the driver stop the truck and allow the container handler to pass. When it is safe to continue, the drone must project the information onto the ground.				x	x	x		
REQ_UC1.1	Asset Integrity & Authentication	The mobile application must notify the driver to return to the correct route if they turn in the wrong direction at an intersection.				x	x			
REQ_UC1.10	Situational Awareness & Alerts	The system must ensure asset integrity using cryptographic methods.	x				x	x		
REQ_UC1.12	Secure Communication for Forklifts	A predictive analytics model must be integrated for risk assessment.	x							
REQ_UC1.13	Web Platform and Graphical Visualization	Secure Machine-to-Machine (M2M) communication must be provided between autonomous forklifts, IoT platforms, and warehouse systems.							x	
REQ_UC1.15	Secure IoT Data Infrastructure	A web platform should be developed with a user-friendly interface for monitoring forklifts and visualizing real-time data.	x							
		A robust data infrastructure must be established for securely storing, processing, and transmitting data from IoT devices. Data produced within the IoT network should be transmitted to the external network within 1 second.								
		An authentication mechanism providing Role-Based Access Control (RBAC) must be developed, and access should be								

Figure 4. A view of the Requ tool, with added requirement grouping columns.

The following is a suggestion of component definition, with all gathered WP2 requirements listed under their respective components. Naturally, some requirements will address more than one component and are in that case included below under all relevant components. In several cases, the requirement to target a specific component was not evident from the requirement text, as the component structure had not yet

¹ Link to the referred requirements excel sheet in the project internal Teams space: [Sa4CPS_requirements_1.0.xlsm](#)

² Link to the deliverable in project internal Teams space: [Sa4CPS_deliverable-D2.1-V1-0.docx](#)

been defined when the requirements were gathered. In these instances, it was assessed whether fulfilling the requirement would likely involve the services of the relevant component.

Finally, while the content of situational awareness blocks concerns WP3 and WP4 more than WP2, quite a few requirements set functional requirements addressing situational awareness capability. These components were included in the component division, but their roles and content are mostly left to WP3 and WP4. Here, the components of Perception, Comprehension, and Projection are included, as their needs are most involved with other components in order to access the necessary data, and controlling and authorizing that access with Zero Trust principles.

3.1 Data Spaces and Communication

Most requirements target data storage and verification of data integrity, or communications for transferring the data. These requirements suggest that, given multiple options for data storage and transfer (including related communications) some kind of harmonized approach to data may be needed. Data storage and transfer can be at least partially implemented by, for example, a digital twin, and a digital twin may also access data from external sources.

Other functionality regarding data (spaces) but not explicitly addressed by any requirement may include such as data discovery, data catalogues / descriptors, and external interfaces (connectors). Table 1 below lists requirements gathered from Use Cases that target or significantly involve Data and / or Communication.

Table 1. WP 2 requirements targeting data and/or communications.

Req Id	Name	Description
REQ_STORY.1	Road and Weather Conditions Information	The system must provide information about road and weather conditions, such as slippery road surfaces, on the way to the port area and within the port area.
REQ_STORY.10	Area Monitoring System	The area monitoring system must track the movements of vehicles, work machines, and people in the area and intervene if necessary.
REQ_STORY.11	Hazard Alert and Communication	The system sends a hazard alert and action instructions to all persons in the area if necessary. The system allows communication and sharing of information about the progress of events.
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_STORY.2	Unloading Time Notification	The application must offer information about the next possible unloading time to the driver.
REQ_STORY.3	License Plate Recognition	The system must recognize the vehicle's license plate at the port gate upon arrival.
REQ_STORY.4	Cabin Occupancy Check	Cameras must check that there are no other persons in the truck cabin at the port gate.
REQ_STORY.5	Vehicle Condition Scanning	Cameras and sensors must scan the vehicle at the port gate to detect any deviations in load securing, load condition, or vehicle condition before arrival.
REQ_STORY.6	Driver Arrival Notification	The identification system must send information about the driver's arrival at the port area to the operator, who has prepared for the arrival based on the advance notice and ensure that the autonomous work machine is waiting in the warehouse.
REQ_STORY.7	Drone Assistance for Route Instructions	The system must enable the driver to order a drone for assistance, which projects route instructions and warnings onto the road.
REQ_UC1.1	Asset Integrity & Authentication	The system must ensure asset integrity using cryptographic methods.

REQ_UC1.12	Secure Communication for Forklifts	Secure Machine-to-Machine (M2M) communication must be provided between autonomous forklifts, IoT platforms, and warehouse systems.
REQ_UC1.15	Secure IoT Data Infrastructure	A robust data infrastructure must be established for securely storing, processing, and transmitting data from IoT devices. Data produced within the IoT network should be transmitted to the external network within 1 second.
REQ_UC1.19	Reliability and Scalability	It must be capable of processing large amounts of data and delivering high performance even in challenging environments.
REQ_UC1.21	Data Processing	Fast and reliable algorithms are required to process the data collected in real-time.
REQ_UC1.25	Zero-trust device identity verification (node authentication)	Must ensure secure Machine-to-Machine (M2M) communication among autonomous forklifts, supply chain vehicles, IoT platforms, and warehouse systems. It should generate one-time passwords suitable for high throughput and be compatible with cyber-physical system devices in accordance with ISO21434. The secure gateway and hardware security module must operate in coordination.
REQ_UC1.26	Accountability of operations over a decentralized network	For data coming from forklift operations and logistics vehicles in the working environment, low latency (under 1 millisecond) must be ensured, and blockchain services suitable for this must be integrated with the cyber-physical system. For accountability, all transaction steps must be recorded by buyers, sellers, and approving authorities in the supply chain (immutability).
REQ_UC1.27	Secure storage of sensitive data	Data received from edge devices (forklifts or vehicles) or services must be securely stored via necessary encryption and hashing. A hardware security module and database must work in coordination for this purpose.
REQ_UC1.28	Secure transmission of sensitive data	Data coming from edge devices (forklifts or vehicles) or services must be securely transmitted end-to-end with encryption and hashing. Hashes of data generated within the decentralized network must be transmitted to or retrieved from the blockchain within a maximum of 1 second. For this purpose, the secure gateway, hardware security module, and database must operate in coordination.
REQ_UC1.3	Asset Integrity & Authentication	Digital twins must be implemented for real-time virtual representation of physical assets.
REQ_UC1.7	IoT-Based Monitorization	IoT gateways must collect and transmit real-time monitoring data.
REQ_UC2.10	Secure communication	Communication between devices and cloud must be secured
REQ_UC2.2	Real-time data collection	It should be possible to collect data from sensors and automatically store them in the cloud
REQ_UC2.3	Data availability	It should be possible to query and retrieve the data in the cloud for ML training
REQ_UC2.4	Manage and visualize the data	There should be an interface to manage and visualize the data
REQ_UC2.5	Data scalability	The system must scale to datasets of at several GBs of data
REQ_UC2.6	Data annotation and pre-processing	Tools should be available that can be used for annotating and pre-processing the collected data. These tools should be specialized for ToF cameras, radar and UWB.
REQ_UC2.9	Online update of ML-based predictors from cloud to the edge	Online update of ML-based predictors from cloud to the edge
REQ_UC3.10	Self-hosted 3D map	3D map needs to be hosted on internal network, no connection to external APIs etc. required
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_UC3.8	Data sources for anomaly detection	Define what sources (e.g. video) to use for data anomaly detection
REQ_UC4.1	Missing data replacement	Missing data needs to be replaced with, e.g. baseline data
REQ_UC4.10	IoT Data Retrieval	Integration of IoT devices

REQ_UC4.13	Energy Community Overview	Community data presentation, including consumption and production information by consumer / prosumer
REQ_UC4.2	Data availability	15-minute energy usage data needs to be available on time
REQ_UC7.1	Cybersecurity	Data should be secure, at least AES-128 level and data transfer messages shall not have overhead
REQ_UC7.10	Multicomunicator radio interface	Radio interface between Multi-communicator and tracking device
REQ_UC7.11	Multicomunicator server interface	Server interface from Multicomunicator to server
REQ_UC7.12	Multicomunicator 3.party interface	Possible radio interface for 3. party devices, e.g. weather sensors
REQ_UC7.2	Data sharing	Data sharing over radio to receiver, from receiver to mobile and from mobile to server and from there to next mobile
REQ_UC7.20	Transfer media Optimized data traffic between components	Optimized and prioritized data connections, which are using reasonable amount of radio resources and battery
REQ_UC7.25	Multiplace Digital Twin Specification	Create Digital Twin, of the device, dog and handheld, which is stored in Mobile app and in Server, and is Read / Write and handles user rights and flow control of the updates and visibility. Remebering that update to
REQ_UC7.27	Multiplace Digital Twin History	Historical stored to view the state of the Digital Twin at any given time. Example you can make search for certain timestamp of the device / dog / person, lets say 10.12.2024 08:24:12 , as a result of the query to twin history, you get situation of the device / dog / person at time stamp, you get all settings, location, all sensor data which was valid in this specific timestamp, taking care also privacy and data level considerations. Digital twin history data vewing rights is dependent on the viewer, time of the data and level of the user
REQ_UC7.29	Send group invitation to single users by the flow user account	From the group solution system, it is possible to invite people to join groups that have a wearable device and a Flow user account. In connection with the invitation, it is checked whether the person is the right person and at the same time the GDPR provides information about the rights and options for sharing data between group members
REQ_UC7.30	Reading and connecting multiple wearable device by synchronization hub	When the operation starts, the synchronization device searches for and establishes a connection to the devices that are connected to the group members devices which are within range of the sync device. If necessary, the synchronization device sets the configurations for the operation of the wearable device, with which it is appropriate to record the activity
REQ_UC7.31	Start recording live data from wearables	The group instructor can start and stop data collection with the synchronization device for each device at the same time or separately by using a dedicated mobile application
REQ_UC7.32	Send wearable data from synchronization hub by SDK interface to mobile device	The wearable devices send biometric and inertial data to the synchronization device in real time via the SDK/BLE interface to the synchronization hub and again by 4-5G/LTE interface to the mobile device's application, where the data can be viewed based on the configurations according to the presets of the instructor's roles or by switching to the views to if person in his role has rights to see the data
REQ_UC7.34	Send wearable data from synchronization hub by LTE interface to Polar Flow backend system	The synchronization device simultaneously opens both the SDK and LTE connection to the Flow backend system and updates the data view of the team members if necessary due to the entries made during the real-time control operation. Otherwise, the synchronization device can be set to send data from wearable devices to the backend system, e.g. Every 5,10,30,60, or 90 seconds, or the entire dataset after the operation has ended and the devices have stopped data collection.

	Gather team related data from Flow backend to Group Solution Service's diary	Data synchronization outside of group activities when an individual independently carries out a pre-planned activity. The person syncs the device after the operation ends, after which the data is automatically transferred from the personal Flow backend account to the Group Solution account. Data transfer is only possible if the person has a) given permission for the data transfer, b) the person is a member of the group and c) the guidance agreement is active. The information is only visible to those persons to whom the individual gives permission a) to view, comment and redesign the activities based on the information and b) to the persons with whom the individual cooperates to carry out the activities
REQ_UC7.35	Data connectivity and cyber-security management	
REQ_UC7.4		SYNCHRONIZATION HUB technical specification for real-time (SDK) data transfer from a wearable device to third-party systems using LTE technology (Perception)
REQ_UC9.1	Secure remote updates	Software on cyber physical systems should be able to be updated remotely, regardless of underlying hardware and OS.
REQ_UC9.2	Secure remote control	Cyber physical systems can be remotely controlled securely, over an unsecure and unstable network.
REQ_UC9.8	Near real-time performance of status updates	Status updates of the state of the cyber physical system should be near real-time if the network connectivity allows it

Summary of Requirements for Data Spaces and Communication

While most operations require some handling of data, there are some specific categories that arise from the requirements that can be addressed by the SA4CPS concept.

Data integrity and performance. There is a need to ensure the authenticity (REQ_UC3.3), integrity (REQ_UC1.1, REQ_UC1.3) and performance (REQ_UC1.3, REQ_UC1.19, REQ_UC1.21) of systems and data.

Data management infrastructure and data availability. Requirements targeting IoT and other data infrastructures (REQ_UC1.15), with monitoring (REQ_UC1.7, REQ_UC4.13, REQ_UC2.4) and data availability (REQ_UC4.1, REQ_UC4.2, REQ_UC4.13, REQ_UC3.10). **Digital Twins** are also explicitly addressed by some requirements (REQ_UC1.3, REQ_UC7.25, REQ_UC2.27).

Data collection and ML readiness. In order to support further machine learning operations and support operational insight creation, several requirements target the data lifecycle: data collection including notifications (REQ_STORY.2, REQ_UC2.2, REQ_UC4.10), making data available for further processing (REQ_UC2.3, REQ_UC2.5) and providing necessary annotation, preprocessing and ML model update functionality (REQ_UC2.6, REQ_UC2.9).

Wearables, synchronization and group data management. Requirements arising from (wearable) devices target data transfers and secure connectivity (REQ_UC1.12, REQ_UC7.1, REQ_UC7.4), data and device synchronization (REQ_UC2.10, REQ_UC9.1, REQ_UC9.2, REQ_UC9.8), and group data sharing (REQ_UC7.29).

3.2 Identity and Access Management (IAM)

From the requirements it becomes evident that there is a repeated need for specifying users and groups that have access to a specific resource. Thus, it is necessary to identify person and non-person entities and define various access policies for those entities into resources. Resources may be defined, for example, in a data space or a digital twin, or a resource can be a communication endpoint. Access to the various resources that, depending on where the resource is located, may be defined in different ways.

Thus, there is a need for some kind of uniform way to manage resource access, with support for multiple access types (role-based, attribute-based, policy-based).

Also, the identification schemes and ways to prove identity differ between enterprise, social and open-source identity providers, which also requires consolidation. Furthermore, to establish the required control for Zero Trust, it is necessary that the IAM system enables continuous evaluation of access conditions (e.g., Verifiable Credentials that prove right to access may expire or get revoked) and changing of the access policies accordingly. Identity and Access Management related requirements from Use Cases are listed below in Table 2.

One example of how Identity and Access management is addressed by use cases is Polar who comply with the general GDPR requirements to protect privacy in the Polar Flow and Flow App services currently in use. The Polar Guidance System, a group management service system that is the subject of research and development in this project, integrates private users with each other in the group formation process through the Polar Flow system. This ensures that each user who has rights within the group has already been identified based on the individual's personal and contact information as well as the unique ID number of the wearable device they are using. Correspondingly, the content of guidance services follows a grouping according to the purpose of use, which can be cognitive, physical, mental or social in nature.

Table 2. WP2 requirements targeting Identity and Access Management.

Req Id	Name	Description
REQ_UC1.16	Role-Based Access Control (RBAC)	An authentication mechanism providing Role-Based Access Control (RBAC) must be developed, and access should be restricted based on user roles.
REQ_UC1.24	Zero-trust user authentication and authorization	Must enable the authentication and authorization of relevant individuals and services in the working environment or during remote access. It should be able to generate one-time passwords suitable for high throughput and operate in compliance with ISO21434 standards, in harmony with services in the cyber-physical system. For this purpose, the secure gateway and hardware security module should work in coordination.
REQ_UC1.3	Asset Integrity & Authentication	Digital twins must be implemented for real-time virtual representation of physical assets.
REQ_UC2.11	Data access control	Access to data should be protected by role-based access control
REQ_UC3.1	Zero trust user authentication and authorization	Apply zero trust principles to user authentication and authorization
REQ_UC3.2	Zero trust device authentication and authorization	Apply zero trust principles to device authentication and authorization
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_UC4.10	IoT Data Retrieval	Integration of IoT devices
REQ_UC4.9	Energy Community Setup	Allow for in-platform community creation
REQ_UC7.2	Data sharing	Data sharing over radio to receiver, from receiver to mobile and from mobile to server and from there to next mobile
REQ_UC7.25	Multiplace Digital Twin Specification	Create Digital Twin, of the device, dog and handheld, which is stored in Mobile app and in Server, and is Read / Write and handles user rights and flow control of the updates and visibility. Remebering that update to

REQ_UC7.27	Multiplace Digital Twin History	Historical stored to view the state of the Digital Twin at any given time. Example you can make search for certain timestamp of the device / dog / person, lets say 10.12.2024 08:24:12 , as a result of the query to twin history, you get situation of the device / dog / person at time stamp, you get all settings, location, all sensor data which was valid in this specific timestamp, taking care also privacy and data level considerations. Digital twin history data vewing rights is dependent on the viewer, time of the data and level of the user
REQ_UC7.28	Create and configure certain goal oriented groups according to the persons bio- and sociographic features	The group owner with user rights in the group management system creates and configures the group or groups that are intended for individuals with a specific group goal, bio and sociographic profile
REQ_UC7.29	Send group invitation to single users by the flow user account	From the group solution system, it is possible to invite people to join groups that have a wearable device and a Flow user account. In connection with the invitation, it is checked whether the person is the right person and at the same time the GDPR provides information about the rights and options for sharing data between group members
REQ_UC7.30	Reading and connecting multiple wearable device by synchronization hub	When the operation starts, the synchronization device searches for and establishes a connection to the devices that are connected to the group members devices which are within range of the sync device. If necessary, the synchronization device sets the configurations for the operation of the wearable device, with which it is appropriate to record the activity
REQ_UC7.32	Send wearable data from synchronization hub by SDK interface to mobile device	The wearable devices send biometric and inertial data to the synchronization device in real time via the SDK/BLE interface to the synchronization hub and again by 4-5G/LTE interface to the mobile device's application, where the data can be viewed based on the configurations according to the presets of the instructor's roles or by switching to the views to if person in his role has rights to see the data
REQ_UC7.33	Person guidance/ coaching according to the live data; mobile device data preview options	Based on the simultaneous monitoring of biometric and inertial data for the entire group, the instructor makes decisions and entries during the activity and, if necessary, intervenes in the activities immediately based on the decisions in accordance with the general guidance plan or special needs. At the same time, other instructors can also make notes and observations of the activities in accordance with their own role, which can be used to specify the need for changes to the action plan to be made later.
REQ_UC7.34	Send wearable data from synchronization hub by LTE interface to Polar Flow backend system	The synchronization device simultaneously opens both the SDK and LTE connection to the Flow backend system and updates the data view of the team members if necessary due to the entries made during the real-time control operation. Otherwise, the synchronization device can be set to send data from wearable devices to the backend system, e.g. Every 5,10,30,60, or 90 seconds, or the entire dataset after the operation has ended and the devices have stopped data collection.
REQ_UC7.35	Gather team related data from Flow backend to Group Solution Service's diary	Data synchronization outside of group activities when an individual independently carries out a pre-planned activity. The person syncs the device after the operation ends, after which the data is automatically transferred from the personal Flow backend account to the Group Solution account. Data transfer is only possible if the person has a) given permission for the data transfer, b) the person is a member of the group and c) the guidance agreement is active. The information is only visible to those persons to whom the individual gives permission a) to view, comment and redesign the activities based on the information and b) to the persons with whom the individual cooperates to carry out the activities

Create and configure activity/ training plans for certain goal oriented groups according to the persons bio- and sociographic features REQ_UC7.36	The group instructor collects the information of the entire group and individual group members separately in the form of a report, prepares a summary of the report and draws conclusions on further measures, which include a) continuing according to the common plan, b) changing and separating the joint plan for some members of the group, or c) amending the joint action plan in part or in full and separating the individual plans that continue according to the previous plan.
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Summary of Requirements for Identity and Access Management

Identity and Access Management are a common requirement for nearly all data usage, and thus implicitly present in all of the above requirements. In addition, there are explicit requirements for **authentication and role-based access control** regarding users, services and devices (REQ_UC1.24, REQ_UC3.1, REQ_UC3.2, REQ_UC2.11). **Digital twins** also require consideration with respect to who is able to access which data (REQ_UC1.3, REQ_UC7.25, REQ_UC7.27).

Group and community data management sets requirements for data access control within groups of users. These requirements involve the creation and configuration of, and notifications to the groups (REQ_UC4.9, REQ_UC7.28, REQ_UC7.29), and controlling data and devices of the groups (REQ_UC7.30, REQ_UC7.33, REQ_UC7.35, REQ_UC7.36).

3.3 Authorization

Several requirements cover actions such as seeing data controlled by others, transferring (uploading) data, and setting / enforcing explicit permissions for certain operations. These actions may further depend on agreements and other conditions that do not necessarily directly translate into access management; while IAM considers *how* identities and roles are defined for an access management system, authorization considers *whether* and under which conditions the access should be granted. In some cases, they are automatically aligned (such as members of an access control group automatically having access to data of that group) but as resources are scattered among heterogeneous systems with different access policies and identities, it becomes increasingly important to be able to specify in detail who is allowed to do what and with which resource. According to Zero Trust principles, authorization should be possible to evaluate continuously.

Access permission should, of course, be possible to define for every resource, not just data stored on servers – for example, a person should have the authority over whether to allow their data to be collected. This also emphasizes the role of authorization in forming communities, where also *trust hierarchies* may become relevant. Trust hierarchies can be used to define relationships between different entities based on proven ownership of the entity (e.g., a company that is a member of a community), and specific definitions of the roles and rights the entity has in said community.

Requirements for Authorization gathered from Use Cases are listed below in Table 3.

As a use case example, in Polar's guidance services, individuals, groups, and content are connected to each other in a way that creates new information about the purpose, goals, needs and development of individuals. In this case, it is possible to create causal rules for the needs of action (operations and behavior) management and create a basis for dealing with the situational awareness of individuals and groups in the manner described in this project. If the information obtained and created during the process is confidentially available to all those who have given permission for the creation and use of the information (establishing a supervision relationship between the supervised and the supervisor/ instructor/ counsellor/ coach), it is possible to use the information

as a kind of "swarm intelligence" that promotes the functional capacity, performance and well-being of each individual much more effectively than the experimental and individual "trial and error" method.

In the process of creating groups, groups are functional (**horizontal**) entities that behave in a way that is typical for them, such as sports (performance of the team), heavy traffic (traffic safety), or fire and rescue services (work and operational capability). Correspondingly, individuals within groups have different roles (player, coach, firefighter, supervisor), which means that tasks, responsibilities, obligations and rights vary when using the Polar Guidance System.

The production of the content needed in the guidance of activities is aimed at developing the functional capacity of the individual and the group and achieving the set goals. Because the group has been formed according to the goal, the guidance of the individuals and the group is also in line with the goal. For this reason, the content of the guidance has also been drawn up from the starting point of individuals, the group and the achievement of goals. The use of **vertical** information is based on data obtained from wearable devices (Perception-Comprehension), which makes it possible to create a feedback guidance process based on the review of activities, content and results (Projection). The guidance process can be both manually implemented with the help of data mining and mass customization principles, as well as guided by biometrics-based machine learning systems.

Table 3. WP2 requirements targeting Authorization.

Req Id	Name	Description
REQ_STORY.10	Area Monitoring System	The area monitoring system must track the movements of vehicles, work machines, and people in the area and intervene if necessary.
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_STORY.6	Driver Arrival Notification	The identification system must send information about the driver's arrival at the port area to the operator, who has prepared for the arrival based on the advance notice and ensure that the autonomous work machine is waiting in the warehouse.
REQ_UC1.16	Role-Based Access Control (RBAC)	An authentication mechanism providing Role-Based Access Control (RBAC) must be developed, and access should be restricted based on user roles.
REQ_UC1.24	Zero-trust user authentication and authorization	Must enable the authentication and authorization of relevant individuals and services in the working environment or during remote access. It should be able to generate one-time passwords suitable for high throughput and operate in compliance with ISO21434 standards, in harmony with services in the cyber-physical system. For this purpose, the secure gateway and hardware security module should work in coordination.
REQ_UC1.25	Zero-trust device identity verification (node authentication)	Must ensure secure Machine-to-Machine (M2M) communication among autonomous forklifts, supply chain vehicles, IoT platforms, and warehouse systems. It should generate one-time passwords suitable for high throughput and be compatible with cyber-physical system devices in accordance with ISO21434. The secure gateway and hardware security module must operate in coordination.
REQ_UC1.26	Accountability of operations over a decentralized network	For data coming from forklift operations and logistics vehicles in the working environment, low latency (under 1 millisecond) must be ensured, and blockchain services suitable for this must be integrated with the cyber-physical system. For accountability, all transaction steps must be recorded by buyers, sellers, and approving authorities in the supply chain (immutability).
REQ_UC1.3	Asset Integrity & Authentication	Digital twins must be implemented for real-time virtual representation of physical assets.

REQ_UC2.11	Data access control	Access to data should be protected by role-based access control
REQ_UC3.1	Zero trust user authentication and authorization	Apply zero trust principles to user authentication and authorization
REQ_UC3.2	Zero trust device authentication and authorization	Apply zero trust principles to device authentication and authorization
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_UC4.9	Energy Community Setup	Allow for in-platform community creation
REQ_UC7.2	Data sharing	Data sharing over radio to receiver, from receiver to mobile and from mobile to server and from there to next mobile
REQ_UC7.25	Multiplace Digital Twin Specification	Create Digital Twin, of the device, dog and handheld, which is stored in Mobile app and in Server, and is Read / Write and handles user rights and flow control of the updates and visibility. Remembering that update to
REQ_UC7.27	Multiplace Digital Twin History	Historical stored to view the state of the Digital Twin at any given time. Example you can make search for certain timestamp of the device / dog / person, lets say 10.12.2024 08:24:12 , as a result of the query to twin history, you get situation of the device / dog / person at time stamp, you get all settings, location, all sensor data which was valid in this specific timestamp, taking care also privacy and data level considerations. Digital twin history data viewing rights is dependent on the viewer, time of the data and level of the user
REQ_UC7.28	Create and configure certain goal oriented groups according to the persons bio- and sociographic features	The group owner with user rights in the group management system creates and configures the group or groups that are intended for individuals with a specific group goal, bio and sociographic profile
REQ_UC7.29	Send group invitation to single users by the flow user account	From the group solution system, it is possible to invite people to join groups that have a wearable device and a Flow user account. In connection with the invitation, it is checked whether the person is the right person and at the same time the GDPR provides information about the rights and options for sharing data between group members
REQ_UC7.30	Reading and connecting multiple wearable device by synchronization hub	When the operation starts, the synchronization device searches for and establishes a connection to the devices that are connected to the group members devices which are within range of the sync device. If necessary, the synchronization device sets the configurations for the operation of the wearable device, with which it is appropriate to record the activity
REQ_UC7.32	Send wearable data from synchronization hub by SDK interface to mobile device	The wearable devices send biometric and inertial data to the synchronization device in real time via the SDK/BLE interface to the synchronization hub and again by 4-5G/LTE interface to the mobile device's application, where the data can be viewed based on the configurations according to the presets of the instructor's roles or by switching to the views to if person in his role has rights to see the data
REQ_UC7.33	Person guidance/ coaching according to the live data; mobile device data preview options	Based on the simultaneous monitoring of biometric and inertial data for the entire group, the instructor makes decisions and entries during the activity and, if necessary, intervenes in the activities immediately based on the decisions in accordance with the general guidance plan or special needs. At the same time, other instructors can also make notes and observations of the activities in accordance with their own role, which can be used to specify the need for changes to the action plan to be made later.

REQ_UC7.34	Send wearable data from synchronization hub by LTE interface to Polar Flow backend system	The synchronization device simultaneously opens both the SDK and LTE connection to the Flow backend system and updates the data view of the team members if necessary due to the entries made during the real-time control operation. Otherwise, the synchronization device can be set to send data from wearable devices to the backend system, e.g. Every 5,10,30,60, or 90 seconds, or the entire dataset after the operation has ended and the devices have stopped data collection.
REQ_UC7.35	Gather team related data from Flow backend to Group Solution Service's diary	Data synchronization outside of group activities when an individual independently carries out a pre-planned activity. The person syncs the device after the operation ends, after which the data is automatically transferred from the personal Flow backend account to the Group Solution account. Data transfer is only possible if the person has a) given permission for the data transfer, b) the person is a member of the group and c) the guidance agreement is active. The information is only visible to those persons to whom the individual gives permission a) to view, comment and redesign the activities based on the information and b) to the persons with whom the individual cooperates to carry out the activities

Summary of Requirements for Authorization

The requirements for authorizing users, devices and services to use a resource can be further categorized: **Authorization with zero trust.** Authorization focusing on security and trustworthiness, with assumption of strong identity verification without implicit trust and need for constant verification (REQ_UC1.24-25, REQ_UC3.1-3.3, REQ_UC1.3), often with compliance and cryptographic requirements.

Role-based operational control. Needs to ensure users are authorized to operations and data access based on their role (REQ_UC1.16, REQ_UC2.11, REQ_UC7.27, REQ_UC7.32, REQ_UC7.33). Also, authorization needs from automated systems in order to coordinate actions across machines, people and services (REQ_STORY.6, REQ_STORY.10, REQ_UC7.30, REQ_UC7.34, REQ_UC7.2)

Digital twins and asset integrity. Secure and authorized use of digital twins for monitoring, history tracking and integrity assurance (REQ_UC1.3, REQ_UC7.25, REQ_UC7.27).

Privacy & Consent. Authorization with respect to user privacy and compliance with data protection regulations (REQ_UC7.29, REQ_UC7.35, REQ_UC7.27).

Accountability and traceability. Means to track and later prove that operations and data access were properly authorized (REQ_UC1.26, REQ_UC7.25, REQ_UC7.35).

3.4 Perception

In the Endsley model, perception covers immediate sensing and data processing to answer the question *what is happening right now*. For example, perception with respect to road conditions could mean sensing the road temperature, air temperature and humidity, vehicle telemetry, and possible precipitation from a weather data source.

Perception establishes the groundwork for comprehension; the following requirements are aimed at making relevant data available and reliable for higher-level reasoning.

For example, Polar addresses Perception from the viewpoint that an individual's direct perception of the external environment occurs through natural senses. Based on the sensations, the person usually creates a real-time situation assessment of the prerequisites for action and tries to anticipate their actions to achieve the goal. Polar also brings a "sixth sense" to its field of observation for observing a person's internal environment, i.e. biometric measurements, which covers both real-time and historical data on (physiological

and psychological) stress and recovery. Together with these senses, individuals and groups can create a broader perspective on the development of capabilities in real time and in the long term, as well as make decisions about the direction of actions.

Table 4. WP2 requirements that target Perception.

Req Id	Name	Description
REQ_STORY.1	Road and Weather Conditions Information	The system must provide information about road and weather conditions, such as slippery road surfaces, on the way to the port area and within the port area.
REQ_STORY.10	Area Monitoring System	The area monitoring system must track the movements of vehicles, work machines, and people in the area and intervene if necessary.
REQ_STORY.3	License Plate Recognition	The system must recognize the vehicle's license plate at the port gate upon arrival.
REQ_STORY.4	Cabin Occupancy Check	Cameras must check that there are no other persons in the truck cabin at the port gate.
REQ_STORY.5	Vehicle Condition Scanning	Cameras and sensors must scan the vehicle at the port gate to detect any deviations in load securing, load condition, or vehicle condition before arrival.
REQ_STORY.7	Drone Assistance for Route Instructions	The system must enable the driver to order a drone for assistance, which projects route instructions and warnings onto the road.
REQ_STORY.8	Stop Sign Projection	The drone or fixed projector must project a stop sign onto the ground to make the driver stop the truck and allow the container handler to pass. When it is safe to continue, the drone must project the information onto the ground.
REQ_STORY.9	Correct Route Notification	The mobile application must notify the driver to return to the correct route if they turn in the wrong direction at an intersection.
REQ_UC1.9	Situational Awareness & Alerts	The system must provide real-time alerts for anomaly detection.
REQ_UC2.1	ML-based inference on the edge	ML-based inference on the edge should be possible to allow for real-time response
REQ_UC2.13	Alerting about platform health	Alerts are to be provided if devices or cloud components malfunction
REQ_UC2.6	Data annotation and pre-processing	Tools should be available that can be used for annotating and pre-processing the collected data. These tools should be specialized for ToF cameras, radar and UWB.
REQ_UC3.14	Object detection camera specs	Determine required technical specifications for cameras used in object detection
REQ_UC3.15	Select object detection model	Select suitable object detection model
REQ_UC3.8	Data sources for anomaly detection	Define what sources (e.g. video) to use for data anomaly detection
REQ_UC4.1	Missing data replacement	Missing data needs to be replaced with, e.g. baseline data
REQ_UC4.11	KPI information	The KPI related to energy aspects to the user to be updated regularly
REQ_UC5.2	Threat analysis per digital moment	We will evaluate potential security threats associated with each identified digital moment, considering factors like user behavior, device vulnerabilities, and data sensitivity.
REQ_UC5.3	Threat Detection Algorithms	We will develop and implement advanced algorithms to detect and respond to security threats in real-time, ensuring the protection of consumer data and digital interactions.
REQ_UC7.21	Autonomous network operation	The devices created in the project must be autonomous and be able to operate without continuous human controlling and monitoring, they could also network themselves automatically and utilize e.g. mesh technologies.
REQ_UC7.3	Sixth Sense Sensor definition	Wearable specification at situational awareness level 1 (Perception)

Summary of Requirements for Perception

The selection of perception-related requirements roughly fall into two categories:

Perception for situational awareness & real-time decision-making. Requirements for perception as source of situational awareness. Focus on gathering and interpreting sensor data to understand the environment and support immediate actions. (REQ_STORY.1, REQ_STORY.10, REQ_STORY.3, REQ_STORY.4-5, REQ_STORY.7-9, REQ_UC1.9, REQ_UC3.8, REQ_UC7.3, REQ_UC5.2, REQ_UC5.3)

Perception infrastructure & data processing. Requirements for the technical foundation for collecting, processing, and interpreting sensor data, including ML model selection and edge processing needs. (REQ_UC2.1, REQ_UC2.6, REQ_UC3.14-15, REQ_UC4.1, REQ_UC4.11, REQ_UC7.21)

3.5 Comprehension

The comprehension part of an Endsley model describes what given data means. This involves interpreting and understanding the significance of perceived elements. Using the road conditions example, comprehension would combine the sensor inputs to determine that road surface is likely wet and below freezing, and that the conditions are conducive to black ice formation.

The following requirements in Table 6 support projection by ensuring the system can derive meaningful context from raw inputs before making forward-looking assessments. When not specifically aimed at comprehension or projection, some requirements – especially those involving machine learning – are listed for both, as machine learning predictors often serve both to contextualize current data and to generate forward-looking insights.

As an example of a use case viewpoint, Polar sees Comprehension so that from the point of view of situational awareness, the individual plays a crucial role in making personal decisions, but decisions made with groups should also be based on the simultaneous examination of the individuals' capabilities and goals. When deciding, the functional conditions of the individual and the group for achieving the goal are emphasized, as well as the goal, which can be both physical and emotional at the same time. In the examination of the operational prerequisites, decision-making criteria are used that have been created in the logic of the guidance of the service system, both in terms of actors and tasks. Criteria are rules in a decision-making machine that are used in different ways for different variations, but the essential thing is that the rules are based on the psychological and physiological laws that are within the scope of the measurement capability of wearable devices.

Table 5. WP2 requirements that target Comprehension.

Req Id	Name	Description
REQ_STORY.1	Road and Weather Conditions Information	The system must provide information about road and weather conditions, such as slippery road surfaces, on the way to the port area and within the port area.
REQ_STORY.10	Area Monitoring System	The area monitoring system must track the movements of vehicles, work machines, and people in the area and intervene if necessary.
REQ_STORY.11	Hazard Alert and Communication	The system sends a hazard alert and action instructions to all persons in the area if necessary. The system allows communication and sharing of information about the progress of events.
REQ_STORY.2	Unloading Time Notification	The application must offer information about the next possible unloading time to the driver.

REQ_STORY.4	Cabin Occupancy Check	Cameras must check that there are no other persons in the truck cabin at the port gate.
REQ_STORY.5	Vehicle Condition Scanning	Cameras and sensors must scan the vehicle at the port gate to detect any deviations in load securing, load condition, or vehicle condition before arrival.
REQ_STORY.8	Stop Sign Projection	The drone or fixed projector must project a stop sign onto the ground to make the driver stop the truck and allow the container handler to pass. When it is safe to continue, the drone must project the information onto the ground.
REQ_STORY.9	Correct Route Notification	The mobile application must notify the driver to return to the correct route if they turn in the wrong direction at an intersection.
REQ_UC1.10	Situational Awareness & Alerts	A predictive analytics model must be integrated for risk assessment.
REQ_UC1.9	Situational Awareness & Alerts	The system must provide real-time alerts for anomaly detection.
REQ_UC2.1	ML-based inference on the edge	ML-based inference on the edge should be possible to allow for real-time response
REQ_UC2.13	Alerting about platform health	Alerts are to be provided if devices or cloud components malfunction
REQ_UC2.3	Data availability	It should be possible to query and retrieve the data in the cloud for ML training
REQ_UC2.6	Data annotation and pre-processing	Tools should be available that can be used for annotating and pre-processing the collected data. These tools should be specialized for ToF cameras, radar and UWB.
REQ_UC2.7	Cloud-based training of ML-based predictors	It should be possible to use cloud-based GPU nodes for ML training over the collected data
REQ_UC2.8	On-demand automatic retraining of ML-based predictors	On-demand automatic retraining of ML-based predictors
REQ_UC3.15	Select object detection model	Select suitable object detection model
REQ_UC3.8	Data sources for anomaly detection	Define what sources (e.g. video) to use for data anomaly detection
REQ_UC5.2	Threat analysis per digital moment	We will evaluate potential security threats associated with each identified digital moment, considering factors like user behavior, device vulnerabilities, and data sensitivity.
REQ_UC5.3	Threat Detection Algorithms	We will develop and implement advanced algorithms to detect and respond to security threats in real-time, ensuring the protection of consumer data and digital interactions.
REQ_UC7.21	Autonomus network opreation	The devices created in the project must be autonomous and be able to operate without continuous human controlling and monitoring, they could also network themselves automatically and utilize e.g. mesh technologies.
REQ_UC7.7	Behavior Change Management tools	Specification of activity control configuration methods for data mining and mass customization needs (Decision)

Summary of Requirements for Comprehension

Comprehension-related requirements can be further divided into subcategories:

Risk and Threat Interpretation. Understanding the implications of perceived anomalies or threats, and determining their relevance to safety, security, or operations. (REQ_UC1.10, REQ_UC1.9, REQ_UC5.2, REQ_UC5.3).

Operational Context Understanding. Interpreting the current state of operations, such as vehicle status, route correctness, and unloading schedules (REQ_STORY.2, REQ_STORY.5 ,REQ_STORY.9, REQ_STORY.10, REQ_UC2.13).

Human and Machine Behaviour Interpretation. Understanding the behaviour of people and autonomous systems, including deviations, intentions, or compliance (REQ_STORY.4, REQ_STORY.8, REQ_UC7.7, REQ_UC7.21).

Data-Driven Model Comprehension. Interpreting data through trained models, enabling the system to derive meaning from complex sensor inputs or historical patterns (REQ_UC2.1, REQ_UC2.3, REQ_UC2.6, REQ_UC2.7, REQ_UC2.8, REQ_UC3.8, REQ_UC3.15).

3.6 Projection

In the Endsley model, projection is the ability to anticipate likely future states or events, based on current understanding. Using the road conditions example, projection could predict, based on weather forecast and traffic volume forecast that road slipperiness will worsen, increasing the risk of accidents.

Table 6 below lists requirements that involve some kind of prediction. Some requirements from comprehension are reinstated here as, e.g., machine learning predictors often serve both to contextualize current data and to generate forward-looking insights.

As a use case viewpoint example, Polar approaches Projection from the viewpoint that a person's ability to make the projection of the situation or to "predict" is based on sensory anticipation of one's own actions in order to achieve the desired and expected results. In this case, when the information provided by wearable devices about a person's functional capacity and information about its continuation (or future duration) is available, a person's functional capacity can be predicted further than what a subjective assessment would indicate.

Further, predictable functional capacity without biometric information can be very low or very high within the limits of mental tolerance/ capacity, depending on the internal motivation to continue the activity. This setting must be examined with parameters related to context and behaviour so that there is a feedback relationship between the collected biometric data and the known content of the activity. By changing the content of the activity, the need for behavioural change can be predicted on the basis of biometric data so that the desired goal could be achieved even in the very long term.

When designing prediction models, Polar sees that the question is how the content of behaviour should be changed in relation to the collected biometric data. In order to make changes, some rule-based functions are needed that describe the action taking place in the context in such a way that the action or its changes have an impact on the individual's ability to function/ competence in the context in question. In this case, it is possible to create a connection between the environment and biometric data in such a way that a motivational potential is created to achieve the predicted result and goal.

The following figure defines from Polar viewpoint the components needed to form situational awareness, which are used in the creation of similar machine learning AI models and control processes:

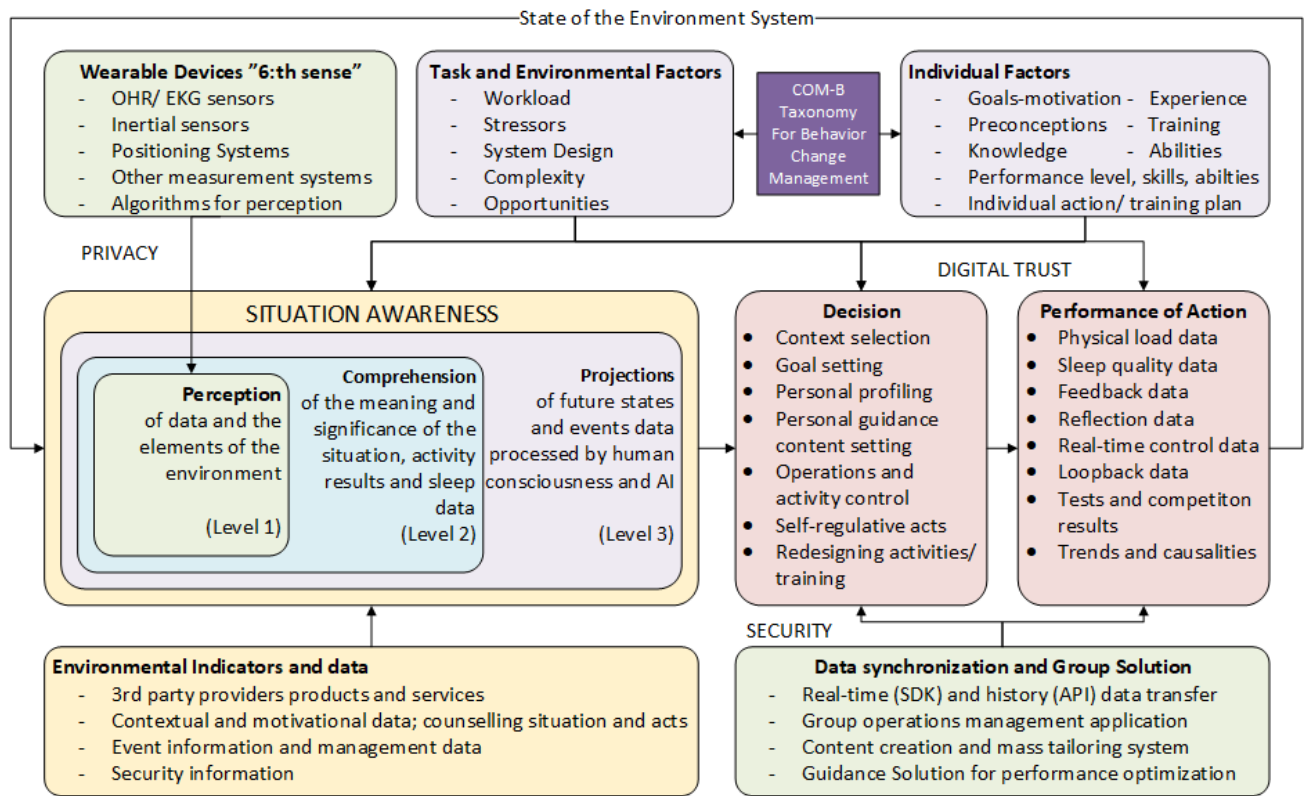


Table 6. WP2 requirements that target Projection.

Req Id	Name	Description
REQ_STORY.11	Hazard Alert and Communication	The system sends a hazard alert and action instructions to all persons in the area if necessary. The system allows communication and sharing of information about the progress of events.
REQ_STORY.2	Unloading Time Notification	The application must offer information about the next possible unloading time to the driver.
REQ_STORY.6	Driver Arrival Notification	The identification system must send information about the driver's arrival at the port area to the operator, who has prepared for the arrival based on the advance notice and ensure that the autonomous work machine is waiting in the warehouse.
REQ_STORY.8	Stop Sign Projection	The drone or fixed projector must project a stop sign onto the ground to make the driver stop the truck and allow the container handler to pass. When it is safe to continue, the drone must project the information onto the ground.
REQ_UC1.10	Situational Awareness & Alerts	A predictive analytics model must be integrated for risk assessment.
REQ_UC2.7	Cloud-based training of ML-based predictors	It should be possible to use cloud-based GPU nodes for ML training over the collected data
REQ_UC2.8	On-demand automatic retraining of ML-based predictors	On-demand automatic retraining of ML-based predictors
REQ_UC2.9	Online update of ML-based predictors from cloud to the edge	Online update of ML-based predictors from cloud to the edge
REQ_UC5.3	Threat Detection Algorithms	We will develop and implement advanced algorithms to detect and respond to security threats in real-time, ensuring the protection of consumer data and digital interactions.

REQ_UC7.37	Analyze and regenerate activity/ training plans according to the training diary and gathered biometrics	The group's instructor performs mass customization and data mining on the data collected on an individual, so that based on experience, it would be possible to get new instructions in accordance with previously successfully implemented action plans when the individual's actions need to be changed
REQ_UC7.6	Guidance Service Application	Group Solution service system specifications for grouping people and content of the goals, from the perspective of physical, psychological and social control (Projection)
REQ_UC7.8	Content creation tools	Specification of goal-driven Guidance service creation tools (Performance of Action)

Summary of Requirements for Projection

Projection related requirements fall roughly into three categories: **Operational forecasting & event anticipation** for projection of future operational states or events to support timely decision making and action (REQ_STORY.2, REQ_STORY.6, REQ_UC7.37, REQ_UC7.6). Another category is **predictive safety & risk management** with focus on projecting potential hazards or to enhance safety and security. (REQ_STORY.11, REQ_UC1.10, REQ_UC5.3, REQ_STORY.8). **Adaptive intelligence & system learning** related requirements need projection of future system states or user needs through machine learning and adaptive systems. (REQ_UC2.7, REQ_UC2.8, REQ_UC2.9, REQ_UC7.8)

3.7 Monitoring

The concept of Zero Trust based authentication and authorization requires the ability to monitor the state of any data source, access rights and permissions in real time, and to generate concise views into the overall system status. The monitored elements could be, for example:

- Access, identity and permission credentials and their status (revocation, expiration, issuer status / evaluation of credibility / trust hierarchy)
- Resource status; access roles, permissions, authorization criteria set for a resource
- Filtered list of users and devices in a group (based on location or any other group definition such as community membership) and their status
- Resource access history, permission / credential status history

Monitoring needs may also include performance (processing / data throughput), system health, etc. Monitoring could also be used for data specific visualization needs. Requirements for Monitoring as recorded by the Use Cases are listed below in Table 7.

As a use case example, the Polar Flow server system currently meets all Zero Trust requirements when it comes to deploying a wearable device, creating a user account, collecting and storing biometric data, and sharing data with the existing Flow for Coach coaching application. During this project, RED and DA compliant requirements will also be researched and developed for both radio technical equipment and web and mobile service systems

Table 7. WP2 requirements that target Monitoring.

Req Id	Name	Description
REQ_UC1.13	Web Platform and Graphical Visualization	A web platform should be developed with a user-friendly interface for monitoring forklifts and visualizing real-time data.
REQ_UC1.17	System Health	The status of network components (including Mesh Routers, Mesh Clients, IoT devices, and Middleware components) should be monitored and evaluated.

REQ_UC1.7	IoT-Based Monitorization	IoT gateways must collect and transmit real-time monitoring data.
REQ_UC2.12	Platform monitoring	The status of devices and cloud components should be monitored.
REQ_UC2.3	Data availability	It should be possible to query and retrieve the data in the cloud for ML training
REQ_UC2.4	Manage and visualize the data	There should be an interface to manage and visualize the data
REQ_UC4.11	KPI information	The KPI related to energy aspects to the user to be updated regularly
REQ_UC4.13	Energy Community Overview	Community data presentation, including consumption and production information by consumer / prosumer
REQ_UC4.14	Energy Community Import / Export	Allow for the visualization of grid consumption and exports
REQ_UC4.9	Energy Community Setup	Allow for in-platform community creation
REQ_UC7.33	Person guidance/ coaching according to the live data; mobile device data preview options	Based on the simultaneous monitoring of biometric and inertial data for the entire group, the instructor makes decisions and entries during the activity and, if necessary, intervenes in the activities immediately based on the decisions in accordance with the general guidance plan or special needs. At the same time, other instructors can also make notes and observations of the activities in accordance with their own role, which can be used to specify the need for changes to the action plan to be made later.
REQ_UC7.36	Create and configure activity/ training plans for certain goal oriented groups according to the persons bio- and sociographic features	The group instructor collects the information of the entire group and individual group members separately in the form of a report, prepares a summary of the report and draws conclusions on further measures, which include a) continuing according to the common plan, b) changing and separating the joint plan for some members of the group, or c) amending the joint action plan in part or in full and separating the individual plans that continue according to the previous plan.

Summary of Requirements for Monitoring

Monitoring-related requirements can be grouped into two main categories based on the type of system state being monitored and the purpose of that monitoring. **System and infrastructure monitoring** includes requirements focused on monitoring the technical health, availability, and performance of systems, platforms, and devices. The goal is to ensure operational continuity, security / trust, data availability, and system reliability. These requirements include (REQ_UC1.13, REQ_UC1.17, REQ_UC1.7, REQ_UC2.12, REQ_UC2.3, REQ_UC2.4).

Another category is **(human-centric) monitoring & decision support** that involves monitoring human activity, energy usage, and community behaviour to support decision-making, coaching, and planning (REQ_UC4.11, REQ_UC4.13, REQ_UC4.14, REQ_UC4.9, REQ_UC7.33, REQ_UC7.36).

3.8 Zero Trust System

Achieving Zero Trust is largely dependent on other components, especially in a heterogeneous environment where different parties may use different systems for authentication, data access and authorization tasks. Thus, the Zero Trust is linked to Identity and Access Management, Authorization and Monitoring components. The Zero Trust system can perform several control and assisting tasks:

- Credential issuance to participants, expiration + revocation control
- Verification of credentials issued within and outside of the system
- Real-time credential state monitoring and notification / automated access control modification to resources and users controlled by the credentials
- Identity harmonization between identities provided by different parties and in different identity systems ("identity anchor")

- Attest to the identity and trust level of all participants, maintaining a federated trust hierarchy of participants (e.g., who can be trusted with which operations, who can issue credentials of which types and for which objects, etc.)

The requirements from Use Cases that address Zero Trust System are listed below in Table 8.

As a use case example, the Polar Zero Trust principle is built on several different stages and system levels: identifying an individual, granting permission to use the Polar Guidance Service, joining groups and building individual guidance relationships.

The Polar Guidance Service is intended for goal-oriented guidance of individuals and groups of them. The Zero Trust security principle is implemented in such a way that an individual identifies themselves as the person they claim to be based on the ID of the wearable device, contact and personal information.

Once the identification has taken place on an individual level, they can join the Guidance Service, either as an independent member, guiding themselves with the system's guidance tools, or they can be invited to join a group that contributes to the achievement of a personal goal. The third option is to join a group (e.g. a sports team or a weight loss group) that has an activity content that interests the individual and a goal that he or she is trying to achieve in the area in question with the group.

When joining the group, the individual goes through a profiling phase in terms of biography, capabilities and goals, during which his or her role in the group is defined (instructor/supervisee). After this, the individual has access inside the organization to the subgroups and tools according to the role, which they can use to promote their own goal-oriented activities and those of the other members of the groups. These measures ensure safe changes, roles, responsibilities and access rights within the organisation and groups.

Within the framework and on the terms of the Polar Guidance Service, a pedagogical dyad, i.e. a guidance relationship, is formed between the instructor and the supervisee with a guidance agreement. The agreement and the arrangement correspond ethically to a similar relationship, such as between a teacher and a student or a patient and a psychotherapist. The agreement creates a relationship of trust and a safe environment for the development of individual identity and functional capacity, because all data collected with wearable devices is health information in both psychological and physiological examination.

All transfers of information and persons from the owner (supervisor or supervisee) to another (supervisor or supervisee) go through the above-mentioned credentials check-out procedures to ensure that there is a permit and a purpose for the use of the data. It is not possible to transfer people from one group to another without the permission of the owner/ supervisor of the group or person being transferred. The transfer also goes through the above-mentioned credentials check-out procedures to ensure that there is both permission and a request for the transfer of the person.

Table 8. WP2 requirements that target Zero Trust System.

Req Id	Name	Description
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_UC1.16	Role-Based Access Control (RBAC)	An authentication mechanism providing Role-Based Access Control (RBAC) must be developed, and access should be restricted based on user roles.

REQ_UC1.24	Zero-trust user authentication and authorization	Must enable the authentication and authorization of relevant individuals and services in the working environment or during remote access. It should be able to generate one-time passwords suitable for high throughput and operate in compliance with ISO21434 standards, in harmony with services in the cyber-physical system. For this purpose, the secure gateway and hardware security module should work in coordination.
REQ_UC1.25	Zero-trust device identity verification (node authentication)	Must ensure secure Machine-to-Machine (M2M) communication among autonomous forklifts, supply chain vehicles, IoT platforms, and warehouse systems. It should generate one-time passwords suitable for high throughput and be compatible with cyber-physical system devices in accordance with ISO21434. The secure gateway and hardware security module must operate in coordination.
REQ_UC1.26	Accountability of operations over a decentralized network	For data coming from forklift operations and logistics vehicles in the working environment, low latency (under 1 millisecond) must be ensured, and blockchain services suitable for this must be integrated with the cyber-physical system. For accountability, all transaction steps must be recorded by buyers, sellers, and approving authorities in the supply chain (immutability).
REQ_UC1.28	Secure transmission of sensitive data	Data coming from edge devices (forklifts or vehicles) or services must be securely transmitted end-to-end with encryption and hashing. Hashes of data generated within the decentralized network must be transmitted to or retrieved from the blockchain within a maximum of 1 second. For this purpose, the secure gateway, hardware security module, and database must operate in coordination.
REQ_UC3.1	Zero trust user authentication and authorization	Apply zero trust principles to user authentication and authorization
REQ_UC3.2	Zero trust device authentication and authorization	Apply zero trust principles to device authentication and authorization
REQ_UC3.3	Zero trust authentication and authorization of video and audio material	Use zero trust principles to verify authenticity of recorded material and authorize its usage
REQ_UC4.9	Energy Community Setup	Allow for in-platform community creation
REQ_UC7.2	Data sharing	Data sharing over radio to receiver, from receiver to mobile and from mobile to server and from there to next mobile
REQ_UC7.27	Multiplace Digital Twin History	Historical stored to view the state of the Digital Twin at any given time. Example you can make search for certain timestamp of the device / dog / person, lets say 10.12.2024 08:24:12 , as a result of the query to twin history, you get situation of the device / dog / person at time stamp, you get all settings, location, all sensor data which was valid in this specific timestamp, taking care also privacy and data level considerations. Digital twin history data viewing rights is dependent on the viewer, time of the data and level of the user
REQ_UC7.28	Create and configure certain goal oriented groups according to the persons bio- and sociographic features	The group owner with user rights in the group management system creates and configures the group or groups that are intended for individuals with a specific group goal, bio and sociographic profile
REQ_UC7.29	Send group invitation to single users by the flow user account	From the group solution system, it is possible to invite people to join groups that have a wearable device and a Flow user account. In connection with the invitation, it is checked whether the person is the right person and at the same time the GDPR provides information about the rights and options for sharing data between group members

REQ_UC7.34	Send wearable data from synchronization hub by LTE interface to Polar Flow backend system	The synchronization device simultaneously opens both the SDK and LTE connection to the Flow backend system and updates the data view of the team members if necessary due to the entries made during the real-time control operation. Otherwise, the synchronization device can be set to send data from wearable devices to the backend system, e.g. Every 5,10,30,60, or 90 seconds, or the entire dataset after the operation has ended and the devices have stopped data collection.
REQ_UC7.35	Gather team related data from Flow backend to Group Solution Service's diary	Data synchronization outside of group activities when an individual independently carries out a pre-planned activity. The person syncs the device after the operation ends, after which the data is automatically transferred from the personal Flow backend account to the Group Solution account. Data transfer is only possible if the person has a) given permission for the data transfer, b) the person is a member of the group and c) the guidance agreement is active. The information is only visible to those persons to whom the individual gives permission a) to view, comment and redesign the activities based on the information and b) to the persons with whom the individual cooperates to carry out the activities

Summary of Requirements for Zero Trust System§

Zero trust related requirements can be classified into three main categories. One group of requirements emphasizes **continuous identity verification** and strict validation of users, devices, and digital assets (REQ_UC1.16, REQ_UC1.24, REQ_UC1.25, REQ_UC3.1, REQ_UC3.2, REQ_UC3.3, REQ_UC7.28, REQ_UC7.29).

Another set of requirements focuses on **data integrity** and traceability (REQ_UC1.28, REQ_UC1.26, REQ_UC7.2, REQ_UC7.34, REQ_UC7.35).

The final category emphasizes **dynamic access control** based on context (e.g., time, role, data type) and auditable actions (REQ_UC1.26, REQ_UC7.27, REQ_UC7.35).

3.9 Discussion of Requirements

The Sa4CPS concept (WP2) related requirements detected in use cases of the project, have been further analysed and discussed in this chapter. The requirements have been categorized to a set of functional blocks (components), that became obvious during the analysis. Such functional blocks are data spaces and communications, identity and access management, authorization, perception, comprehension, projection, monitoring and zero trust system. It is obvious that this list of functional blocks is preliminary only, because some of the identified requirements were a bit fuzzy enabling also other interpretations. In addition, relationship of each use case requirements towards other use cases may still require further analysis. However, we have now at least some starting point for further works, and description of the preliminary Sa4CPS concept as the starting point.

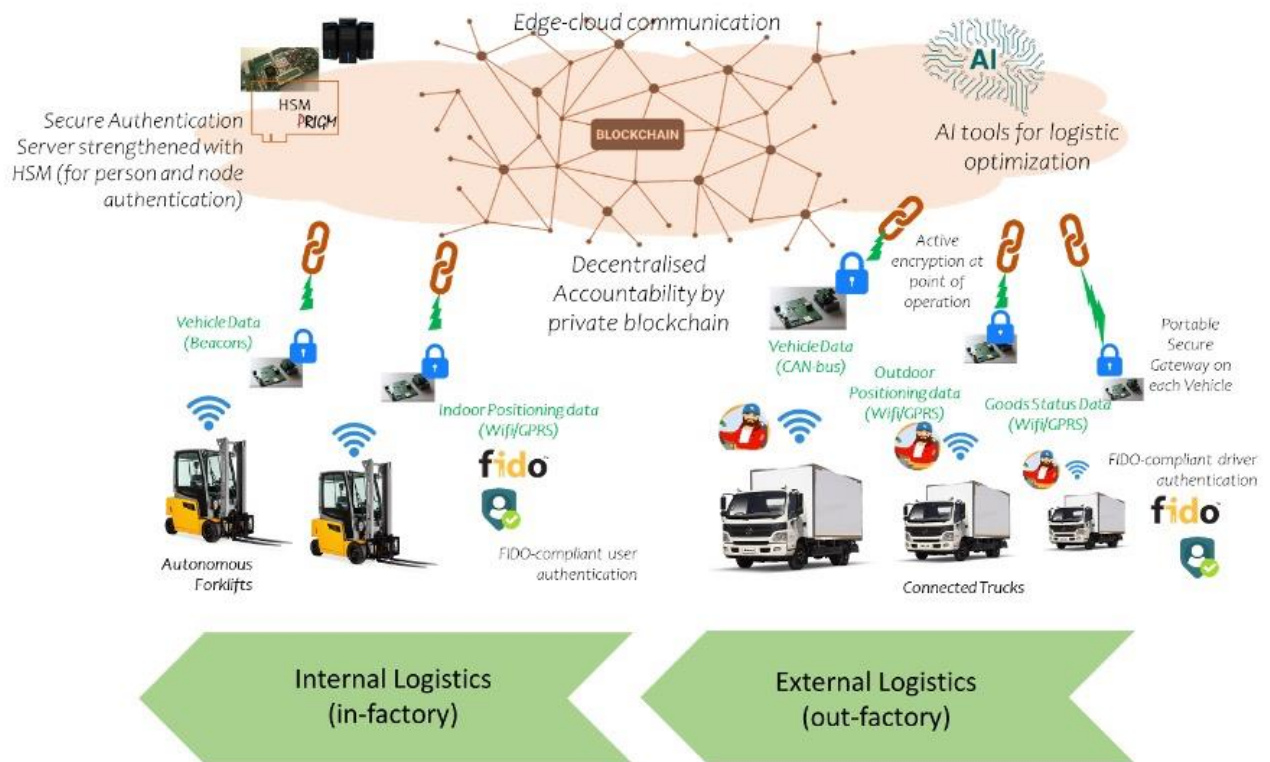
4 Contribution(s) of use cases into the Sa4CPS concept

This section discusses/clarifies the potential contributions of each use case into the Sa4CPS concept.

4.1 Autonomous lightweight logistics (UC1)

Use Case 1 (UC1) plays a crucial role in contributing to the development and validation of the Sa4CPS concept by addressing secure situational awareness challenges in a real-world cyber-physical system (CPS) scenario within the automotive supply chain domain, Figure 5. UC1 spans both cross-

organisational (out-factory) and intra-organisational (in-factory) environments, encompassing a multi-vendor logistics network that includes third-party suppliers and OTOKAR's internal factory logistics.



Connected and Autonomous lightweight logistics for in-factory and out-factory supply chain

Figure 5 UC1 Concept

At the cross-organisational level, UC1 contributes to the perception and comprehension aspects of situational awareness by implementing IoT-based asset monitoring and decentralized traceability solutions. These ensure real-time visibility and digital trust in the movement and integrity of automotive parts across organisational boundaries. This supports secure information sharing and collaboration in zero-trust conditions, a key goal of the Sa4CPS concept.

Within OTOKAR's internal logistics environment, UC1 contributes to the projection, decision, and actuation layers of situational awareness through the deployment of autonomous forklifts. These operate in a connected environment via Wi-Fi and beacon-based infrastructure, supported by IoT gateways. The continuous exchange of location and carriage data ensures operational efficiency while maintaining awareness of dynamic elements such as workers, vehicles, and obstacles—enhancing overall safety and coordination.

UC1 further strengthens the cyber-physical resilience component of Sa4CPS by addressing hardware-level and communication backend vulnerabilities. Through lab-scale test cases and countermeasures, UC1 validates mechanisms for resisting service-level attacks, ensuring the robustness and security of data transmission across both cross-organisational and in-factory contexts.

Overall, UC1 provides a comprehensive testbed that integrates multiple CPS domains—logistics, mobility, and security—thereby demonstrating how the Sa4CPS concept can be instantiated to enable secure collaboration, trusted information exchange, and adaptive control in critical industrial settings.

4.2 People and machine position awareness in production or logistics area (UC2)

As described in Deliverable D1.1, UC2 showcases the project's results in a production/logistics area where people and machines must coexist safely and securely. A high-level overview of UC2 is shown in the following figure.

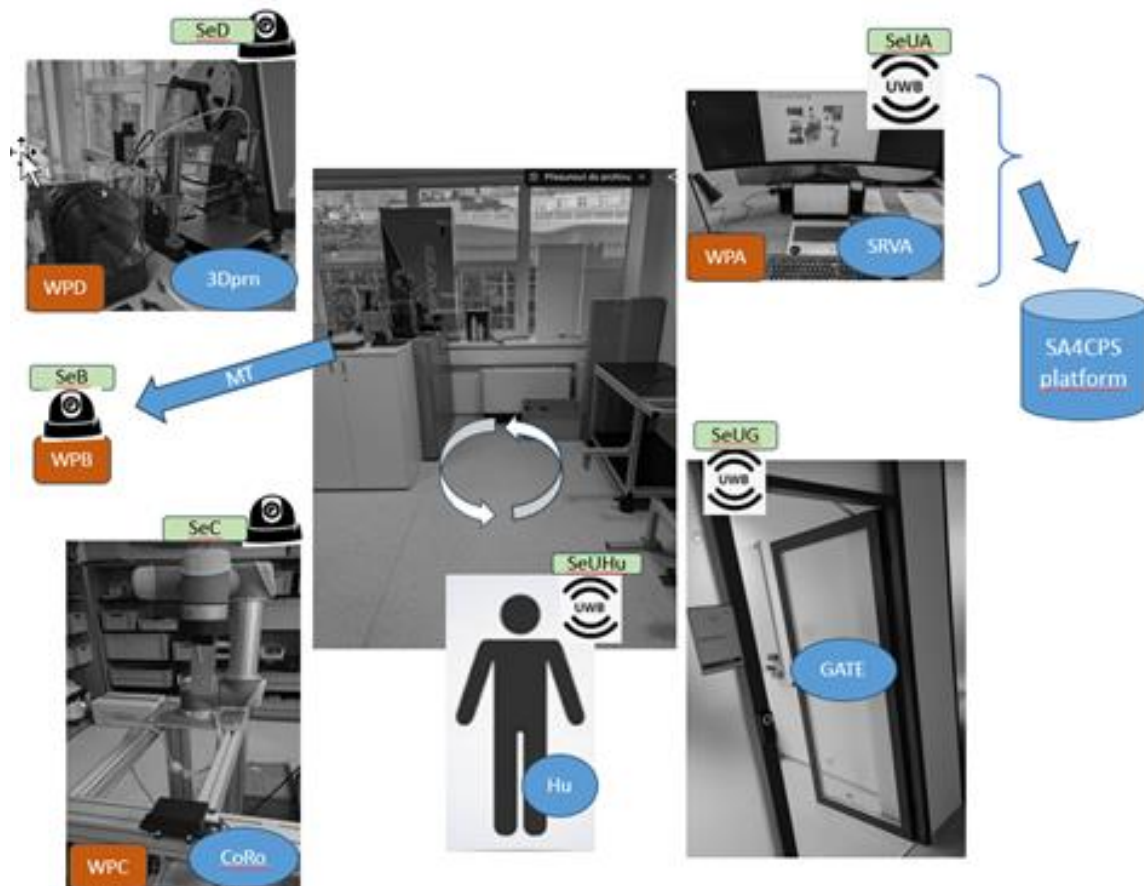


Figure 6. High-level overview of UC2.

Here, a workplace of the production process consists of several work positions (WPs). Each work position is characterized by a specific set of tools and tasks performed by a human (Hu) or a machine (Ma), independently or cooperatively. Within the workspace, humans are not restricted regarding potential collisions.

At work position A (WPA), which includes a work desk with a server (SRVA), activities related to configuration, parameterization, visualization, and control of the entire workspace are carried out. Activities here are performed exclusively by a human, who can move directly to other positions from this location. Work position B (WPB) consists of a table temporarily storing products.

At work position C (WPC), a collaborative robot (CoRo) performs material transfer (MT) actions, either between work positions or in collaboration with a human. CoRo is a stationary operational element. All work positions are within CoRo's reach.

At work position D (WPD), manufacturing equipment (e.g., a 3D printer) is located, which autonomously executes the production process based on the settings defined at WPA. The only human-performed activities in this position involve input (e.g., supplying material, such as filament) and output (e.g., removing the finished product). Here, humans act as operational elements responsible for input and output actions. The removed product is handed over by the human to the collaborative robot (human-machine interaction), which then transfers it between WPD and WPB.

All work positions are spatially bounded, and the entry gate to this workspace (GATE) is equipped with an active UWB sensor (SeUG) operating in ranging mode to monitor authorized access to the production area. Entry to the area is possible only through the GATE, with authorization ensured by an access control system based on UWB sensors.

The production process at all work positions is monitored by a sensor system that collects data to track the position and trajectory of all operational elements, verify activities within all work positions, and ensure human safety during tasks. In this regard, the sensor system is responsible for preventing collisions between humans and the collaborative robot. The sensor system consists of three time-of-flight cameras, one passive UWB sensor (SeUA) located at WPA, and one active UWB sensor (SeUHu) carried by the human. This active sensor authorizes the person to perform tasks at the workstation and grants access through the GATE.

The ToF cameras are installed at a height of 2.5 meters, positioned above each work position (SeB, SeC, SeD). The passive SeUA sensor is located on the PCA monitor. The field of view zones of SeB through SeD is configured to avoid any blind spots. All sensors communicate online in real-time with the server SRVA at WPA. Sensors SeB through SeD provide precise positional data of operational elements at all positions, while sensor SeUA provides data on the presence of a human at WPA in front of the computer monitor, which activates immediately upon detecting a human presence. Sensors SeB through SeD monitor the positions of all elements and objects, both stationary and mobile, in a 3D coordinate system with a temporal resolution of 100+ ms.

The Figure 6 shows the deployment architecture of UC2. Software components are divided between two execution environments – the cloud and the edge of the cloud. Individual components were identified based on scenarios and data flow described in Deliverable D1.1.

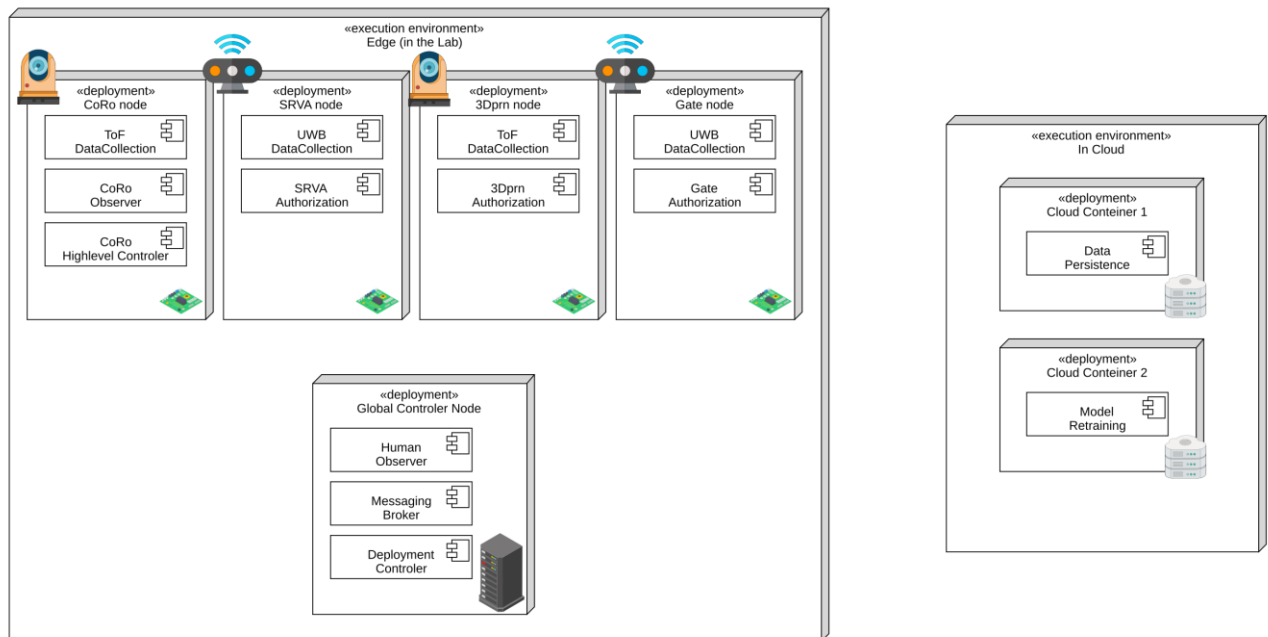


Figure 7. Deployment architecture of UC2

CUNI and IMA are developing a dedicated development platform for the edge nodes. On the edge, the UC will employ a number of sensor nodes (SN). Each sensor node consists of a board that contains an NXP S32-based chip connected to ToF sensor. This takes care of real-time control. The board further integrates with the Raspberry Pi compute module, which provides WiFi-based internet connectivity and acts as a gateway to the cloud. The Figure 8 below illustrates all the parts of SN.

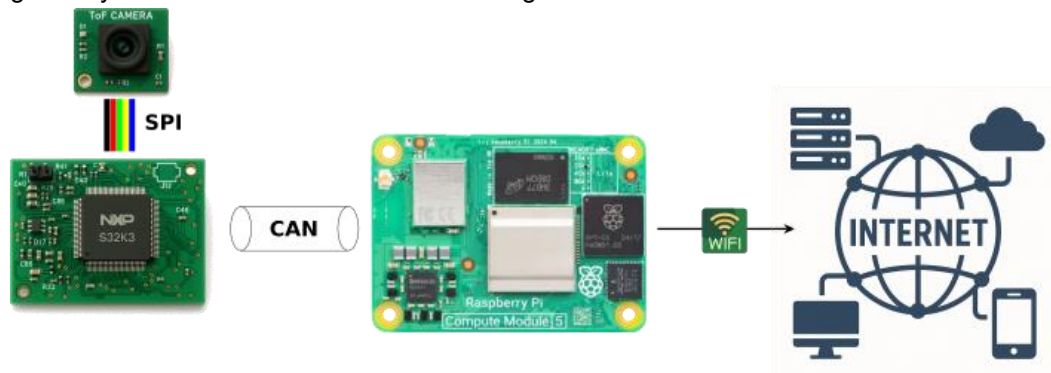


Figure 8. The parts of the sensor nodes.

On SNs, there is always the DataCollection component (specialized based on the particular attached sensor, i.e., UWB or ToF ones), which reads sensor data and sends them to the DataPersistence components deployed in a container in the cloud. Also, the data are sent to the HumanObserver component.

Additionally, on SNs, there are components that control a particular device that is observed by the sensor. These components, in cooperation with the HumanObserver component, control the specific device (CoRo, etc.). Importantly, these components (and the HumanObserver) employ AI models trained based on the collected data from the sensors (the AI training pipeline described below).

The HumanObserver component synthesizes a picture of the overall situation in the workplace and coordinates components controlling the devices.

HumanObserver is deployed on a computer (edge central node – ECN) that also runs at the edge (i.e., on-premises of the workspace). It is a “regular” server-like computer.

For mutual communication, all the components use messages. As a particular messaging technology, we will employ MQTT. The MessagingBroker (i.e., the software component that distributes the messages) will also be deployed on ECN. From the semantic point of view, there will be multiple message topics (queues), one for each message type (i.e., messages from the particular sensor, messages to the specific controller, etc.).

Finally, there is the DeploymentController component (also deployed on ECN), which monitors all the nodes on the edge, (re)deploys individual components and provides monitoring and controlling UI to humans.

The DataPersistence and ModelRetraining components are deployed in the cloud as they need large data storage (to store all the sensor data) resp. computing power (to (re)train AI models). For the AI training pipeline, we plan to use and adapt an existing framework for MLOps, but a particular one has not been selected yet. The DataPersistence and ModelRetraining components will communicate together via a shared database. The trained AI models will be delivered to specific controllers again via messages (as the AI models must be able to run on S32 ECUs, it is expected they will be rather small).

From the development point of view, all the components will be built over a common basis (a library), which will be developed within the project. This library, together with the DeploymentController component, UI of the DeploymentController and AI training pipeline will be developed as use-case neutral and thus directly contribute to the project results. Together, they will provide the basis for the perception (sensor data collection and AI model training) of the current situation in a workspace and comprehension (employment of the AI models and controlling the specific device) of the situation.

4.3 Application of advanced maps in public safety (UC3)

NSC3 will serve as the mapping component within the SA4CPS (Situational Awareness for Cyber-Physical Systems) project, transitioning from the current 2D mapping solution to provide enhanced 3D visualization and spatial awareness capabilities to the overall system (see Figure 9).

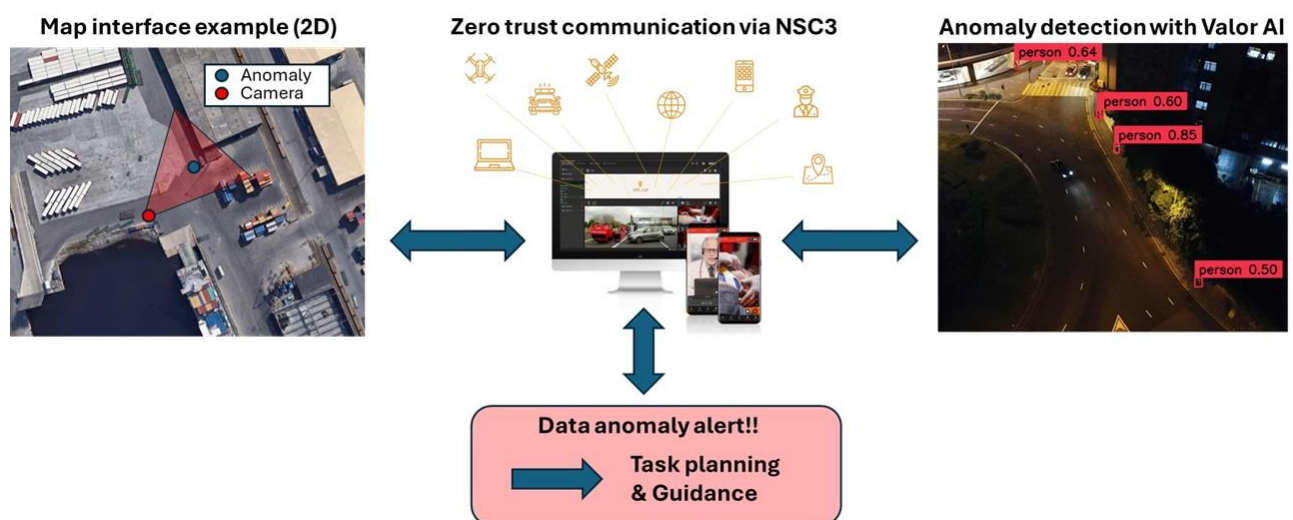


Figure 9. An example how NSC3 will operate under Zero trust communication with other devices (camera in this example) and how this data will be utilized when planning further actions, if facing unwanted anomalies/objects.

The NSC3 platform represents a comprehensive situational awareness solution that will introduce zero trust principles during the project to enhance its security. Based on the set requirements earlier, Zero trust will be implemented on user authentication, strict device validation, and recorded material integrity. The security architecture is designed to maintain high performance. These could be presented in more detail as follows:

User Authentication (Never Trust a User by Default)

- MFA (Multi-Factor Authentication): Every user must verify identity using multiple factors (e.g., password + hardware token or biometrics).
- Role-Based Access Control (RBAC): Users only see data relevant to their role (e.g., operator vs. commander).
- Session Monitoring: Each session is continuously monitored for abnormal behavior (e.g., unusual access times or location).

Strict Device Validation (Trust No Device Without Proof)

- Device Certification: Only approved, hardened devices can connect to NSC3 (e.g., pre-registered laptops or tablets with secure boot).
- Health Checks: Devices must pass compliance checks (e.g., no outdated software or known vulnerabilities) before access is allowed.
- Dynamic Risk Assessment: A device used in a different location or under strange conditions may trigger additional checks or be denied access.

Recorded Material Integrity (Trust No Data Without Verification)

- Digital Signatures: All video, audio, and sensor data is cryptographically signed at the source to prevent tampering.
- Chain-of-Custody Logs: Any access or modification to recorded material is logged with a secure audit trail.
- Hashing for Integrity Checks: Periodic verification of recorded material against known-good hashes ensures no undetected tampering.

Security + High Performance

- Edge Processing:** Data filtering and initial validation occur at the edge (if possible) to reduce latency.
- Smart Caching + Streamlined Checks:** Security checks are optimized to avoid bottlenecks without weakening protection.

The platform's mapping system will support both low-fidelity and high-fidelity mapping solutions, each tailored for different levels of detail and use cases.

The low-fidelity solution is designed to provide broad 2D situational awareness across the entire environment (e.g., the Port area) using existing traditional map data. This approach ensures efficient monitoring of large areas with sufficient accuracy for general operational awareness.

In contrast, **the high-fidelity solution** focuses on delivering detailed, high-resolution information for specific, critical areas—such as inside buildings or complex infrastructure zones—where more granular data is essential. Achieving this level of detail will require novel mapping approaches, including the accurate modeling of both interior and exterior building dimensions.

As a long-term goal, the project will evaluate the most effective ways to utilize a complete 3D model of the Port of Oulu. A detailed 3D representation will allow the platform to use accurate building and object dimensions, enhancing the localization and contextual understanding of detected objects, particularly in outdoor environments. The system will also support visual layer management, enabling users to toggle

specific information on or off, such as 3D building models, to customize the view according to operational needs.

Real-time target/anomaly/event detection and tracking within the mapping system will be developed using state-of-the-art YOLO object detection model architectures provided by Ultralytics. To ensure the model meets the specific requirements of the platform, such as detecting people and vehicles at defined distances, we will collect and annotate custom training and validation datasets. Utilizing commercially free annotated datasets will be also considered. The placement and configuration of detection cameras will be strategically planned using insights from both 2D and 3D map data, in order to maximize coverage across indoor and outdoor environments. This includes a visual representation of static camera positions and their corresponding fields of view within the mapped environment.

Furthermore, the system's operational capabilities are enhanced through advanced geofencing features, supporting both restricted and allowed area configurations, while maintaining comprehensive surveillance through high-precision cameras and optimized detection models. All components are designed to operate within the internal network, with efficient data processing and storage mechanisms ensuring minimal latency in real-time operations.

4.4 Critical energy distribution (UC4)

A distribution network operated by an energy community aggregator, as a cooperative, wants to develop the means to use flexibility in case of reliability issues in the distribution grid or when asked by the upstream network operator. At the same time, this operator is managing an energy community composed of a part of the consumers in the area, targeting to optimize the energy contracts definition considering user's flexibility, energy prices, and operation costs. A flexibility aggregator gets involved to collect and provide this flexibility.

Main contribution area to the Sa4CPs concept (see Figure 10): Distributed AI/explainable AI, digital twin. Several initiatives in the field focus on energy aspect, ignoring the learning of user's comfort and flexibility.

Targeted innovation boosted by the Sa4CPS concept: Use of Explainable AI (XAI) for situational awareness; Activation of Flexibility with XAI; Use of federated learning models for energy communities
Target business impact: Use of flexibility; Optimization of resource use.

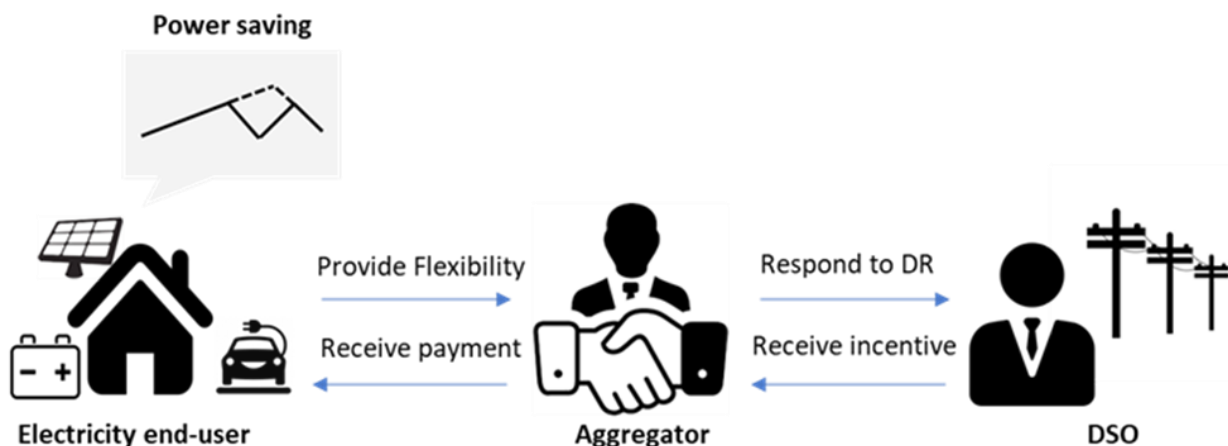


Figure 10. UC4 concept.

4.5 Security and location awareness of persons (UC5)

The consumer cybersecurity ecosystem illustrated in Figure 11 comprises six key stakeholders working together to protect personal devices and data. At the center are consumers who need protection for their digital lives, supported by cybersecurity technology providers who develop specialized security solutions and other technology providers who supply necessary software infrastructure. Government regulatory bodies establish and enforce cybersecurity standards, while threat actors represent the adversarial forces that create the need for protection in the first place. Research institutions play a crucial collaborative role, partnering with companies to advance cybersecurity technologies and methodologies through ongoing research and development.

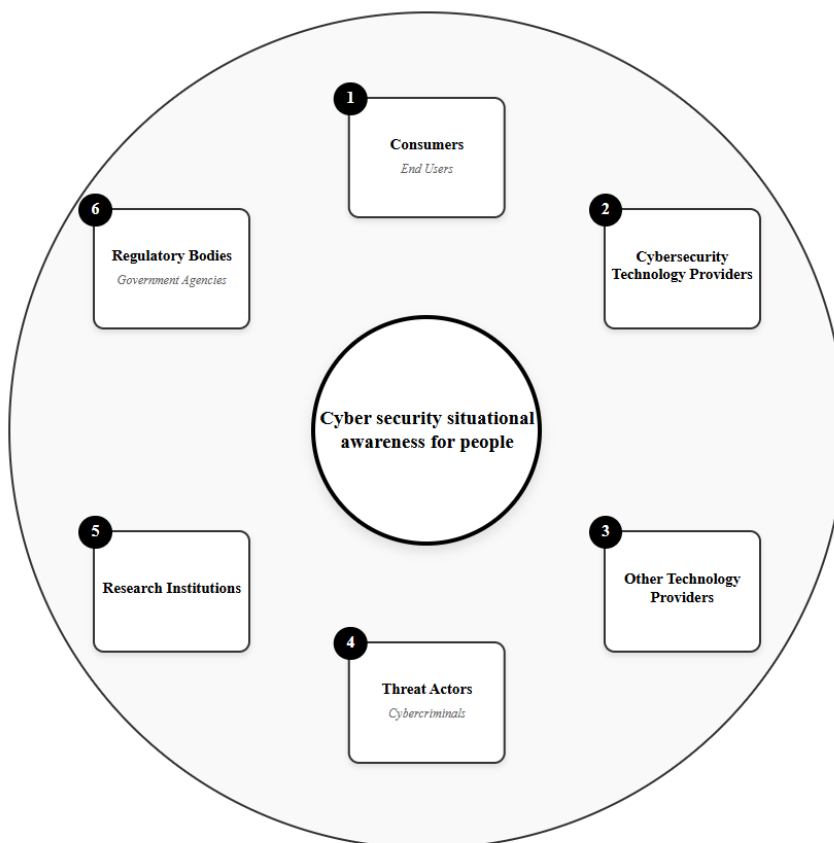


Figure 11. UC5 targeted ecosystem

This ecosystem operates through six interconnected value chains that create a comprehensive security framework. The process begins with threat intelligence and research, where emerging threats are identified and analyzed to develop effective countermeasures. This intelligence informs product development, where cybersecurity solutions including software and hardware are designed and created. The solutions then move through implementation phases, deployed and integrated seamlessly into consumer devices. Once operational, continuous monitoring and operations maintain protection levels, while incident response capabilities ensure rapid mitigation when security breaches occur. Throughout this entire process, customer training and support provides the essential human element, educating consumers on cybersecurity best practices and offering ongoing assistance to maximize the effectiveness of security products.

The UC5 contributes to the Sa4CPS concept using a new model of a contextually relevant and proactive engagement that promotes cyber security situational awareness for people in the context of Sa4CPS framework. Figure 12 presents the initial research setup for the new model of a contextually relevant and proactive engagement that adjusts with what consumers need in moments requiring protection and when they expect a sense of security. It incorporates the following stages of the Sa4CPS concept: perception, comprehension, projection and decision by including the stages and the loop of situation-sensitive protection and situation-sensitive education as illustrated in Figure 12.

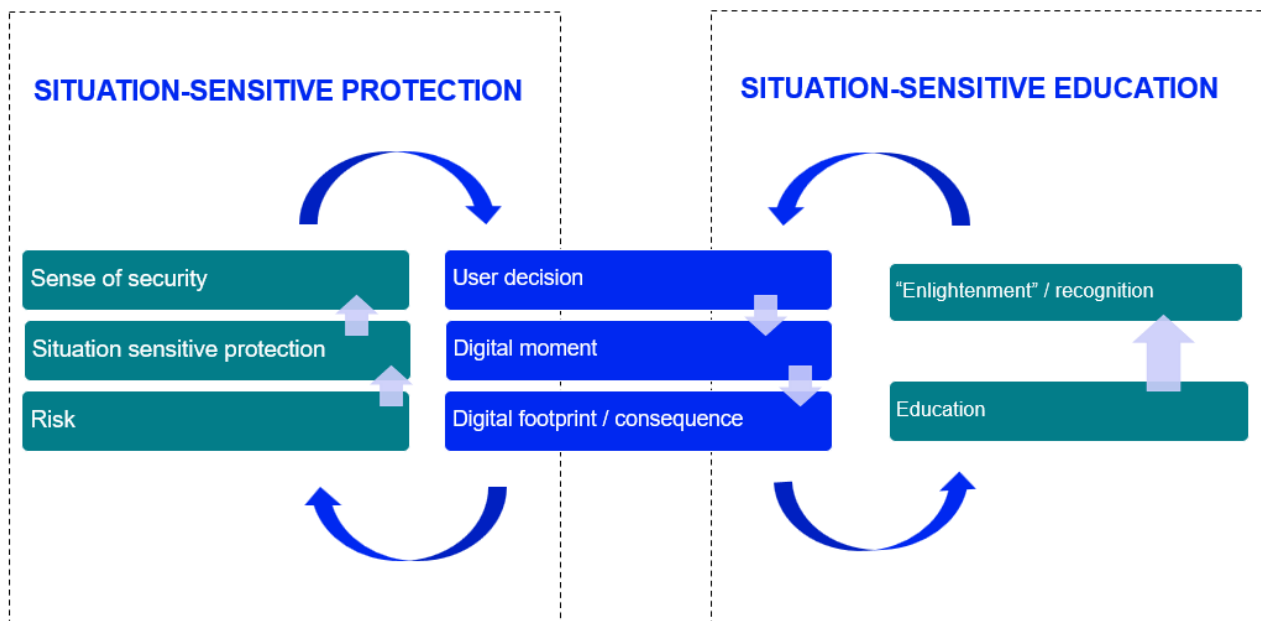


Figure 12. UC5 approach mapped in the research frame: perception, comprehension, projection & decision

4.6 Multirouting secure awareness system for outdoors (UC7)

4.6.1 Natlink & Tracker: Enabling Technologies and Cross-Use Case Contributions

Natlink and Tracker, through Use Case 7 (UC7), will deliver an advanced multirouting secure awareness system tailored for outdoor and mission-critical environments, Figure 13. The UC7 concept is designed to maximize system resilience, operational flexibility, and security, providing a robust reference architecture that can be leveraged by other Sa4CPS use cases. Key findings, technical solutions, and research results will be published to the Sa4CPS consortium, and active collaboration during research and implementation phases is encouraged and preferred to accelerate progress and maximize impact.

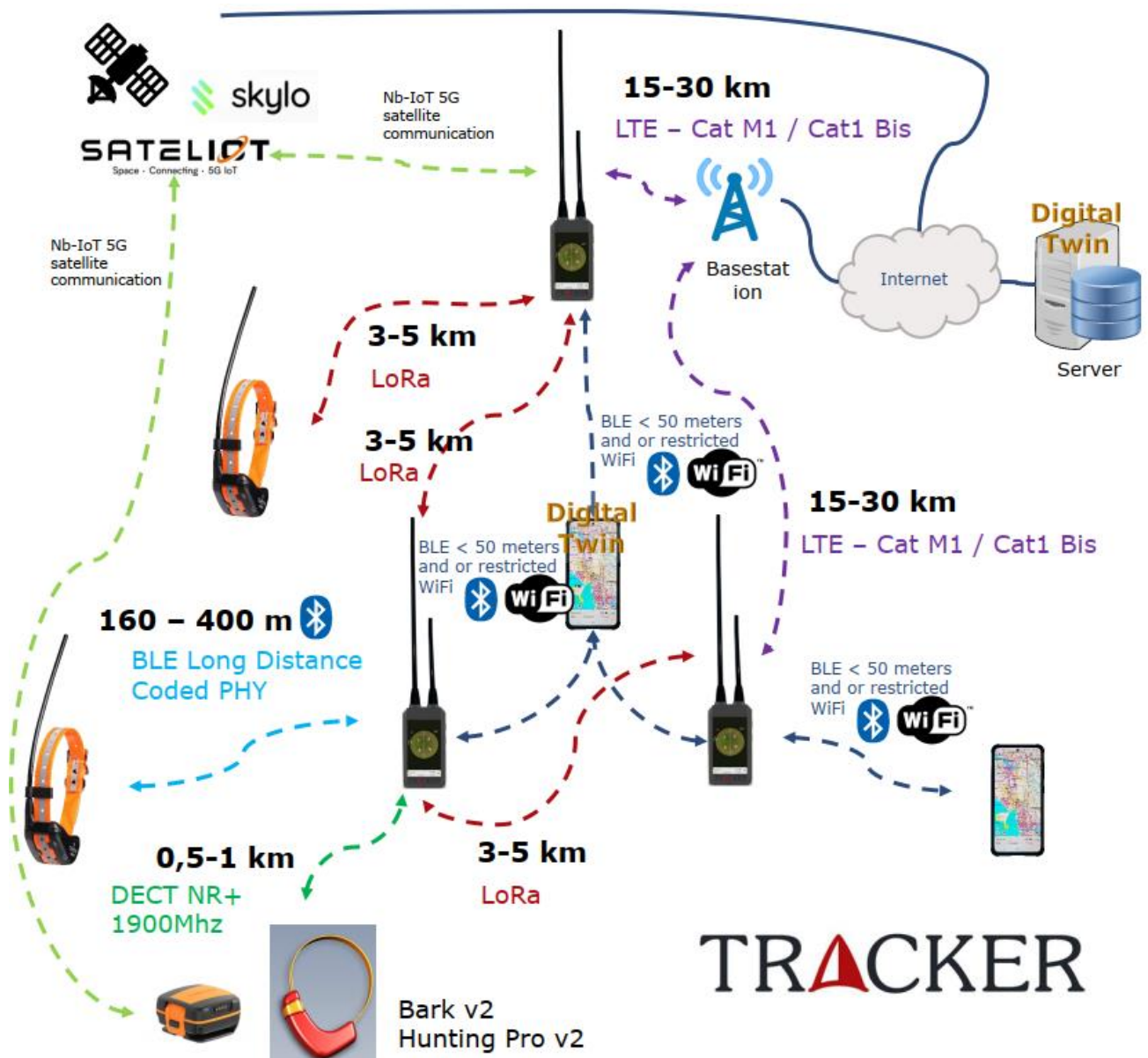


Figure 13. Natlink view to the UC7 concept.

1. Phased Technology Development and Evaluation

UC7 will begin with a comprehensive evaluation of radio technologies, including LoRa, LoRaWAN, DECT NR+, Bluetooth Low Energy (BLE) Long Range, LTE Cat-M1, LTE CAT 1 bis, and satellite communications (3GPP Rel 17, with providers such as Sateliot and Skylo). Each technology will be assessed for its RF propagation characteristics, link budget, interference resilience, power efficiency, and practical usability in the field. The evaluation will also include a detailed study of radio Power Classes, with a particular focus on the feasibility and practical implications of using Power Class 1 (31 dBm) radios to extend uplink range and reliability. The research will consider device size, battery life, heat dissipation, and regulatory requirements, ensuring that the selected solutions are both effective and compliant.

Antenna design and integration will be a major research focus. The project will investigate multi-band, MIMO (Multiple Input Multiple Output), and diversity antenna configurations, aiming to optimize simultaneous multi-radio performance and minimize intermodulation, desensitization, and cross-talk effects. Special attention will be paid to maximizing antenna efficiency through correct coupling resistances and impedance matching, and to exploring the potential to extract additional output power from modem modules via specific AT commands or firmware settings, leveraging the flexibility allowed by 3GPP Power Class 3 specifications. All antenna solutions will be validated through extensive laboratory and field trials to ensure optimal performance under real-world conditions. The research will not only focus on technical performance but also on regulatory and operational constraints, such as duty cycle limitations, frequency allocations, and device certification processes. Collaboration with other consortium partners is encouraged to share test results, harmonize design choices, and accelerate the adoption of best practices across use cases.

2. Dog Tracking Devices: Communication, Antenna Research, and Security

Dog tracking devices are a cornerstone of the UC7 system, enabling continuous, reliable monitoring of animals in challenging outdoor scenarios. These devices will be equipped with multiple radio interfaces, including LoRa, LoRaWAN, BLE Long Range, LTE Cat-M1, and NB-IoT. LTE Cat-M1 and NB-IoT will provide continuous, low-power cloud connectivity, supporting real-time tracking, health monitoring, and remote configuration. LoRa, in contrast, will be used specifically for direct point-to-point communication between the dog tracking collars and the Multicomunicator, ensuring that critical data can be exchanged even when cellular connectivity is unavailable or unreliable. LoRaWAN may be used as an alternative network solution where infrastructure is available.

Antenna and device size optimization will be approached systematically. By utilizing existing mechanical designs from Tracker Artemis, Luna, and Bark products, the project will conduct comparative studies of various antenna configurations, device enclosures, and placement strategies. The goal is to identify the best trade-offs between device size, antenna efficiency, battery life, and user ergonomics. Field trials will be conducted in diverse environments to assess radio range, signal robustness, and device durability under real operational conditions. LoRa communication presents unique challenges, including limited available frequencies, strict duty cycle regulations, and the constraint of a single 0.5-watt channel. To address these, the data protocol for LoRa (and BLE Long Range) will be highly optimized and compressed, with typical payloads limited to around 20 bytes to maximize transmission range and reliability. Advanced data compression, protocol minimization, and efficient error correction will be explored. All data transmitted over these links will be encrypted to ensure confidentiality and integrity, even in the most constrained environments. Security is a fundamental requirement across the entire system. All dog tracking devices, the Multicomunicator, the mobile application, and backend services will be developed within a unified security framework. This includes the implementation of strong symmetric encryption (AES-128), secure key management, mutual authentication, and compliance with

relevant EU and international standards (RED, CRA, GDPR, Data Act, ETSI EN 303 645). Security features will be validated in each device and at every communication interface, ensuring end-to-end protection of data and user privacy throughout the ecosystem. The project will also investigate secure device provisioning, over-the-air (OTA) firmware updates, and secure device addition/removal workflows, ensuring that security is maintained even in offline or low-connectivity scenarios.

aMulticomunicator: Versatile Communication Hub

The Multicomunicator is envisioned as a central hub and gateway within the UC7 system, providing seamless connectivity, data aggregation, and network extension capabilities:

- **Network Extender & Sharing:** The Multicomunicator will aggregate connectivity from multiple uplinks (LoRaWAN, LTE Cat-M1, CAT 1 bis, NB-IoT, satellite) and share this boosted network locally to users' mobile phones. Network sharing will be realized through technologies such as WiFi, WiFi Direct, Bluetooth, or other suitable methods. A key research objective is to identify the most effective and user-friendly technology for this purpose, balancing performance, energy consumption, and ease of use. A critical requirement is that network sharing must not block or interfere with normal phone operation: when Natlink-related or other selected data is delivered via the Multicomunicator, other mobile applications and data (such as WhatsApp or general internet access) must continue to function through the phone's own native network connection. This ensures seamless user experience and system integration.
- **kMulti-Radio Gateway:** The Multicomunicator will function as a direct multi-radio receiver and router, aggregating, multiplexing, and routing data from dog collars (via LoRa), environmental sensors, and handheld devices across different PHY/MAC layers. Protocol translation, redundancy, and data prioritization will be implemented to ensure reliable and efficient data flow. LoRaWAN will be one of the key technologies researched and potentially utilized for both uplink and downlink communication between devices and the Multicomunicator, where infrastructure is available.
- **nSatellite Uplink Node:** For satellite communication, the Multicomunicator will leverage 3GPP Release 17 NTN (Non-Terrestrial Networks) capabilities, enabling direct device-to-satellite connectivity. Providers such as Sateliot and Skylo will be evaluated, with openness to integrating other suitable satellite IoT service providers as the technology landscape evolves. The system will explore fallback strategies, multi-path routing, and dynamic link selection to maximize availability and resilience.

All system components—including dog tracking devices, Multicomunicator, mobile applications, and backend—are designed with extensibility in mind. This means they will support third-party connectivity, allowing integration of accessories, other vendor devices, and external services. This open architecture approach is aligned with the main goal of the SA4CPS project: to improve secure situational awareness through interoperable, multi-vendor, and collaborative solutions. Secure APIs, modular interfaces, and standardized protocols will be developed to facilitate this integration while maintaining robust security and privacy controls.

eAll functionalities will be implemented with modular, extensible hardware and firmware, supporting dynamic reconfiguration, remote management, and future expansion. The project will also investigate power management strategies, ruggedization, and environmental protection to ensure reliable operation in harsh outdoor conditions.

3. Mobile Application, Offline Operation, and Visualization

A core component of UC7 will be a mobile application that provides users with a comprehensive situational overview and full control of the system. The application will feature an interactive map interface powered by a

3D map engine, enabling real-time visualization of tracked assets, device statuses, and environmental data. Users will have access to versatile visualizations, including 3D terrain, object overlays, historical playback, event notifications, and predictive analytics. The mobile app will serve as the primary control interface, allowing users to configure devices, manage system settings, and issue commands remotely—enabling full system control and situational management directly from a smartphone or tablet.

A critical research and development focus will be to ensure that the mobile application and system are fully operational offline, without continuous IP network connectivity. The system will be designed to handle all essential functions—including secure data encryption and decryption, device addition and removal, and local data storage and synchronization—entirely on the mobile device. This ensures that users can maintain security, privacy, and operational control even in remote areas or during network outages, with all cryptographic operations and device management use cases functioning independently of the backend. The app will also support seamless synchronization and conflict resolution when connectivity is restored. Like other components, the mobile application will be designed to support third-party accessories and services, further enhancing the ecosystem's flexibility and reach.

Backend: Digital Twin and Visualization Dashboards

The backend architecture will center around a Digital Twin system, providing real-time and historical monitoring, predictive analytics, and operational dashboards. The Digital Twin will maintain a synchronized, structured representation of all devices, assets, and environmental data, supporting advanced analytics, anomaly detection, and decision support. Visualization dashboards will be available for mobile, server, automotive (Android Auto/Apple CarPlay), and wearable platforms, enabling users and support teams to monitor system status, analyze trends, and respond to events.

The backend will leverage containerized microservices (e.g., Kubernetes, Docker), time-series databases (e.g., MongoDB Atlas), and event-driven data pipelines (e.g., Google DataFlow) for scalability, reliability, and maintainability. The architecture will support bidirectional synchronization, role-based access control (RBAC), and secure APIs for integration with third-party systems and services. Key architectural patterns, APIs, and research results will be documented and shared with the consortium to facilitate reuse and cross-use-case adoption.

Cybersecurity, Ciphering, and Serialization

UC7 will implement a multi-layered cybersecurity framework aligned with EU and international standards and regulations:

- **Encryption & Key Management:** AES-128 (Advanced Encryption Standard, FIPS 197) will be used for all sensitive data, with secure key provisioning, storage (e.g., HSM, TPM), and rotation. Asymmetric cryptography (ECC, RSA) will be evaluated for mutual authentication, OTA firmware signing, and secure device onboarding.
- **Protocols & Compliance:** All communications and data handling will comply with the Radio Equipment Directive (RED, 2014/53/EU), Cyber Resilience Act (CRA), General Data Protection Regulation (GDPR, EU 2016/679), Data Act (EU 2023/2854), and ETSI EN 303 645 (IoT cybersecurity baseline). The NSTP protocol will provide secure session management, mutual authentication, integrity verification (e.g., HMAC, SHA-256), and comprehensive audit logging.
- **Serialization:** Serialization protocols under consideration include JSON, Protocol Buffers (Protobuf), CBOR (RFC 7049), custom binary formats, and MessagePack. MessagePack is favored for resource-

constrained and low-latency environments due to its compactness and speed. The project will evaluate serialization choices for efficiency, security, and interoperability.

- **Access Control & Privacy:** The backend and mobile application will implement RBAC, fine-grained access policies, pseudonymization/anonymization for personal data, and full audit trails to support regulatory compliance, incident response, and user trust.

4. Cross-Use Case Value

The multirouting secure awareness system, Multicomunicator, dog tracking devices, and mobile application developed in UC7 will provide a comprehensive, resilient, and secure reference architecture for situational awareness in demanding environments. The approach is modular and extensible, allowing other Sa4CPS use cases to adopt, adapt, or integrate key components and research findings. Key findings, technical solutions, and research results will be published to the Sa4CPS consortium, and active collaboration and co-operation during research and implementation phases is encouraged to maximize impact, accelerate progress, and ensure that the consortium as a whole benefits from shared expertise and innovation.

4.6.2 Polar data synchronization device for multiple wearables data collection

Polar Electro's research in improving situational awareness and data connectivity is focused on the development of a biometric data synchronization device that can be used to collect data from several wearable devices at the same time during activities. The purpose of the synchronization device is to serve the real-time monitoring of the information recorded about the group at the individual level, to create the conditions for anticipating control measures and making immediate decisions if the data obtained from the individual and the group so requires, for example, due to a decline in functional capacity.

After the data transfer, the situational awareness information is used in real time on the mobile devices of several different operators separately for each purpose of use and control. The historical data and its statistical processing, analysis and replanning of operations are done in the Polar Guidance Service system.

Table 9. Main use scenarios of the Polar's view to the UC7 concept.

Data transfer method	User group	Context and Use case	Data items	Power (electrical/battery)
Real-time via BLE	Gym (Group sessions)	HR data visualization during gym training (body pump, spinning)	HR	Battery
	Schools	HR data visualization during physical education	HR	Battery

	Sport club premises	Data live stream during practices and matches	HR Training load (Muscle/Cardio) Power Speed Location	Battery Electrical
	Firefighters , First responses, Police	Monitoring Firefighters, polices and first responses physical load during missions	HR Training load (Muscle/Cardio) Power Speed Location	Battery
	Clinics and Hospitals	Monitoring patients' data during and after care	HR RRi/PPi Skin temp ACC GYRO	Battery Electrical
	Military	Collecting and visualizing live data during military field events	HR RRi/PPi ACC GYRO Glucose Skin/Body temp GNSS	Battery Electrical / Power supply
Offline data sync via BLE	Military: Barracks, Vehicles, Base camps	Synchronizing Offline data recordings 24/7 Smart Coaching algorithm data 24/7 sample data	24/7 Sample data (HR, PPi, MET, TRIMP, Skin temp) Offline data (HR, PPi, ACC, GYRO, BARO, Skin/Body temp, Glucose) Sleep Activity Training sessions Training load Performance tests Recovery Stress	Electrical / Power supply

Offline/Training data sync via USB	Sport performance labs	Synchronizing Training session data after test	Training sessions	Electrical
	Military: Data collection events, Barracks	Synchronizing offline recordings and training session data	Offline recordings Training sessions	

Synchronize device model:

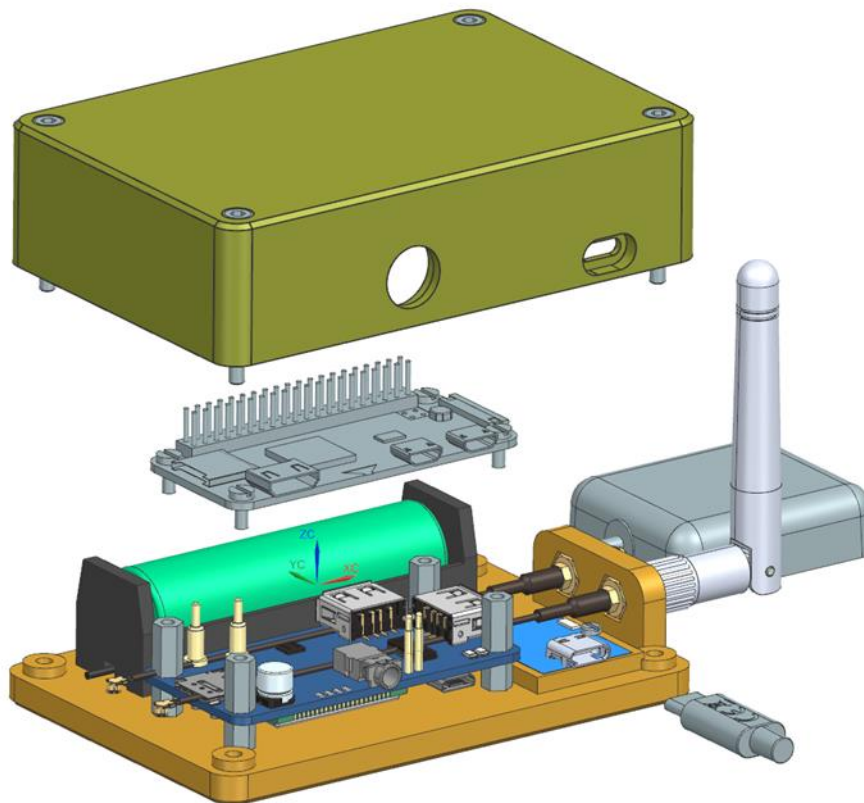


Figure 14. Setup for Raspberry PI Zero W / Raspberry PI 5 + 4G Module

Prerequisites

One of

- Raspberry PI Zero W: <https://www.raspberrypi.com/products/raspberry-pi-zero-w/>
- Raspberry PI 5
- Sandisk Extreme Pro 64GB microsd memory card
<https://www.verkkokauppa.com/fi/product/794368/SanDisk-64-Gt-Extreme-Pro-SDXC-UHS-I-muistikortti>

- 4G pHAT for Raspberry Pi - LTE Cat-4/3G/2G with GNSS Positioning
<https://thepihut.com/products/4g-phat-for-raspberry-pi-lte-cat-4-3g-2g-with-gnss-positioning>
 - Contains SIM7600G-H circuit. Official manual homepage:
[https://www.waveshare.com/wiki/SIM7600G-H_4G_HAT_\(B\)](https://www.waveshare.com/wiki/SIM7600G-H_4G_HAT_(B))
- A SIM Card for 4G connection
- Raspberry PI Imager v.18.5
- 5V USB power source.
- HDMI Mini/micro (pi 5) connector, keyboard, mouse and a display for local development mode

Indicator LEDS

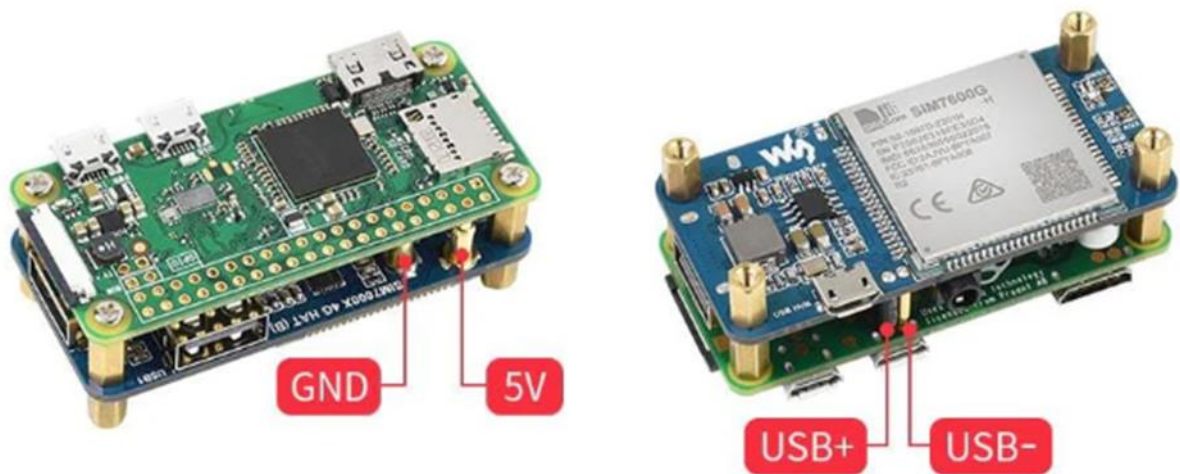


Figure 15. Structure of the experimental prototype.

4.7 Drone operation control system (UC8)

The Figure 16 represents the core concept of the intended system developed in UC 8: a situational awareness platform for aerial drone operations, receiving and processing all relevant flight information, assessing an operation's current state, and enabling various outputs and feedback loops enhancing operational safety and automation.

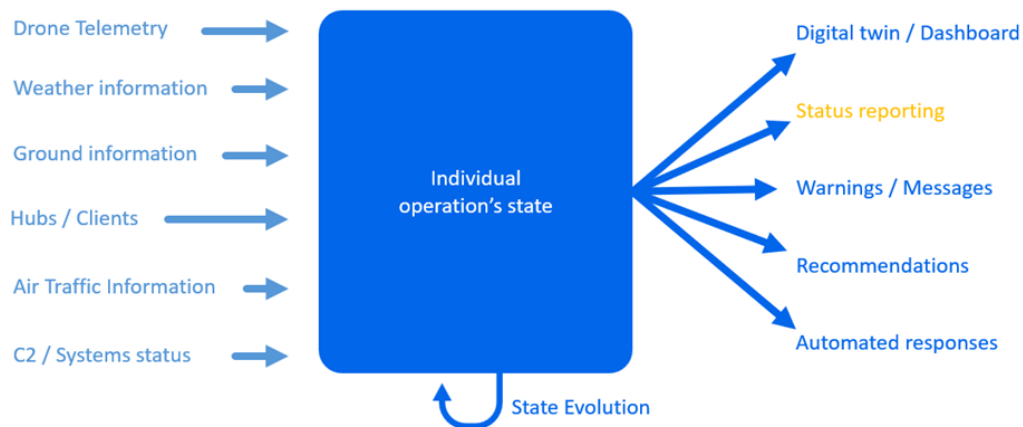


Figure 16. A view to the UC8 concept.

This platform will at first be used to enhance ADLC's flight operations as part of its growing drone transportation service, but could also become useful for other aerial drone operators (including for other use cases like inspection), and could easily be adapted for other types of remotely monitored automated vehicles.

Key members of this use case ecosystem are:

- Regulatory authorities: Regulations for the remote monitoring for a large scale fleet of unmanned aerial vehicles do not exist or are unclear at this moment, but as it has a safety impact the use of the platform will need to be approved for some of its outputs. The Belgian Civil Aviation Authority is already kept informed of the project.
- Drone manufacturers: They usually design flight control systems to be used directly and continuously during the flight. The developed platform enables control on a higher layer, but does require their cooperation and understanding.
- Existing operational staff: Drone pilots will remain in the loop of the system, and their feedback is key to enable safe high volume operations where they can focus on solving situations which can not be safely automatically resolved.

Use Case 8 contributes to the Sa4CPS project as it explores:

- Situational awareness in the case of automated vehicles, imposing the taking into account of movement in its projections.
- Perception and comprehension of more than the physical system itself, but of all external factors that affect its operation.
- Perception and comprehension through a global control system, not only on the device itself, and therefore including as core components communication links wrt. Reliability and security.
- The inclusion of a human in the loop of the system, also needing to reach comprehension via the system and able to conduct actions and respond to incidents.
- The prospective utilization of the system in high-risk environments, leading to important requirements on reliability and sufficiently low latency.
-

4.8 Management of CPS assets (UC9)

Vintecc uses the Capture platform to monitor remote devices (in factories, driving vehicles, etc), Figure 17. The contribution of UC9 is to use Capture to provide the possibility to remotely update the OS and application software of remote devices in a uniform way, regardless if they are running in Windows, Linux or Docker.

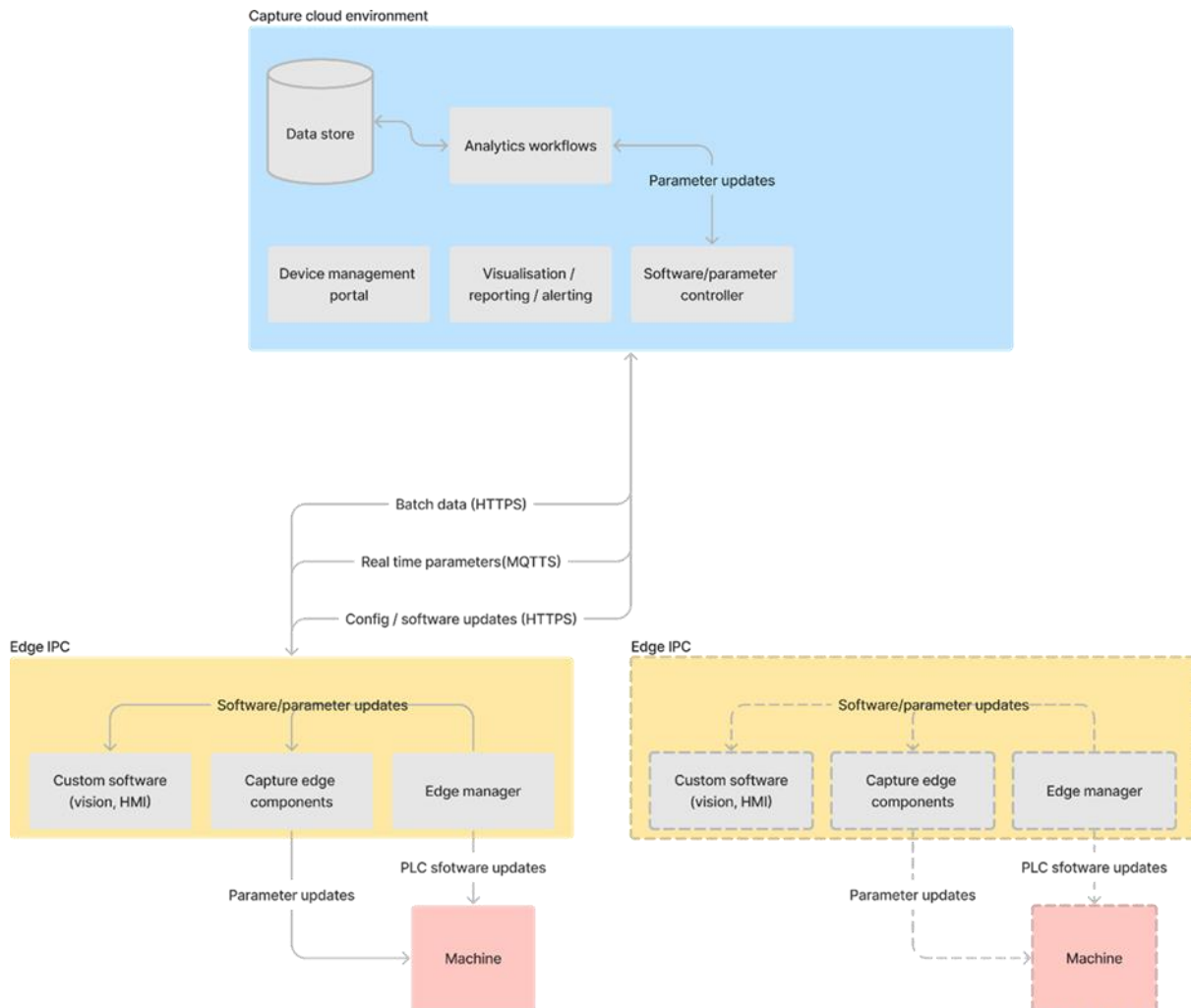


Figure 17. A view to the UC9 concept.

A second contribution is the situational awareness of cyber physical systems regarding updates and device writes (for example configuration of physical machines from the cloud). The situational awareness should allow the system to automatically block actions that could result in a dangerous situation (physical harm to machines, animals or people). This situational awareness can be situated in the cloud (blocking a dangerous action before it is sent to the CPS), or on the CPS itself. It should also allow the CPS to schedule updates in a time when this update can safely be executed, without intervention of a human.

5 Contribution(s) of each technical work package into the Sa4CPS concept

This section discusses / clarifies potential contributions of each technical work package WP3-5 into the Sa4CPS concept.

5.1 Contributions related to perception and comprehension to the Sa4CPS concept (WP3)

5.1.1 Perception and comprehension

WP3 focuses on the R&D of the technical enablers for perception and comprehension.

The perception means sensing the environment using sensors, cameras, available radios etc. The comprehension is built by fusing the data from the various sensors and by data processing (including AI/ML based approaches) to provide more accurate and comprehensible reading of the situation.

While the sensing happens at the level of the devices, the sensor fusion may span multiple devices and is potentially realized on the edge or in the cloud.

The aim of the work package is to provide trustworthy data, on which situations can be projected in WP4.

5.1.2 Potential contributions from physical environment sensing (task 3.1)

Physical environment sensing [Natlink, IMA, Polar, Digitalmente, CEL, ACD, VTT, Otokar, Enerim] (all UCs)

Partners in T3.1 will provide contributions related to

- Dog tracking devices - not only for GPS-based location but also for monitoring specific behaviors such as pointing, barking, or detection responses, effectively turning dogs into mobile environmental sensors. By interpreting these behaviors—such as a detection dog indicating the presence of contraband or a hunting dog locating pests—the system can deliver actionable, real-time situational awareness to users, enhancing exploitation of physical environment information across diverse scenarios.
- Human presence and activity tracking in the workplace environment. This will comprise detection of authorized and unauthorized personnel at a workplace, detection of potential hazards around a moving robot arm, and comprehension of gesture-based interaction with the system.
- Biometric Wearables for Environmental Monitoring. This will provide real-time biometric and location monitoring in sport, rescue, or hazardous environments.
- IoT-based Environmental Sensing in Energy Communities. This comprises sensing data about grid status and flexibility assessment in distributed energy communities.

5.1.3 Potential contributions from Interaction based sensing of information context (task 3.2)

Interaction based sensing of information context [Otokar, Natlink, Polar, Digitalmente, ACD, Vintec, VTT, Enerim] (all UCs) Technical requirements and R&D of the means for Acquisition of information from other computers nearby, information discovery, and exploitation from cloud environments of different stakeholders

Partners in T3.2 will provide contributions related to:

- Context-aware Security Moments Detection
AI-driven identification of digital moments where users are at heightened risk, enabling adaptive, situation-aware cyber-protection measures that respond to the user's context and activity.
- Collaborative Environment Sensing
Fusion of human presence sensors, machine activity, and access control to build a dynamic picture of the operational context in shared workplaces, supporting proactive safety and interaction-based event detection.
- Multi-actor Field Sensing (Dogs, Hunters, Devices)
Use of interconnected field devices (hunters, dogs, portable radios) to sense not only environmental data but also social and collaborative context (e.g., team location, interaction patterns) in low-connectivity areas.

5.1.4 Potential contributions from information and sensor fusion (task 3.3)

Information & sensor fusion [Polar, Natlink, ISEP, Digitalmente, Natlink, ACD, CEL, INOVASYON, ADLC, Vintec, VTT, Enerim] (all UCs) Technical requirements and R&D of the technical means for Smart fusion of information from various sources and sensors (sensor fusion). Algorithms and tools for enabling such processes

Partners in T3.3 will provide contributions related to:

- Edge/Cloud AI-Powered Sensor Fusion
- Multi-modal Biometric and Movement Data Fusion for Human State Assessment. Polar's wearable devices non-invasively capture ECG and optical heart rate signals, and fuse these with GNSS and inertial sensor data to build a comprehensive profile of a person's physiological load, recovery, and activity level. Mathematical algorithms will be used to interpret changes in heart rate variability, movement, and location to provide nuanced insights into both physical strain and psychological stress.
- Algorithmic Interpretation and Integration Across Devices. Algorithms that aggregate and process raw sensor data to derive actionable metrics—such as activity type, quality of rest, or levels of stress—enabling both direct and indirect assessment of human performance and wellbeing in demanding environments. These metrics can be made available as raw or processed data for integration into broader situational awareness systems.
- Device Network Fusion and Operational Efficiency Assessment. Wearable sensors can be seamlessly integrated with other devices used during human activity—forming an operational device network. This integration allows for holistic analysis of both individual and collective efficiency, skill, and performance, by combining biometric and operational data streams in real time.
- Secure Multi-device Communication and Data Transfer. The architecture includes secure radio protocols and data transfer technologies, ensuring that sensitive physiological and operational data

from wearables are protected during transmission between devices and information systems. This is critical for maintaining data privacy and reliability in large-scale field operations.

- Algorithmic Situational Image Generation and Operational Control. Sensor fusion algorithms synthesize biometric and operational device data to produce situational images across vast environments—potentially covering thousands of square kilometers. The resulting situational awareness supports operational control systems in anticipating and managing diverse scenarios, enabling informed decision-making and optimal activity coordination.

5.1.5 Potential contributions from modelling and tools for comprehension (task 3.4)

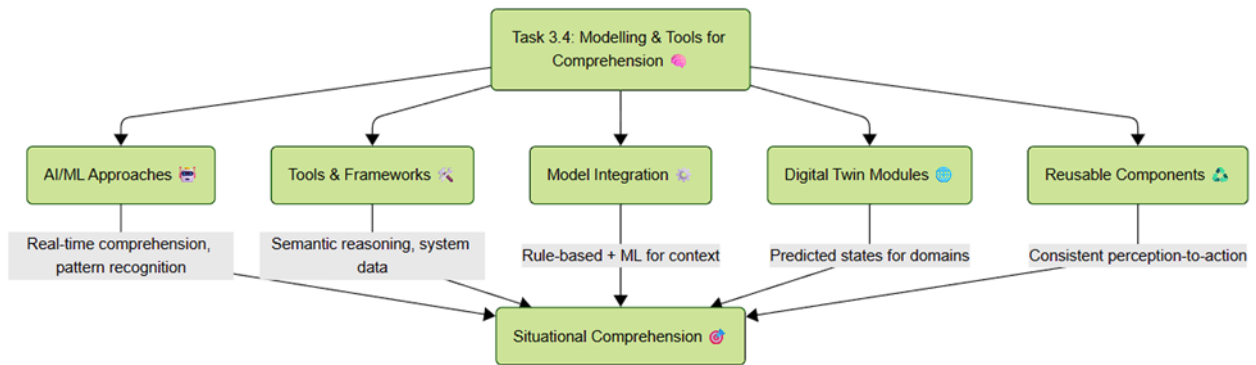
Modelling and Tools for comprehension [INOVASYON, CUNI, ISEP, CEL, Digitalmente, Natlink, VTT,] (all UCs) Technical requirements and R&D of the technical means for Modelling and tools, e.g. AI/machine learning, for enabling smart comprehension of the specific situations in the practical targeted use cases

Partners in T3.4 will provide contributions related to:

- Semantic Situation Modelling and Ontologies
- Development of ontologies and high-level situation models for dynamic environments and CPS interactions.
- Modelling Human/Machine Interactions in Production. ML and logical comprehension models will be developed for representing and evaluating human/machine interactions, for ensuring safety, and for detecting unauthorized personnel within secure areas.

Contribution of task 3.4 towards SA4CPS platform will be performed over the following achievements:

- Development of AI/ML-based modelling approaches to enable real-time comprehension of complex and dynamic situations, including uncertainty handling and pattern recognition from multimodal sensor inputs,
- Creation of tools and frameworks to support semantic interpretation and reasoning over environmental and system data, allowing for deeper situational understanding in cyber-physical environments,
- Integration of machine learning models with rule-based systems to describe context-dependent behaviours and relationships, especially for scenarios involving human machine interaction and dynamic operational risks,
- Support for building digital twin-based comprehension modules that reflect current and predicted system states for energy, logistics, and mobility domains,
- Providing reusable modelling components for situation understanding, which can be applied across different use cases to enable consistent and scalable perception-to-action pipelines.



5.2 Contributions related to status, projections and actions to the Sa4CPS concept (WP4)

The Sa4CPS concept emphasizes the importance of achieving secure situational awareness for critical cyber-physical systems (CPS), particularly in response to unforeseen events that could compromise safety, operations, or societal stability. Work Package 4 (WP4) plays a pivotal role in realizing this vision by developing the core technical enablers necessary to support real-time status tracking, future projections, and context-aware actions across logistics, mobility, energy, and security domains. WP4 contributes to the Sa4CPS concept in the following ways:

- **Status Awareness:** By integrating digital twins, object tracking mechanisms, real-time data collection, and user-facing dashboards, WP4 provides a comprehensive understanding of the current situation across diverse CPS environments. This includes mapping ongoing activities, resources, and potential anomalies in both physical and virtual domains.
- **Projections:** WP4 enables predictive modeling and future-state estimation based on contextual data streams. Through AI-supported reasoning engines and simulation tools embedded within the digital twin frameworks, it becomes possible to anticipate potential events such as resource shortages, equipment failure, or safety breaches.
- **Actions:** Building on real-time insights and projections, WP4 supports autonomous and semi-autonomous decision-making and execution of corrective actions. This encompasses responses to emergencies, rerouting in logistics, fail-safes in energy systems, and intervention protocols in security scenarios. AI-assisted reasoning is used to ensure appropriate responses are derived, maintaining control in dynamic and potentially high-risk environments.
- **User-Centric 3D Visualization:** To support operators and stakeholders from various disciplines, WP4 introduces 3D visual interfaces that ensure clarity of overlapping views, enhancing the ability of human users to interpret and respond to complex situations effectively.

Overall, WP4's contributions serve as the technological backbone enabling the Sa4CPS concept to meet its objectives of zero trust, multi-stakeholder awareness, and resilience enhancement.

5.2.1 Status, projections and actions

Status: The real-time condition or state of a CPS environment, including location, performance, operational parameters, or external influences.

Projections: Future predictions derived from current data using models or AI reasoning to forecast potential events or conditions.

Actions: Context-aware, possibly autonomous responses to status and projections in order to maintain or restore desired system behavior.

Digital Twin: A virtual representation of a physical system that mirrors its state, behavior, and processes.

Zero Trust: A security principle that assumes no implicit trust; every access or transaction is continuously verified.

3D Maps/Views: Visualizations providing spatial and temporal context to improve understanding and decision-making.

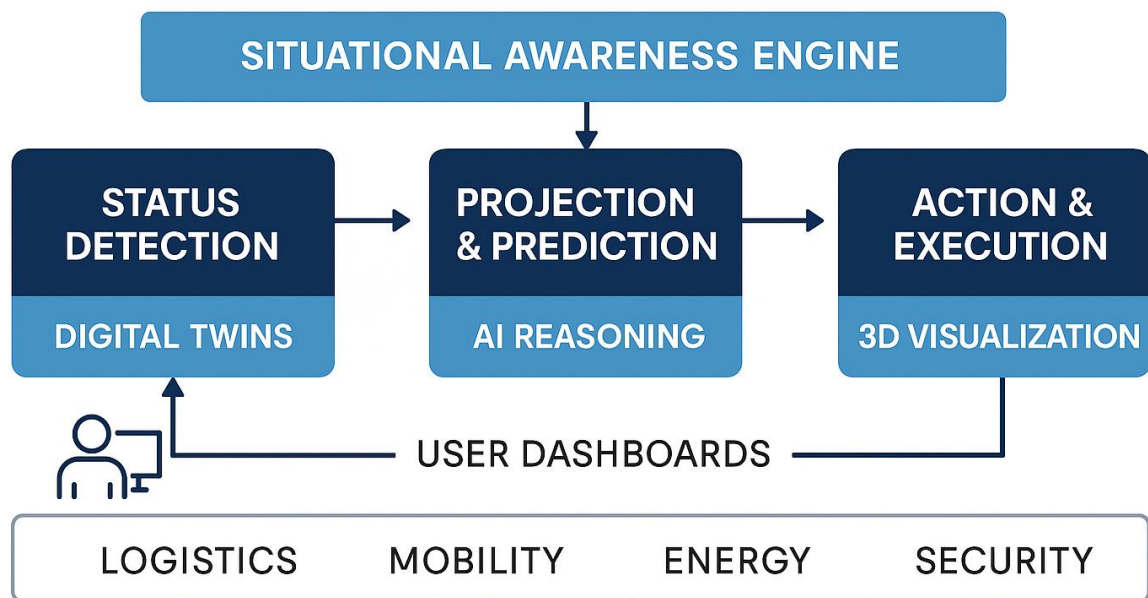


Figure 18. Scope and Integration of Status, Projections, and Actions in WP4 for Sa4CPS.

This figure illustrates the key scope areas of WP4 within the Sa4CPS framework. WP4 serves as the enabler that connects real-time sensing and tracking, predictive analytics, and automated or human-triggered responses, all while ensuring that information is presented clearly and securely to diverse stakeholders. The integration of digital twins and AI reasoning ensures a proactive and resilient CPS ecosystem.

5.2.2 Potential contributions from digital twins and dashboards (task 4.1)

Digital twins & dashboards [ISEP, Digitalmente, CEL, Natlink, Port of Oulu, ACD, Inovasyon, F-Secure, ADLC, VTT, Vintec, Otokar] (all UCs) Technical requirements and R&D of the digital twins, technologies, dashboards for various purposes ex. visualizations

5.2.3 Potential contributions from positioning and tracking of objects (task 4.2)

Partners in task 4.2 will

- follow the sensor system development performed in WP3. The systems will perform collection of data about objects' and humans' position as well as activities of the entities,
- use sensor and sensor fusion to provide situation awareness of user position and status. Location can be based on GPS, inertial sensors, and radio awareness (BLE, WiFi (5G)),
- track the targets, utilized different methods,
- Integrate location data from different sensors and sources of the specific location.

Data will be provided to Sa4CPS system by means of defined and designed protocols.

5.2.4 Potential contributions from autonomous operations (task 4.3)

Autonomous operations [Polar, ISEP, Digitalmente, CEL, Natlink, Modirum Defence, Port of Oulu, ACD, Inovasyon, , VTT, Otokar] (all UCs) Technical requirements and R&D of the Autonomous operations required to happen locally in real-time or near real-time manner

5.2.5 Potential contributions from detection and responses (task 4.4)

Contribution of task 4.4 towards SA4CPS platform will be performed over the following achievements

- development of techniques to model and detect potentially hazardous situations. This will consist of a formal model to describe the scope of the situation and tools to use the model to identify such situations at runtime. The model will explicitly integrate ML-models, which will be used to describe the hard-to-model and uncertain aspects of the potentially hazardous situations,
- Location detection devices will automatically trigger operations like data collection or transmissions
- enabling the use of digital twins, autonomous operation, and detection and response in the context of energy communities
- Use of AI models to detect events which require response from authorities like accidents, [trespassing](#) or fires from surveillance camera feeds.

5.2.6 Potential contributions from real-time deep 3D maps and tools (task 4.5)

Real-time deep 3D maps & tools [Modirum | Gespi, Natlink, Port of Oulu, VTT, Vintec, Otokar] (all UCs) Technical requirements and R&D of the means for Real-time deep 3DP maps & tools.

Modirum | Gespi leads the development of real-time 3D perception (3DP) mapping and related tools within the scope of the project, with the aim of integrating the 3D mapping capabilities as a functional component of the NSC3 system. The overarching objective is to deliver accurate and actionable 3D situational awareness by combining real-time sensor data with advanced AI-driven analytics and interactive visualization capabilities. This capability is intended to enhance both operational decision-making and environmental understanding in dynamic and security-sensitive areas, such as ports and restricted zones.

At the current stage, Modirum | Gespi provides 2D mapping functionality that incorporates real-time object detection overlays, utilizing ValoR AI's computer vision engine (Figure 9, Section 4.3). This solution supports the detection and classification of objects such as people, vehicles, and other anomalies, which are then georeferenced and displayed on 2D situational maps. This forms the basis for expanding into 3D mapping and visualization.

The development approach draws on existing 3D mapping solutions and tools produced by project partners, including VTT and Natlink, who have already made initial progress in the modeling and visualization of the Port of Oulu. These existing assets and methods will be integrated into the broader 3D mapping framework, serving as a foundation for real-time augmentation with live detection overlays and tracking information provided by Modirum | Gespi's NSC3 system.

An additional area of focus is the ability to deploy and operate the system across a variety of edge devices and data sources, including UAVs, ground-based surveillance cameras, and other autonomous sensors. The system is designed to process and analyze video or image data (regardless of whether it is captured from the air or from the ground) for the detection of anomalies such as people or vehicles. This requires optimized processing methods that can operate within the computational limitations of onboard platforms, while ensuring synchronization and data fidelity with central command systems.

The intended outcome of this task is to provide a real-time, interactive 3D mapping system of the Port of Oulu area, capable of visualizing live anomaly detections with high spatial accuracy. The solution aims to demonstrate significant operational value in domains such as surveillance, reconnaissance, and mission planning.

5.3 Contributions related to trust, security and privacy to the Sa4CPS concept (WP5)

The Sa4CPS concept emphasizes the importance of achieving secure situational awareness for critical cyber-physical systems (CPS), particularly in response to unforeseen events that could compromise safety, operations, or societal stability. The role of Work Package 5 (WP5) in realizing this vision is in developing the core technical requirements for trust, security and privacy across logistics, mobility, energy, and security domains within the Sa4CPS framework.

WP5 contributes to the Sa4CPS concept by, first, defining the risks, challenges and technical requirements for trust, security and privacy in the UCs. These are discussed in D5.1. Second, R&D of the technical enablers required in the Sa4CPS concept for digital trust, security and privacy for increasing reliability of the situational awareness for the devices, people and digital twins. This includes aspects of identification of objects, owners & stakeholders, verification, access control, secure information spaces, sharing and overlayed routing, zero trust services and applications with distributed ledger technologies (DLT), and sensing security situation.

5.3.1 Trust, security and privacy

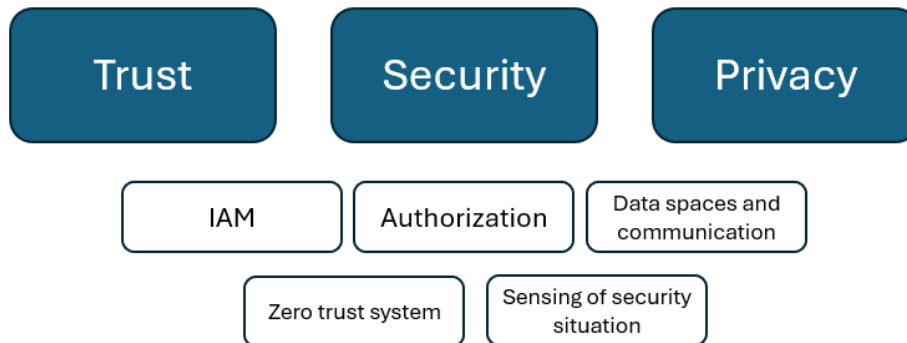


Figure 19. A view to the WP5 content.

Work Package 5 (WP5) addresses the fundamental pillars of trust, security, and privacy that are essential for realizing secure situational awareness in critical cyber-physical systems.

Trust refers to the confidence in the reliability and integrity of cyber-physical system components, digital identities, data sources, and communication channels to operate as expected and provide accurate information for situational awareness. In the Sa4CPS framework, trust is not assumed but must be continuously established and verified through technical mechanisms.

The Zero Trust System functional block directly addresses this challenge by implementing a "never trust, always verify" approach. Rather than relying on perimeter-based security models that implicitly trust entities within organizational boundaries, the zero trust system requires continuous verification of all participants, devices, and data sources. This is particularly crucial in the multi-stakeholder environments that characterize the Sa4CPS use cases.

The Identity and Access Management (IAM) functional block supports trust by providing uniform mechanisms for identifying and authenticating both person and non-person entities across heterogeneous systems. This ensures that trust relationships can be established and maintained even when different stakeholders use different identity systems, enabling the secure cross-organizational collaboration that is central to the Sa4CPS concept.

Security encompasses the protection of cyber-physical systems against threats, attacks, and unauthorized access that could compromise system functionality, data integrity, or operational safety. Given the critical nature of the targeted application domains, security failures could have serious consequences for human safety.

The Authorization functional block provides granular control over who can access what resources under which conditions. This goes beyond simple access control to include dynamic, context-aware authorization that can adapt to changing situations.

Data Spaces and Communication security ensures the integrity and authenticity of information flowing between different stakeholders and systems. This is essential for maintaining the trustworthiness of situational awareness information.

The Sensing of Security Situation enables proactive threat detection and response, ensuring that security incidents can be identified and addressed before they compromise situational awareness capabilities.

Privacy involves the protection and controlled disclosure of sensitive information collected and processed within cyber-physical systems, ensuring unauthorized parties cannot access confidential or personally identifiable data.

All above mentioned functional blocks contribute to privacy protection through their implementation of privacy-by-design principles. The IAM and Authorization blocks ensure that personal and sensitive data is only accessible to authorized entities with legitimate need-to-know. The Data Spaces and Communication block implements secure information sharing protocols that protect data in transit and at rest. The Zero Trust System provides continuous verification that privacy policies are being enforced correctly.

These trust, security, and privacy capabilities are not standalone features but are deeply integrated into the perception, comprehension, projection, and action phases of situational awareness. For instance:

- Perception phases require trusted sensors and secure data collection mechanisms
- Comprehension depends on verified data sources and authenticated information fusion processes
- Projection capabilities must operate on trusted models and secure historical data
- Actions require authorized decision-making processes and secure command execution

Security of edge management system, Sirris contributes the following aspects:

- **Reference architecture for user authentication** for secure remote firmware update for edge/end devices with trace and tracking ability. Ensuring that access to edge/end device(s) is securely managed and monitored. The proposed architecture works both for connected and office edge/end devices. We envisage using two factor authentication method with the ability to centrally record access activities of user/application to edge/end node during the firmware update process.
- **Firmware update with code signing techniques**. To ensure trust and authenticity of the firmware to be installed on edge/end device(s), the software should be checked before allowing it to be installed on the device. Sirris is investigating the integration of code signing techniques into the remote firmware update process for edge/end device(s). In addition to that, we also look into signature key management, especially the security of the key when storing it on edge/end device(s).
- **Cybersecurity assessment for edge device management system**. Sirris is working on the design of a general process and guideline for cyber security assessment of edge/end device management system, based on ISO 27001/2 and Mitre framework. The main goal is to identify and document different security elements that need to comply with the ISO 27001/2 and assess the security posture of edge device management systems.

Security awareness of drone operation/flight control system, Sirris contributes the following aspects:

- **Cybersecurity assessment for drone flight control systems.** Sirris is working on the design of a general process and guideline for cyber security assessment of drone flight control systems, based on ISO 27001/2 and Mitre framework. The main goal is to identify and document different security elements that need to comply with the ISO 27001/2 and assess the security posture of the drone flight control management systems.
- Design of reference architecture for situational awareness for drone flight control systems, allowing pilots to operate more than one flight at the same time.
- Design algorithm for security risk assessment and classification of each flight based on real-time and historical data. This risk classification is important for pilots to decide which flight should be monitored closely in an event where an anomaly is detected.

5.3.2 Potential contributions from identification, verification and access control (task 5.1)

Identification, verification & access control [Polar, ISEP, Digitalmente, CEL, Natlink, ACD, , ADLC, VTT, Sirris, Vintec, Otokar, Enerim,] (all UCs) Technical requirements and R&D of the enablers and means for identification, verification and access control

The identity and access management (IAM) functional block (sec 3.2)

it is necessary to identify person and non-person entities and define various access policies for those entities into resources. Resources may be defined, for example, in a data space or a digital twin, or a resource can be a communication endpoint. Access to the various resources that, depending on where the resource is located, may be defined in different ways. Thus, there is a need for some kind of uniform way to manage resource access, with support for multiple access types (role-based, attribute-based, policy-based).

Identity and Access Management are a common requirement for nearly all data usage, and thus implicitly present in all of the above requirements. In addition, there are explicit requirements for **authentication and role-based access control** regarding users, services and devices (REQ_UC1.24, REQ_UC3.1, REQ_UC3.2, REQ_UC2.11). **Digital twins** also require consideration with respect to who is able to access which data (REQ_UC1.3, REQ_UC7.25, REQ_UC7.27).

Group and community data management sets requirements for data access control within groups of users. These requirements involve the creation and configuration of, and notifications to the groups (REQ_UC4.9, REQ_UC7.28, REQ_UC7.29), and controlling data and devices of the groups (REQ_UC7.30, REQ_UC7.33, REQ_UC7.35, REQ_UC7.36)

Authorization functional block (sec 3.3)

The requirements for authorizing users, devices and services to use a resource can be further categorized: **Authorization with zero trust.** Authorization focusing on security and trustworthiness, with assumption of strong identity verification without implicit trust and need for constant verification (REQ_UC1.24-25, REQ_UC3.1-3.3, REQ_UC1.3), often with compliance and cryptographic requirements.

Role-based operational control. Needs to ensure users are authorized to operations and data access based on their role (REQ_UC1.16, REQ_UC2.11, REQ_UC7.27, REQ_UC7.32, REQ_UC7.33).

Digital twins and asset integrity. Secure and authorized use of digital twins for monitoring, history tracking and integrity assurance (REQ_UC1.3, REQ_UC7.25, REQ_UC7.27).

Privacy & Consent. Authorization with respect to user privacy and compliance with data protection regulations (REQ_UC7.29, REQ_UC7.35, REQ_UC7.27)

5.3.3 Potential contributions from secure information spaces, sharing and overlayed routing (task 5.2)

Secure information spaces, sharing & overlayed routing [ISEP, Digitalmente, CEL, Natlink, ACD, , ADLC, VTT, Sirris, Vintec, Otokar, Enerim,,] (all UCs) Technical requirements and R&D of the enablers and means for secure information spaces, sharing and overlayed routing

Data spaces and communications functional block (sec 3.1)

While most operations require some handling of data, there are some specific categories that arise from the requirements that can be addressed by the SA4CPS concept.

Data integrity and performance. There is a need to ensure the authenticity (REQ_UC3.3), integrity (REQ_UC1.1, REQ_UC1.3) and performance (REQ_UC1.3, REQ_UC1.19, REQ_UC1.21) of systems and data.

Data management infrastructure and data availability. Requirements targeting IoT and other data infrastructures (REQ_UC1.15), with monitoring (REQ_UC1.7, REQ_UC4.13, REQ_UC2.4) and data availability (REQ_UC4.1, REQ_UC4.2, REQ_UC4.13, REQ_UC3.10). **Digital Twins** are also explicitly addressed by some requirements (REQ_UC1.3, REQ_UC7.25, REQ_UC2.27).

Data collection and ML readiness. In order to support further machine learning operations and support operational insight creation, several requirements target the data lifecycle: data collection including notifications (REQ_STORY.2, REQ_UC2.2, REQ_UC4.10), making data available for further processing (REQ_UC2.3, REQ_UC2.5) and providing necessary annotation, preprocessing and ML model update functionality (REQ_UC2.6, REQ_UC2.9).

Wearables, synchronization and group data management. Requirements arising from (wearable) devices target data transfers and secure connectivity (REQ_UC1.12, REQ_UC7.1, REQ_UC7.4), data and device synchronization (REQ_UC2.10, REQ_UC9.1, REQ_UC9.2, REQ_UC9.8), and group data sharing (REQ_UC7.29).

5.3.4 Potential contributions from zero trust services and applications with DLT infrastructures (task 5.3)

Zero trust services and applications with DLT infrastructures [Èrange, ISEP, Digitalmente, CEL, Natlink, ACD, , ADLC, VTT, Otokar, Enerim,] (all UCs) Technical requirements and R&D of the enablers and means for zero trust services and applications with DLT infrastructures.

Zero trust system functional block (sec 3.8)

Achieving Zero Trust is largely dependent on other components, especially in a heterogeneous environment where different parties may use different systems for authentication, data access and authorization tasks. Thus, the Zero Trust is linked to Identity and Access Management, Authorization and Monitoring components. The Zero Trust system can perform several control and assisting tasks:

- Credential issuance to participants, expiration + revocation control
- Verification of credentials issued within and outside of the system
- Real-time credential state monitoring and notification / automated access control modification to resources and users controlled by the credentials
- Identity harmonization between identities provided by different parties and in different identity systems ("identity anchor")
- Attest to the identity and trust level of all participants, maintaining a federated trust hierarchy of participants (e.g., who can be trusted with which operations, who can issue credentials of which types and for which objects, etc.)

5.3.5 Potential contributions from sensing of security situation (task 5.4)

Sensing of security situation [F-Secure, Polar, ISEP, Digitalmente, CEL, Natlink, ACD, , ADLC, VTT, Otokar, Enerim,] (all UCs) Technical requirements and R&D of the enablers and means for sensing of the security situation

Potential Technical Requirements for Sensing of Security Situation:

Detection and Monitoring Capabilities:

- Real-time threat detection systems for critical infrastructure
- Multi-layer sensing mechanisms across physical and cyber domains
- Integration of IoT sensors with AI-driven analytics
- Anomaly detection algorithms for identifying security incidents

Distributed Ledger Technology Integration:

- Implementation of zero-trust security models
- Blockchain-based trust mechanisms for multi-stakeholder environments
- Secure data sharing protocols between different organizational entities
- Immutable logging of security events and responses

AI and Machine Learning Enablers:

- Intelligent pattern recognition for threat identification

- Predictive analytics for proactive security measures
- Automated response systems for incident management
- Adaptive learning capabilities for evolving threat landscapes

Communication and Interoperability:

- Secure communication protocols between sensors and control systems
- Standardized interfaces for cross-sector information sharing
- Real-time data exchange mechanisms for situational awareness
- Integration with existing security management systems

6 Enhanced project story

This chapter presents the enhanced project story, originally introduced in D1.1. The Figure 20 summarizes the updated story by illustrating its main phases and highlighting the key actions within each phase.



Journey to the port	Arrival and guidance	Unloading	Emergency	Resolution & Recovery	Departure from the port
• Driver confirms arrival at the port using the mobile app • Driver receives information about road conditions and chooses a safer route • The driver receives real-time, pre-arrival data on congestion levels at the truck unloading zone and optimizes her arrival timing accordingly	• Driver drives through the portal with license plate and QR code recognition • Cameras and radars check the vehicle and load condition • Identification system notifies the operator of driver's arrival • Drone assists with route direction and warnings	• Driver drives to the designated unloading area where the autonomous forklift is waiting for her • Forklift unloads timber load into the warehouse • Electric truck charges at the unloading area while unloading	• Alarm triggered due to smoke detection in a nearby warehouse • Monitoring system identifies potential fire and sends alerts and notification with danger zone and evacuation route • Drone projects guidance signs to a safe area • Control center monitors real-time locations of all entities • Autonomous work machines move to safe standby positions	• Automatic fire suppression system contain the spread of fire • Emergency services arrive and ensure that the fire no longer poses any danger • Control center updates situational view and reclassifies danger zone • Mobile app notifies that normal operations can resume • Forklift resumes unloading timber into the warehouse	• Driver receives notification that unloading is complete • Drone guidance is stopped, and drone returns to charging station • Driver chooses the shortest route back to the sawmill to pick up a new load • Mobile app corrects driver's wrong turn and offers drone guidance • Surveillance cameras and sensors monitor driver's exit

Figure 20. Summarized description of the enhanced project story.

The story of the operation in the port area in simple terms is described in the following:

“Truck driver Ella was driving towards the port of Oulu, carrying fresh northern Finnish sawn timber destined for Egypt. Before departing, Ella had confirmed her arrival in the PORT OULU Smarter app, from which she also received information that the road surface on the way to the port was particularly slippery and there had just been an accident on the shortest route, so Ella chose a slightly longer but less congested route. Additionally, the app informed her that there were currently several trucks unloading similar sawn timber loads at the port, and Ella's unloading time would be in two hours. Ella decided to stop for lunch on the way.

Upon arriving at the port, she drove through the portal, where her vehicle's license plate number and the QR code received from the app were recognized. Cameras also checked that there were no other people in the truck's cab. The portal's cameras and radars also scanned the combination to detect any deviations in load securing, load condition, or vehicle condition or temperature before arriving at the port.

The identification system sends information about Ella's arrival at the port area to the operator, who was already prepared for the arrival based on the advance notice, and an autonomous work machine is already waiting for Ella at the warehouse. Ella is arriving at the port of Oulu for the first time, so she has also ordered a drone to assist her, which projects the route direction and any warnings onto the road. Ella has checked the route guidance and the port's weather and road conditions in advance from the app, as well as the safety instructions video for the port. The route is also visible on the guide board inside the port gate, where Ella can also check that the load documents' codes and schedule match.

The drone projects a stop sign onto the ground, causing Ella to stop her truck and let container handler pass. The same information also appeared as a warning sound on the mobile app. When it is safe to continue, the drone projects the information onto the ground, and the app stops the warning sound.

Ella drives under the safe guidance of the drone to the designated unloading area, where the port operator's autonomously guided forklift is already waiting. When Ella's truck is positioned in the required spot markers and Ella opens the truck's side doors, the forklift unloads the timber load into the warehouse. At the same time, Ella's electric truck is charging at the charging station at the unloading area.

Suddenly, an alarm is triggered in the port area: smoke has been detected in a nearby warehouse. The port's monitoring system, which utilizes thermal cameras and air quality sensors, has automatically identified a potential fire and sent an alert to all operators and personnel in the area. Ella immediately receives a notification on her mobile app, showing a map with the danger zone and a safe evacuation route. At the same time, the drone that had previously guided Ella reactivates and begins projecting visible guidance signs onto the ground, directing her to a safe area. The port's digital twin displays real-time locations of all vehicles, machinery, and people, allowing the control center to ensure that everyone is moving away from the danger zone. Autonomous work machines halt their operations and move to safe standby positions.

Emergency services arrive quickly, and the automatic fire suppression system in the warehouse has already managed to contain the spread of the fire. The situation is brought under control in about ten minutes. The control center updates the situational view in the digital twin in real time, and the danger zone is reclassified as safe. The mobile app notifies all operators in the area that the situation is resolved, and normal operations can resume. Ella receives a notification in the app that unloading can now be completed. The autonomous forklift resumes its task and transfers the remaining timber into the warehouse.

When the unloading is complete, Ella thinks she knows the way back and has selected from the app that drone guidance can be stopped, so the drone returns to its charging station. The app informs her that the road is still slippery, but all routes are open, so Ella chooses the shortest route back to the sawmill to pick up a new load. Since Ella is in the area for the first time, she turns in the wrong direction at an intersection. The mobile app gives a notification directing Ella back to the correct route. Additionally, the mobile app asks if Ella wants drone guidance again. However, Ella now trusts that she knows the way and declines the request.

Throughout Ella's visit, the area surveillance cameras, radars, and sensors have been monitoring Ella's movements and ensuring that Ella finds the right route and exits the area. From the digital twin of the area surveillance, the movements of vehicles, work machines, and people can also be monitored in the area and intervene if necessary."

7 Links of contributions into the enhanced story

7.1 Potential links of use case contributions into the project story

The potential links of the use case contributions into the project story are discussed in the Table 10.

Table 10. Potential links of the use case contributions into the project story.

UC1: Connected & Autonomous lightweight logistics for in-factor/out-factory supply chain
Text in the storyline 1. Connected Logistics Network (Out-factory Integration) Use Case Goal: Enable continuous, connected monitoring and control across external logistics boundaries (e.g., vendors, roads, ports). Story Alignment: Ella receives real-time route updates and hazard information via the Smarter Port App, representing a connected supply chain outside the factory perimeter. The system integrates traffic data, unloading schedules, and environmental conditions, allowing for dynamic decision-making (e.g., alternate routing, stopovers). The truck's entry into the port is digitally registered and validated, illustrating secure cross-organizational handovers in logistics chains. 2. Autonomous Lightweight Logistics (In-factory Handling) Use Case Goal: Automate the movement of goods within controlled environments using lightweight and safe transport systems. Story Alignment: At the port, autonomous forklifts and container handlers are used to unload and transport sawn timber — a direct parallel to in-factory logistics automation. These machines act on pre-planned logistics flows and safety protocols, showcasing resilient, automated intra-facility movement. Ella's vehicle is guided into position by both the drone and the port's digital system, highlighting precise coordination between autonomous and human actors. 3. Situational Awareness for Safety and Efficiency Use Case Goal: Enhance human and system awareness to reduce risk and ensure responsive operations. Story Alignment: The use of drones projecting guidance cues, AR visualizations, and safety videos reflects enhanced human-centric situational awareness.

When a fire is detected, the real-time evacuation alerts and drone guidance enable safe and coordinated responses.

Continuous monitoring of worker locations, vehicle states, and environmental risks mirrors the intended safety infrastructure of your use case.

4. Cyber-Physical Resilience and Real-time Coordination

Use Case Goal: Ensure logistics continue safely and reliably under various cyber-physical disruptions or anomalies.

Story Alignment:

The system detects a potential fire using sensor fusion, triggers a multi-channel alert, and autonomous machines halt operations — all without human initiation.

The digital twin updates the danger zones, manages emergency flow, and resumes operations post-validation — a showcase of resilient digital infrastructure.

These features align with self-adaptive logistics systems that continue functioning through situational disruptions.

5. Cross-organizational Logging and Interoperability

Use Case Goal: Support secure data sharing and event logging across organizational boundaries.

Story Alignment:

Vehicle and cargo information (e.g., license plate, QR code, loading schedule) is recognized and logged automatically upon Ella's arrival.

The system notifies multiple stakeholders — including the operator, autonomous machines, and emergency services — via integrated data sharing protocols.

The unloading confirmation, drone commands, and app notifications represent a synchronized multi-party workflow, consistent with your cross-organizational objectives.

Visualization of the situation

<picture showing what could happen + clarification>

Identification of the related technical contributions (refs. e.g. to the sections 4 and 5)

- Decentralized Asset Integrity: Using blockchain + IoT integrity checks for supply-chain traceability
- Situational Awareness: Beacon-based positioning integrated with autonomous mobile robots and sensor fusion for real-time tracking.
- Cyber-Physical Resilience: Multi-path communication (Wi-Fi + 4G fallback) and self-healing connectivity
- Safety & Alerting Mechanisms: Collision prevention system; worker proximity alerts; emergency braking protocols for autonomous forklifts.
- Cross-Organizational Logging: Shared ledger across vendors and plant; supports service-level assurances traceability.
- Lab-Scale Testing Framework: Framework for controlled cyber-network drill, integrated digital twin of factory logistics, and performance measurement under adversarial conditions.

UC2: People and machine position awareness in production or logistics area

Text in the storyline

<copy or propose text into the storyline>.

UC2 will contribute to the following parts of the project story:

- Perception (sensing the environment and humans using sensors, cameras, available radios etc.)

"Upon arriving at the port, she drove through the portal.... The portal's cameras and radars also scanned the combination to detect any deviations in load securing, load condition, or vehicle condition or temperature before arriving at the port."

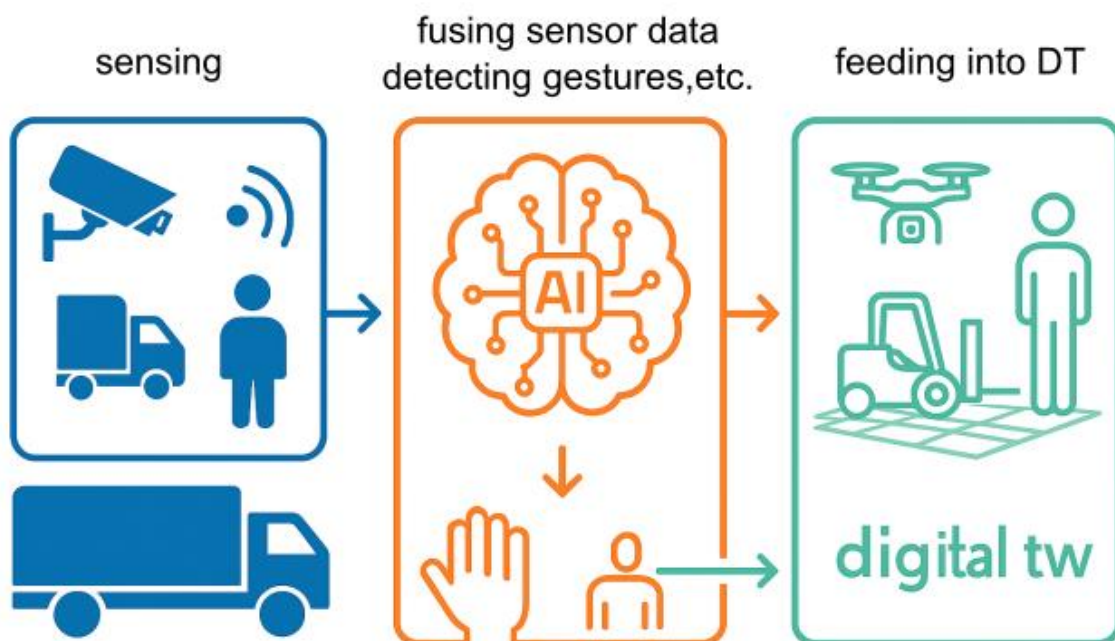
"Throughout Ella's visit, the area surveillance cameras, radars, and sensors have been monitoring Ella's movements"

- Comprehension (built by fusing the data from the various sensors and by data processing including AI/ML based approaches to provide accurate and comprehensible reading of the situation, e.g., human gestures, machines to close to humans)

"When Ella's truck is positioned in the required spot markers and Ella opens the truck's side doors, the forklift unloads the timber load into the warehouse."

"Cameras also checked that there were no other people in the truck's cab."

Visualization of the situation



Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

UC2 provides a multi-modal sensing layer—UWB radios, ToF cameras, and radars—that can scan trucks and personnel at the port gate for load, vehicle, and occupancy checks.

An AI/ML-driven fusion engine processes this data to verify cab occupancy, detect gestures, detect autonomous machines dangerously close to humans, etc.

All raw sensor data and also fused data can be fed into the port's digital twin for real-time situational awareness and continuous model retraining.

UC3: Map at the centre of situational awareness in addition to real time video [modirum]

Text in the storyline

Modirum Gespi will provide technical solutions supporting this part of the storyline:

Suddenly, an alarm is triggered in the port area: smoke has been detected in a nearby warehouse. The port's monitoring system, which utilizes thermal cameras and air quality sensors, has automatically identified a potential fire and sent an alert to all operators and personnel in the area. Ella immediately receives a notification on her mobile app, showing a map with the danger zone and a safe evacuation route. At the same time, the drone that had previously guided Ella reactivates and begins projecting visible guidance signs onto the ground, directing her to a safe area. The port's digital twin displays real-time locations of all vehicles, machinery, and people, allowing the control center to ensure that everyone is moving away from the danger zone. Autonomous work machines halt their operations and move to safe standby positions.

Emergency services arrive quickly, and the automatic fire suppression system in the warehouse has already managed to contain the spread of the fire. The situation is brought under control in about ten minutes. The control center updates the situational view in the digital twin in real time, and the danger zone is reclassified as safe. The mobile app notifies all operators in the area that the situation is resolved, and normal operations can resume. Ella receives a notification in the app that unloading can now be completed. The autonomous forklift resumes its task and transfers the remaining timber into the warehouse.

Visualization of the situation

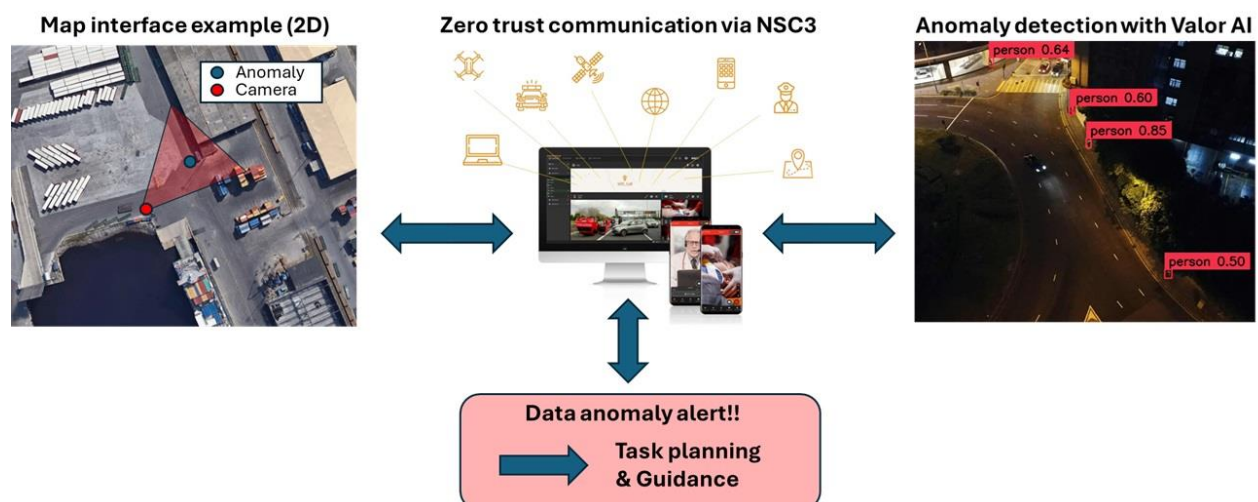


Figure 21. NSC3 with ValoR AI in project context.

Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

NSC3 platform and ValoR AI will provide the mapping, anomaly detection, alerting and task planning solutions for the overall concept. 3D mapping will be utilized for Port of Oulu area, for enhanced localization and contextual awareness particularly in outdoor environments. Within the mapping system, real-time solutions for target, event and anomaly detection will be developed using cameras and Yolo object detection model.

For more details, see section 4.3.

UC4: Critical electricity distribution grid operation in energy communities

Text in the storyline

UC4 is related to the energy sector, with focus on energy communities. It contributes to the technical and economic operation of the electricity distribution grid. It will improve the operation of the port distribution grid. Artificial intelligence will be used to handle energy data. This UC provides situational awareness of the electricity consumers in order to rise their participation in demand response programs. It will improve the process of charging vehicles in the port area.

Visualization of the situation

When a truck is ready for charging in the port area and a demand response event is declared, the truck may be scheduled to charge later in time. Even when there is no demand response event, according to the community needs, it may be scheduled to charge later. The truck owner will receive a compensation for this change in the schedule.

Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

This UC4 is related with the machine learning and explainable AI algorithms developed in WP3 for user perception and comprehension of energy decisions. Moreover, security on access to private user data will be handled. In this UC according to the developments of WP2.

UC5: Security and location awareness of persons

Text in the storyline

“Truck driver Ella was driving towards the port of Oulu, carrying fresh northern Finnish sawn timber destined for Egypt. Before departing, Ella had confirmed her arrival in the PORT OULU Smarter app, from which she also received information that [...]”

“Ella has checked the route guidance and the port's weather and road conditions in advance from the app, as well as the safety instructions video for the port.”

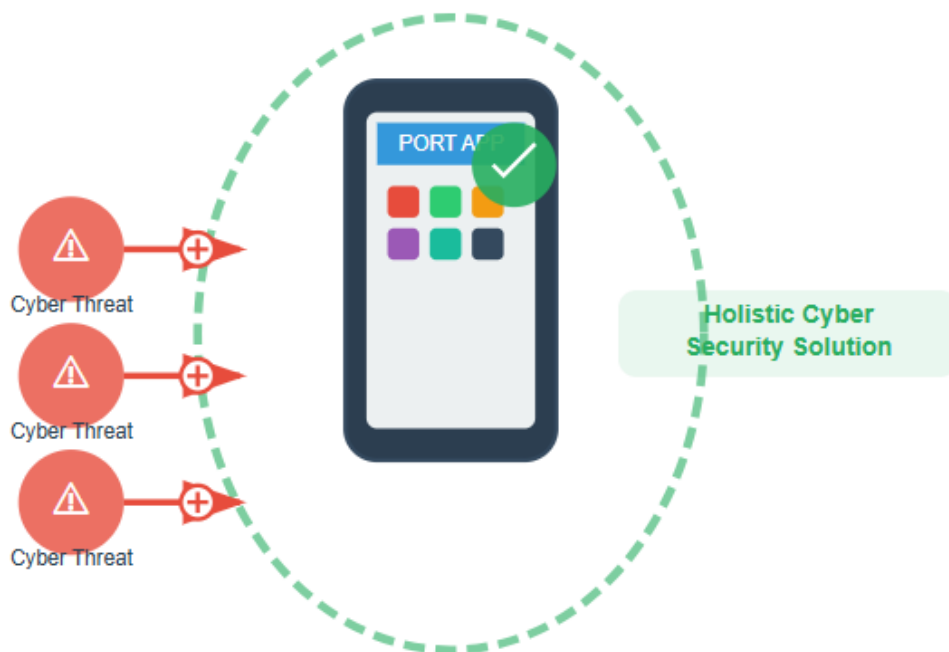
“Ella immediately receives a notification on her mobile app, showing a map with the danger zone and a safe evacuation route.”

“The mobile app notifies all operators in the area that the situation is resolved, and normal operations can resume. Ella receives a notification in the app that unloading can now be completed.”

“When the unloading is complete, Ella thinks she knows the way back and has selected from the app that drone guidance can be stopped, so the drone returns to its charging station. The app informs her that the road is still slippery, but all routes are open, so Ella chooses the shortest route back to the sawmill to pick up a new load.”

Since Ella is in the area for the first time, she turns in the wrong direction at an intersection. The mobile app gives a notification directing Ella back to the correct route. Additionally, the mobile app asks if Ella wants drone guidance again. However, Ella now trusts that she knows the way and declines the request.”

Visualization of the situation



Truck driver Ella uses her mobile device many times during the story to check information on a mobile app. The mobile device Ella uses also contains variety of other apps for both work and personal use and is vulnerable to cyber threats. However, Ella has a holistic cyber security solution on her mobile device. A contextually relevant and proactive engagement protection model adjusts with what Ella need in moments requiring protection and when she expects a sense of security.

Potential cyber threats associated with the described story that would need contextual and proactive engagement model based protection:

- Cybercriminals could create fake versions of the PORT OULU Smarter app to steal Ella's credentials or inject malware onto her device. If she downloads from unofficial sources, she might unknowingly install a compromised version.
 - Protection: URL protection
- When Ella connects to PORT OULU (public) Wi-Fi networks at truck stops or the port, attackers could intercept her communications with the app, potentially stealing sensitive information or manipulating the data she receives about routes, weather, or safety instructions.

- Protection: Wi-Fi protection
- If the app tracks cargo information, criminals could use this data to plan thefts, knowing exactly when and where valuable timber shipments will be located.
 - Protection: privacy protection
- Malicious actors could exploit the app's location tracking capabilities to monitor Ella's patterns, potentially for criminal purposes or unauthorized surveillance and collection of sensitive information
 - Protection: privacy protection
- Since Ella's device contains both work and personal apps, cross-app data sharing vulnerabilities could expose personal information through the work application.
 - Protection: privacy protection

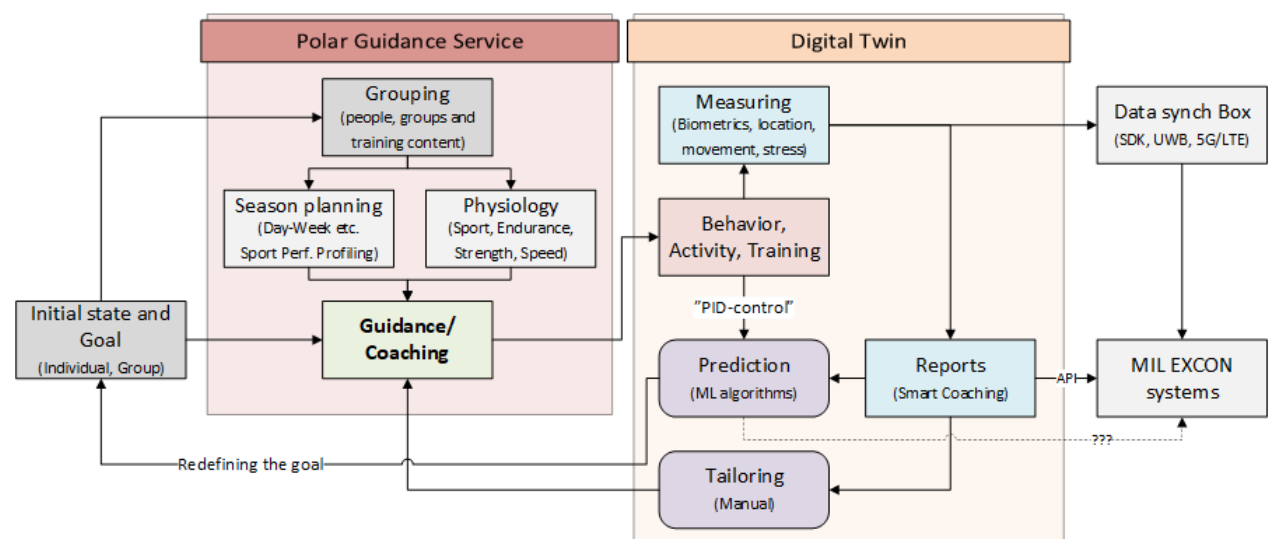
Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

UC7: Multirouting secure awareness system for hunting and outdoors

Text in the storyline

In the project, Polar Electro will implement a device and system design that is tasked with collecting biometric data from wearable devices in real time, transferring the data to people for reviewing the operations and analysis and making a decision on action. The processing of information is carried out in a service system that makes it possible to master the pedagogical methods of guidance that are used to promote physical and mental functional capacity and in the production of guidance content intended for this purpose.

Visualization of the situation



Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

Research and development of radio technical equipment for two-way communication and configuration of wearable devices. Research and development of a service system for operational management and data processing

In Natlink system

There is no direct contribution in to main scenario, but there is situational awareness contribution in to Surrounding ecosystem

In the vicinity of the port area, hunters and their dogs move around, and inside the port area, customs official dogs examine cargoes, sometimes hunters also come inside the port area to tax the increased wild animal population. Some of the port area and ships are areas with poor mobile phone coverage, and it must be possible to maintain the most important functions even if there is a disturbance in the mobile network in the event of an accident, for example. Civil servants may move in dangerous conditions

Potential contribution area in the port environment

The system is able to share the situational information with users' phones even if no mobile phone connection is available because alternative radios are used. In addition, when the connection is incomplete or exists at some point, a situation and selected parts of the twins can be transmitted to external systems. Hunters and authorities working in the area receive information about not safe places and working dogs. The vital signs of the civil service can be monitored remotely, in which case unconsciousness caused by, for example, anoxic situations can be detected.

Potential contribution area into the Sa4CPS concept

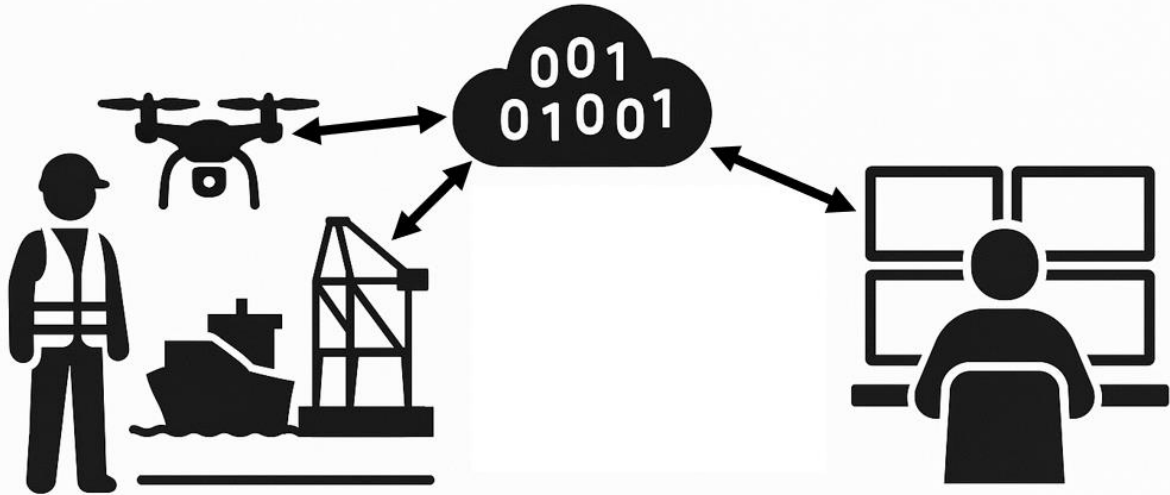
UC7 focuses on improving and connecting devices even if the cellular network is lost or unavailable, as well as on collecting and controlling data from various sensors without direct internet access.

UC8: Secure, trustworthy and reliable drone operation control system

Text in the storyline

The drone guiding Ella is monitored remotely by a remote pilot simultaneously in charge of many similar guidance missions. Through collection of the drone's telemetry as well as integration with other relevant sources of information on the drone's flight, a system actively monitors the operations's progress, and is ready to immediately land the drone at a safe location and warn the pilot if safety is compromised, avoiding risks to Ella, assets of the Port of Oulu, and the drone itself.

Visualization of the situation



Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

The platform centralising flight information, providing operational status to a remote pilot and detecting situations to react to will be developed by ADLC as part of UC 8.

UC9: Situational awareness and secure management of digital twin assets [vintecc]

Text in the storyline

<copy or propose text into the storyline>.

Visualization of the situation

<picture showing what could happen + clarification>

Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

<>.

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7.2 Potential links of WP3-5 contributions into the project story

The potential links of the WP contributions into the project story are discussed in the Table 11.

Table 1112. Potential links of the WP contributions into the project story.

WP3 perception and comprehension
Text in the storyline
T3.1 Physical Environment Sensing

Environmental Sensing with Cameras, Radars, and Sensors

"Upon arriving at the port, she drove through the portal, where her vehicle's license plate number and the QR code received from the app were recognized. Cameras also checked that there were no other people in the truck's cab. The portal's cameras and radars also scanned the combination to detect any deviations in load securing, load condition, or vehicle condition or temperature before arriving at the port."

Area-wide Threat and Incident Sensing

"The port's monitoring system, which utilizes thermal cameras and air quality sensors, has automatically identified a potential fire and sent an alert to all operators and personnel in the area."

T3.2 Interaction-based Sensing of Information Context

Gesture, App, and Interaction Sensing for Adaptive Guidance

"Ella is arriving at the port of Oulu for the first time, so she has also ordered a drone to assist her, which projects the route direction and any warnings onto the road. Ella has checked the route guidance and the port's weather and road conditions in advance from the app..."

T3.3 Information & Sensor Fusion

Sensor and Data Fusion for Real-time Situational Awareness

"The port's digital twin displays real-time locations of all vehicles, machinery, and people, allowing the control center to ensure that everyone is moving away from the danger zone."

"Throughout Ella's visit, the area surveillance cameras, radars, and sensors have been monitoring Ella's movements and ensuring that Ella finds the right route and exits the area."

T3.4 Modelling and Tools for Comprehension

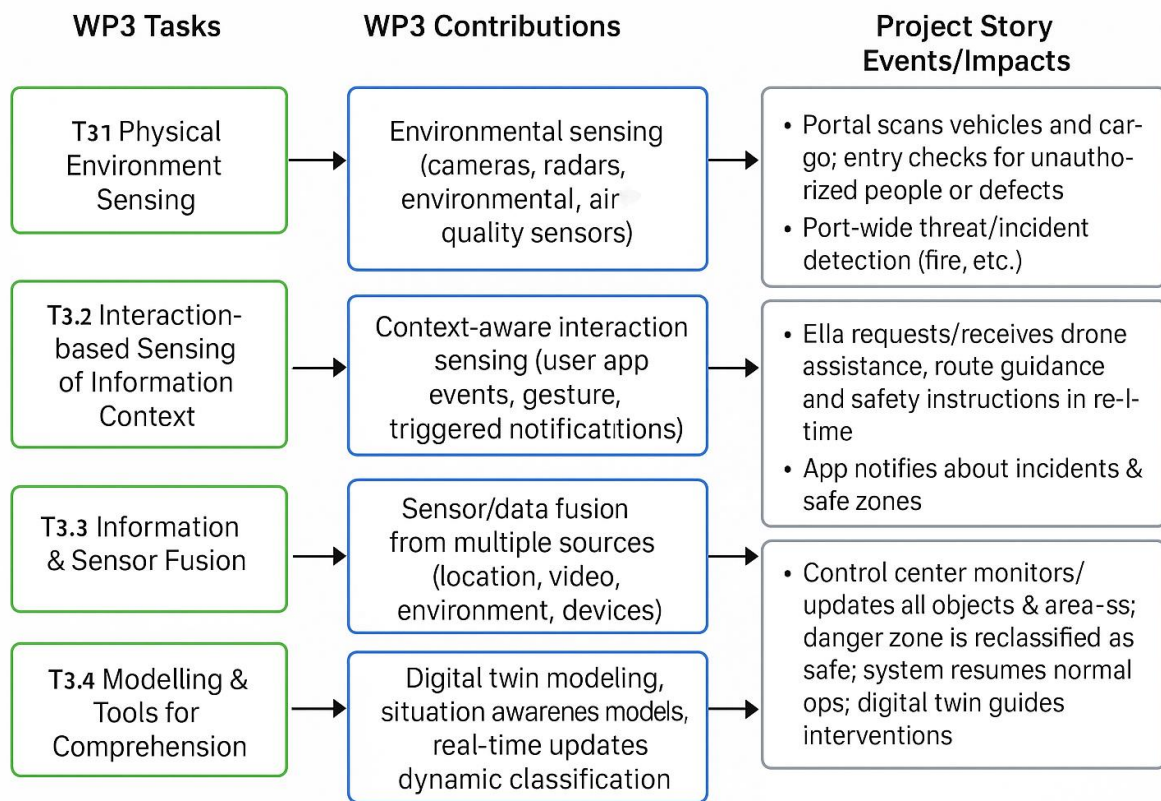
Digital Twin and Situation Modelling for Decision Support

"From the digital twin of the area surveillance, they can also monitor the movements of vehicles, work machines, and people in the area and intervene if necessary."

Dynamic Re-classification and Recovery Modelling

"The control center updates the situational view in the digital twin in real time, and the danger zone is reclassified as safe. The mobile app notifies all operators in the area that the situation is resolved, and normal operations can resume."

Visualization of the situation



Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

T3.1 Physical Environment Sensing

Environmental Sensing with Cameras, Radars, and Sensors

WP3 can contribute by developing and integrating a network of cameras, radars, and environmental sensors for real-time detection and monitoring of vehicles, cargo, and personnel entering and moving within the port.

Area-wide Threat and Incident Sensing

WP3 can provide sensor technologies for environmental monitoring.

T3.2 Interaction-based Sensing of Information Context

Gesture, App, and Interaction Sensing for Adaptive Guidance

WP3 can develop systems that sense and interpret user interactions (requests for guidance, confirmation of information).

T3.3 Information & Sensor Fusion

Sensor and Data Fusion for Real-time Situational Awareness

WP3 can contribute by integrating data from multiple sensor modalities (location, cameras, environmental sensors) and fusing them via AI/ML techniques.

T3.4 Modelling and Tools for Comprehension

Digital Twin and Situation Modelling for Decision Support

WP3 can contribute advanced modelling tools that allow operators to comprehend, simulate, and predict the evolution of situations, supporting effective intervention and incident management.

Dynamic Re-classification and Recovery Modelling

WP3 can enable dynamic updating and visualization of situational models, ensuring all stakeholders are accurately and promptly informed as the environment and threat levels change.

WP4 status, projections and actions

Text in the storyline

Task 1: Digital Twin&Dashboard

"The port's digital twin displays real-time locations of all vehicles, machinery, and people, allowing the control center to ensure that everyone is moving away from the danger zone."

"The control center updates the situational view in the digital twin in real time, and the danger zone is reclassified as safe."

"From the digital twin of the area surveillance, they can also monitor the movements of vehicles, work machines, and people and intervene if necessary."

Task2: Positioning & tracking of objects

"Ella drives under the safe guidance of the drone to the designated unloading area, where the port operator's autonomously guided forklift is already waiting."

"Upon arriving at the port, she drove through the portal, where her vehicle's license plate number and the QR code received from the app were recognized."

"The port's monitoring system... displays real-time locations of all vehicles, machinery, and people."

Task3: Autonomous operations

"An autonomous work machine is already waiting for Ella at the warehouse."

"The port operator's autonomously guided forklift unloads the timber load into the warehouse."

"Autonomous work machines halt their operations and move to safe standby positions."

Task4: Detection and responses

"The port's monitoring system, which utilizes thermal cameras and air quality sensors, has automatically identified a potential fire and sent an alert to all operators and personnel in the area."

"Emergency services arrive quickly, and the automatic fire suppression system in the warehouse has already managed to contain the spread of the fire."

"Cameras also checked that there were no other people in the truck's cab."

Task5: Real-time deep 3D maps & tools

"The drone projects a stop sign onto the ground... The same information also appeared as a warning sound on the mobile app."

"The drone... begins projecting visible guidance signs onto the ground, directing her to a safe area."

"The app shows a map with the danger zone and a safe evacuation route."

"The route is also visible on the guide board inside the port gate."

Visualization of the situation

Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

1. Digital Twin and Operational Dashboard Integration

Development and real-time synchronization of a digital twin that provides a live visual representation of port operations, including dynamic updates of vehicle positions, alerts, and status changes. Implementation of a centralized dashboard enabling human operators to interact with situational data, override autonomous systems if needed, and manage responses to anomalies or emergencies. Contribution to context-aware system architectures capable of integrating static infrastructure with mobile assets and environmental sensors.

2. High-Precision Positioning and Real-Time Object Tracking

Use of multi-modal positioning technologies, including GNSS, radar, computer vision, and QR-based recognition, to track trucks, autonomous machines, and personnel throughout the port. Development of interoperable tracking mechanisms that enable both internal (intra-organizational) and external (cross-organizational) assets to be localized in a secure and verifiable manner. Advancement in temporal-spatial correlation frameworks to associate events, assets, and operational phases in real time.

3. Autonomous Operations and Fleet Coordination

Deployment of autonomous forklifts and container handlers integrated into a smart logistics flow, capable of executing delivery, unloading, and repositioning tasks without human intervention. Design of machine-to-infrastructure (M2I) communication layers, allowing autonomous systems to synchronize with digital infrastructure (e.g., route markers, gates, beacons). Contribution to the safety-aware autonomy domain, where robotic systems not only plan optimal paths but also react to human presence, emergency conditions, and operational uncertainties.

4. Automated Event Detection and Emergency Response Systems

Implementation of sensor fusion methods (thermal imaging, air quality monitoring, visual analytics) to detect early-stage anomalies such as fires or unauthorized personnel. Integration of automated alerting chains, ensuring that both human stakeholders and autonomous systems are notified and guided in real time during critical situations. Contribution to the development of self-healing infrastructure protocols, where systems dynamically adapt and reroute workflows during safety or operational interruptions.

5. Immersive 3D Spatial Tools and Augmented Guidance

Development of interactive, real-time 3D mapping tools that inform users of spatial context, danger zones, route changes, and object positions.

Integration of augmented reality (AR) projection systems, such as drones projecting stop signs and directional cues, to support human operators in complex outdoor environments.

Contribution to human-centric CPS interfaces, where the combination of spatial mapping and multimodal feedback (visual + acoustic) enhances situational awareness and trust in automation.

WP5 trust, security and privacy

Text in the storyline

Potential threats related to trust, security and privacy in the Sa4CPS storyline and the framework for situational awareness for cyber physical systems.

Mobile Device and App-Based Threats

Malicious App Impersonation: Cybercriminals could create fake versions of the PORT OULU Smarter app to steal Ella's credentials or inject malware onto her device. If she downloads from unofficial sources, she might unknowingly install a compromised version.

Man-in-the-Middle Attacks: When Ella connects to public Wi-Fi networks at truck stops or the port, attackers could intercept her communications with the app, potentially stealing sensitive information or manipulating the data she receives about routes, weather, or safety instructions.

GPS Spoofing: Attackers could send false GPS signals to make Ella's device think it's in a different location, potentially misdirecting her to dangerous areas or causing her to miss critical safety notifications about the evacuation zone.

Data and Communication Threats

False Emergency Notifications: Cybercriminals could send fake emergency alerts through compromised systems, causing unnecessary panic or drawing Ella away from safe areas. Conversely, they could block legitimate emergency notifications.

Route Manipulation: Attackers could compromise the app's routing system to direct Ella through unsafe areas, cause delays, or lead her to locations where she might be vulnerable to physical crimes.

Cargo Tracking Exploitation: If the app tracks cargo information, criminals could use this data to plan thefts, knowing exactly when and where valuable timber shipments will be located.

Infrastructure and IoT Threats

Drone Hijacking: The guidance drones mentioned in the story could be compromised, potentially providing false navigation information or being used for surveillance of Ella's movements and cargo.

Port System Infiltration: Attackers could breach the port's operational systems to create fake "all clear" notifications after emergencies, potentially putting Ella and others at risk by resuming operations prematurely.

Device Sensor Manipulation: Sophisticated attacks could target the sensors that detect road conditions, weather, or other environmental factors, providing Ella with inaccurate safety information.

Privacy and Surveillance Concerns

Location Tracking Abuse: Malicious actors could exploit the app's location tracking capabilities to monitor Ella's patterns, potentially for criminal purposes or unauthorized surveillance.

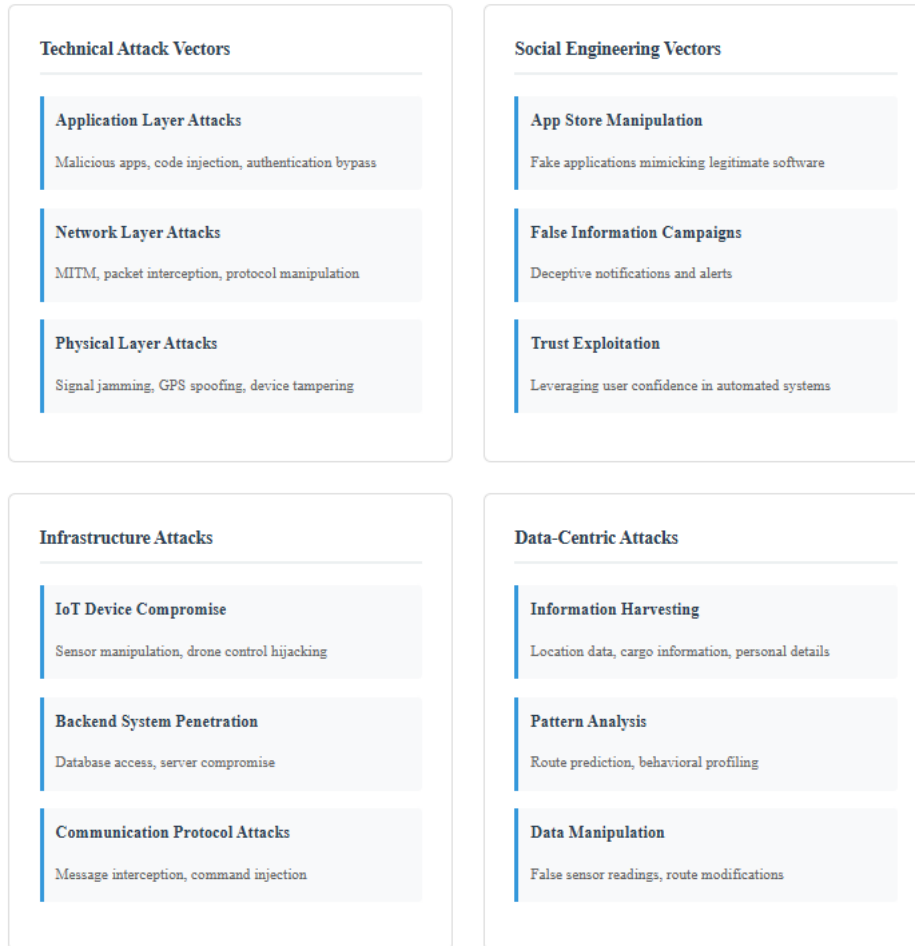
Personal Data Harvesting: Since Ella's device contains both work and personal apps, cross-app data sharing vulnerabilities could expose personal information through the work application.

Visualization of the situation

Figure 1: Operational Timeline and Threat Exposure Points



Figure 2: Attack Vector Classification



Identification of the related technical contributions (refs. e.g. from sections 4 and 5)

The feasible technical requirements should include:

1. Zero-Trust Architecture
2. Multi-layered Authentication (biometric + certificate + behavioral)
3. Industry-Specific Compliance (automotive, energy, maritime)
4. AI-Enhanced Security Monitoring with situational awareness
5. Cross-Domain Interoperability (logistics, energy, transportation)
6. Privacy-by-Design with GDPR compliance
7. Real-time Performance
8. Digital Twin Security for cyber-physical system protection

8 Preliminary Sa4CPS concept framework description

A preliminary view to the main functional blocks (FB) of the Sa4CPS concept framework is depicted in the Figure 22. The reasoning of these functional blocks have been clarified in the section 3 of this report. Short descriptions of the FBs, related tasks of the WP3-5 and main contribution areas of the UCs into the FBs are overviewed in the Table 13. The UC contributions into the tasks are also further depicted in the Table 14

Preliminary view into the Sa4CPS Concept

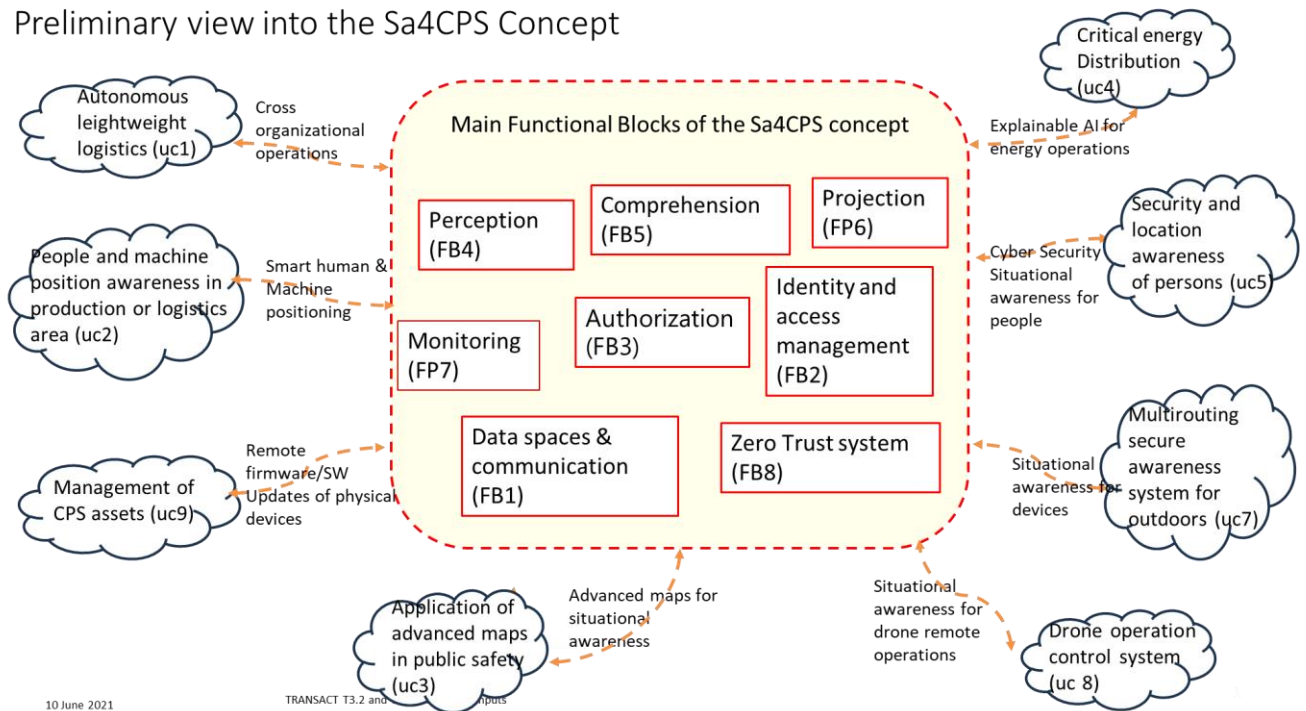


Figure 22. A preliminary view to the Sa4CPS concept framework – main functional blocks.

Table 13. Description of the FBs, related tasks of the WP3-5 and contribution areas of the UCs into the FBs.

	Short description of the FB	Related tasks of the WP3-5	Contribution area of the UC into the FB
FB1 Data Spaces and Communication	Harmonized approach(es) to provisioning, storing and connecting to sensor and other data, including related communications and data integrity.	T3.1 Physical environment sensing T5.2 Secure information spaces, sharing and overlayed routing	Location & carriage data [UC1] Human presence [UC2] Sync hub [UC7] Multimode radio [UC7] Secure & trusted comms [UC1]
FB2 Identity and Access Management	Means to incorporate varying identification and access management schemes into the overall system, and managing group and community data access.	T5.1 Identification, verification and access control T5.2 Secure information spaces, sharing and overlayed routing	Community mgt [UC4] CPS asset mgt [UC9]
FB3 Authorization	Fine-grained control mechanisms for requesting, verifying and granting access to various types of resources, including evaluation of trust hierarchies.	T5.1 Identification, verification and access control T5.3 Zero trust services and applications with DLT infrastructures	Asset monitoring & traceability [UC1]
FB4 Perception	Infrastructure and data processing for collecting, processing, and interpreting sensor data as a source of situational awareness.	T3.1 Physical environment sensing T3.2 Interaction based sensing of information context T3.4 Modelling and tools for comprehension T4.2 Positioning and tracking of objects T4.5 Real-time deep 3D maps and tools	Community mgt [UC4] CPS asset mgt [UC9]
FB5 Comprehension	Infrastructure to enable risk and threat interpretation and operational context understanding, including human and machine behaviour interpretation.	T3.2 Interaction based sensing of information context T3.3 Information and sensor fusion T3.4 Modelling and tools for comprehension T3.5 Detection and responses	Security moments detection [UC5] AI sensor fusion edge/cloud, Dig twin comprehension support, Model&detect hazardous situations[UC2] Trigger events [UC3] Explainable distributed AI, Event detection + response wrt energy communities [UC4] Human (stress) state assessment UC7 Harmful CPS update prevention [UC9]
FB6 Projection	Infrastructure for operational forecasting & event anticipation, predictive safety & risk management using adaptive intelligence and system learning.	T4.2 Positioning and tracking of objects T4.3 Autonomous operations T4.4 Detection and responses	Movement & projection of automated vehicles [UC8] Event detection + response wrt energy communities [UC4] Object/target tracking [UC3]
FB7 Monitoring	Support for system and infrastructure monitoring and decision support, including status of permissions / authorizations, resources, credentials and their history.	T4.1 Digital twins and dashboards T5.4 Sensing of security situation	Asset monitoring & traceability [UC1] Data sharing API + security awareness [UC4]
FB8 Zero Trust system	Functionality to support credential issuance and verification, identity harmonization between multiple identity systems, continuous identity and access control verification with support for trust hierarchies, data integrity, and traceability.	T5.3 Zero trust services and applications with DLT infrastructures	Asset monitoring & traceability [UC1] (Zero Trust communication via NSC3 [UC3])

Table 14. Contribution areas of the use cases into the functional blocks in different tasks of the WPs3-5.

	T3.1 Physical environment sensing	T3.2 Interaction based sensing of information context	T3.3 Information and sensor fusion	T3.4 Modelling and tools for comprehension	T4.1 Digital twins and dashboards	T4.2 Positioning and tracking of objects	T4.3 Autonomous operations	T4.4 Detection and Responses	T4.5 Real-time deep 3D maps and tools	T5.1 Identification, verification and access control	T5.2 Secure information spaces, sharing and overlayed routing	T5.3 Zero trust services and applications with DLT infrastructures	T5.4 Sensing of security situation
UC1 Autonomous lightweight logistics	FB1 (loc & carriage data)				FB7 (asset monitoring & traceability)	FB4 (loc & carriage data)					FB1 (Robust backends, Secure data transmission & trusted comms)	FB3, FB8 (Asset monitoring & traceability)	
UC2 People and machine position awareness in production or logistics area	FB1 (human presence)	FB4 (human presence, machine activity)	FB5 (AI sensor fusion edge/cloud)	FB4 (Human / machine interaction modelling) FB5 (Dig twin comprehension module support)		FB4 (human presence)		FB5 (model & detect hazardous situations)					
UC3 Application of advanced maps in public safety	FB4 (Object,event detection)					FB6 (object/target tracking)		(Object,event detection, events, triggers)	FB4 (3D maps, RT object detection & classification)			FB8 (Zero Trust communication via NSC3)	
UC4 Critical energy distribution								FB5, FB6 (digital twins, autonomous operation, and detection and response in the context of energy communities)					FB7 (data sharing api, awareness of data security)
UC5 Security and location awareness of persons		FB4, FB5 (Security moments detection)		FB5 (explainable / distributed AI)	FB2?						FB2 (CIS/community management)		
UC7 Multirouting secure awareness system for outdoors	FB1 (sync hub) FB1 (multimode radio access, multicomunicator)	FB5 (human state assessment, e.g. stress) FB1 (secure multi-device comms)				FB4 (positioning via gnss, inertia)			FB4 (maps?)				
UC8 Drone operation control system	FB4 (Perception of object & surroundings)						FB6 (movement + projection of automated vehicles)						
UC9 Management of CPS assets								FB5 (CPS asset management / harmful update prevention)		FB2 (CPS asset management / harmful update of edge devices)			

9 Discussion and concluding remarks

This deliverable was based on the results of the parallel requirement analysis process, which was carried out focusing on the needs of the project use cases and the project story. 82 of the detected 123 technological requirements were categorized to the Sa4CPS concept architecture level requirements³. When analysing the commonalities in these requirements, we have identified 8 functional blocks clarified in the section 3 of this deliverable. These FBs have been used to structure the preliminary Sa4CPS concept in this report, so that the contributions of UCs and technological tasks towards the concept is a bit more clear. However, it is important to point out that these FBs are still quite abstract in nature, and plan is to continue and open their internal functional structure, relationships and interfaces to practical objects in which the real secure situational awareness capabilities are needed.

The content, focus and plans of the UCs have been described in this report so that their potential contribution areas into the FBs of the Sa4CPS concept are clarified. It is clear now that each UC has a specific role in the development of Sa4CPS concept, its FBs and links to the technological tasks of the project WPs3-5. However, there is still needs to continue this work to further clarify the content of the identified contribution areas of the UCs into the concept, see Table 13 and Table 14.

The technological tasks of WPs3-5 have been quite much still in the background, and therefore their relationships to the FBs are clarified in this report only in preliminary way. It is expected that this deliverable provides a solid starting point and working framework for the R&D works in these technological tasks. In addition, the links to the project UCs and the Sa4CPS concept is now a bit more clear.

The project story has been structured and enhanced, and the links of the UCs and WP3-5 contributions towards solving the visions and challenges indicated by the story is now a bit more obvious. This is envisioned to help all of the project partners to adjust the focus and plans so that the resulting actions are beneficial to their business cases but also to the results deployment in the port environment.

The preliminary Sa4CPS concept description clarified in the section 8 depicts a view to the targets so that the technological WPs3-5 and the UCs of the project can now proceed in the technological R&D towards specification and design of the solutions. The plan is to let them now proceed more deeper in the technological development, and then in the next iteration of the project come back and make a new version of the Sa4CPS concept description.

³ D2.1 Technological requirements architecture.

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